Chapter 2

Foundations

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2.1 Set Theory

2.1.1 Relations

Definition 2.1.1 (Binary Relation)
*A **binary relation** (or just **relation**) R on a pair of sets (X, Y) is subset of the cartesian product $X \times Y$.*

We write xRy to mean precisely $(x, y) \in R$.

Definition 2.1.2 (Converse Relation)
Let R be a binary relation on (X, Y) then we the converse relation R^T on (Y, X) given by

$$yR^T x \iff xRy$$

Definition 2.1.3 (Domain and Range)
*Let R be a relation on (X, Y). We define the **domain** of R to be*

$$\text{dom}(R) \coloneqq \{x \in X \mid \exists y \in Y \text{ s.t. } xRy\}$$

*and the **range** of R*

$$\text{range}(R) \coloneqq \{y \in X \mid \exists x \in X \text{ s.t. } xRy\}$$

Definition 2.1.4 (Equivalence Relation)
Let R be a binary relation on (X, X). It is said to be

*a) **reflexive** if xRx for all $x \in X$*
*b) **symmetric** if $xRy \implies yRx$ for all $x, y \in X$*
*c) **transitive** if $xRy \wedge yRz \implies xRz$ for all $x, y, z \in X$*

*A relation which satisfies all these properties is called an **equivalence relation** on X. In this case we would write*

$$x \sim y$$

*instead of xRy. For an element $x \in X$ denote the **equivalence class** of x by*

$$[x]_R = \{y \mid xRy\}$$

Note that $R^T = R$.

Definition 2.1.5 (Partition)
*Let X be a set an $\mathcal{F}$ a family of subsets of X. It is said to be a **partition** if*

a) $X = \bigcup_{A \in \mathcal{F}} A$
b) $A, B \in \mathcal{F} \implies A = B$ or $A \cap B = \emptyset$

Proposition 2.1.6 (Equivalence Classes form a Partition)
Let E be an equivalence relation on X. The family

$$\mathcal{F} = \{[x]_E \mid x \in X\}$$

*forms a partition of X. Denote by X/E the family of equivalence classes, called the **quotient** of X with respect to E.*

Proof. It's clear that

$$X = \bigcup_{A \in \mathcal{F}} A$$

because by reflexive-ness $x \in [x]_E$ for all $x \in X$.

We claim that for any $z \in [x]_E$ we have $[z]_E = [x]_E$. Suppose $y \in [x]_E$ then xRz and xRy. By symmetry and transitivity we then have zRy which implies $y \in [z]_E$. In other words $[x]_E \subseteq [z]_E$. By symmetry of R we have $x \in [z]_E$, so by the same token $[z]_E \subseteq [x]_E$, which shows they are equal.

Therefore it's clear that $[x]_E \cap [y]_E \neq \emptyset \implies [x]_E = [y]_E$ and thus $\mathcal{F}$ forms a partition. ∎

Definition 2.1.7 (Composition of Relation)
Suppose R is a relation on (X, Y) and S a relation on (Y, Z). We define the composition $S \circ R$ on (X, Z)

$$S \circ R = \{(x, z) \mid \exists y \in Y \text{ s.t. } xRy \text{ and } yRz\}$$

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2.1.2 Functions

Definition 2.1.8 (Function)
*A **function** $f : X \to Y$ consists of a binary relation $\Gamma(f)$ on (X, Y) such that*

- $\mathrm{dom}(f) = X$
- $\Gamma(f)$ *is* ***single-valued*** *that is* $x\Gamma(f)y \wedge x\Gamma(f)y' \implies y = y'$

Equivalently for all $x \in X$ there exists precisely one $y \in Y$ such that $x\Gamma(f)y$.

We write $f(x) = y$ for the unique element $y \in Y$ such that $x\Gamma(f)y$.

Proposition 2.1.9 (Equality of Functions)
Two functions $f, g : X \to Y$ are equal if and only if $f(x) = g(x)$ for all $x \in X$.

Proposition 2.1.10 (Composition of Functions)
Let $f : X \to Y$ and $g : Y \to Z$ be functions then the composition $\Gamma(g) \circ \Gamma(f)$ is still a function, which we write $g \circ f$, and

$$(g \circ f)(x) = g(f(x))$$

Furthermore composition is ***associative*** *in the sense that*

$$(h \circ g) \circ f = h \circ (g \circ f)$$

Definition 2.1.11 (Injective, Surjective and Bijective)
Let $f : X \to Y$ be a function then we say

- *f is* ***injective*** *if $f(x) = f(x') \implies x = x'$*
- *f is* ***surjective*** *if for all $y \in Y$ there exists x such that $f(x) = y$*
- *f is* ***bijective*** *if it is both injective and surjective*

Definition 2.1.12 (Inverse Function)
Let $f : X \to Y$ and $g : Y \to X$ be functions. We say

- *g is a* ***left inverse*** *for f if $g \circ f = 1_X$*
- *g is a* ***right inverse*** *for f if $f \circ g = 1_Y$*
- *g is a* ***two-sided inverse*** *for f if it is both a left and right inverse*

Proposition 2.1.13
Let $f : X \to Y$ be a function then

- *f is* ***injective*** *if and only if it has a* ***left inverse***
- *f is* ***surjective*** *if and only if it has a* ***right inverse***
- *f is* ***bijective*** *if and only if it has a* ***two-sided inverse***

Definition 2.1.14 (Idempotent Function)
A function $p : X \to X$ is ***idempotent*** *$p \circ p = p$.*

Lemma 2.1.15 (Idempotent Criterion)
Let $p : X \to X$ be a function. Then $\mathrm{Fix}(p) \subseteq \mathrm{Im}(p)$ and these are equal if and only if p is idempotent.

2.1.3 Partial Orders

Definition 2.1.16 (Preorder)
A binary relation $\leq$ on (X, X) is a ***preorder*** *if*

- ***reflexivity*** *$x \leq x$*
- ***transitivity*** *$x \leq y$ and $y \leq z \implies x \leq z$*

We may refer to $(X, \leq)$ as a ***preordered set*** *or* ***preorder****.*

Definition 2.1.17 (Poset)
A binary relation $\leq$ on (X, X) is a ***partial order*** *if*

- ***reflexivity*** *$x \leq x$*

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- ***antisymmetry*** $x \leq y$ and $y \leq x \implies y = x$
- ***transitivity*** $x \leq y$ and $y \leq z \implies x \leq z$

*We may refer to $(X, \leq)$ as a **partially ordered set** or **poset**. Equivalently, a poset is a preorder which is additionally antisymmetric.*

Definition 2.1.18 (Dual Poset)
*Given a poset $(X, \leq)$ denote the set X with the converse relation by $\left(X, \overset{d}{\leq}\right)$. This is the **dual poset** to $(X, \leq)$.*

Example 2.1.19
Let $\mathcal{F}$ be a family of subsets of a fixed set E. Then $(\mathcal{F}, \subseteq)$ is a poset ordered under inclusion.

Definition 2.1.20 (Top and Bottom)
*Let $(X, \leq)$ we say $\top$ (resp. $\bot$) is a **top element** (resp. **bottom element**) if it is greater than (resp. less than) every element of x. In this case it is unique.*

Definition 2.1.21 (Monotone/Antitone Function)
Let $(X, \leq)$ and $(Y, \leq)$ be posets. A function $f : X \to Y$ is

- ***monotone / order-preserving*** *if $x \leq y \implies f(x) \leq f(y)$*
- ***antitone / order-reversing*** *if $x \leq y \implies f(y) \leq f(x)$*
- *a **monotone embedding** if $x \leq y \iff f(x) \leq f(y)$*
- *an **order isomorphism** if it is bijective and monotone*
- *a **dual isomorphism** if it is bijective and antitone*

Proposition 2.1.22
Let $f : X \to Y$ be a monotone function. Then it is an embedding if and only if it is injective.

In what follows the notion of closure and kernel operator will be important.

Definition 2.1.23 (Closure operator)
*Let $(X, \leq)$ be a partially ordered set. A function $c : X \to X$ is a **closure operator** if it is*

a) ***extensive*** $x \leq c(x)$
b) ***monotone*** $x \leq y \implies c(x) \leq c(y)$
c) ***idempotent*** $c(c(x)) = c(x)$

Definition 2.1.24 (Kernel operator)
*Let $(X, \leq)$ be a partially ordered set. A function $\kappa : X \to X$ is a **kernel operator** if it is*

- ***co-extensive*** $\kappa(x) \leq x$
- ***monotone*** $x \leq y \implies \kappa(x) \leq \kappa(y)$
- ***idempotent*** $\kappa(\kappa(x)) = \kappa(x)$

Note these definitions are "dual" with respect to the ordering on X.

2.1.4 Lattices

Certain families of subsets of algebraic structures (e.g. ideals, subgroups, normal subgroups, submodules) form a "sublattice" of the power set. Certain operations on, and results about, these subsets share common features regardless of the type of algebraic structure. Therefore we detail some elements of "Lattice Theory" (see Birkhoff) which may clarify the exposition.

Definition 2.1.25 (Upper and Lower Bounds)
*Let $(X, \leq)$ be a poset and $S \subseteq X$. Define the set of **upper bounds** for S by*

$$S^{\uparrow} = \{x \in X \mid s \leq x \quad \forall s \in S\}$$

*and the set of **lower bounds** for S by*

$$S^{\downarrow} = \{x \in X \mid x \leq s \quad \forall s \in S\}$$

Note by convention $\emptyset^{\uparrow} = \emptyset^{\downarrow} = X$. Furthermore

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$$X^{\uparrow} = \begin{cases} \{\top\} & X \text{ has a top element} \\ \emptyset & \text{otherwise} \end{cases}$$

and

$$X^{\downarrow} = \begin{cases} \{\bot\} & X \text{ has a bottom element} \\ \emptyset & \text{otherwise} \end{cases}$$

Lemma 2.1.26 (Upper/Lower bounds are antitone maps)
Let $(X, \leq)$ be a poset and S, T subsets of X then

- ***antitone*** $S \subseteq T \implies T^{\uparrow} \subseteq S^{\uparrow}$ *and* $T^{\downarrow} \subseteq S^{\downarrow}$
- ***unit-counit relations*** $S \subseteq S^{\uparrow\downarrow}$ *and* $T \subseteq T^{\downarrow\uparrow}$
- ***triangular identities*** $S^{\uparrow} = S^{\uparrow\downarrow\uparrow}$ *and* $T^{\downarrow} = T^{\downarrow\uparrow\downarrow}$

Proof. We prove only the first triangular identity as the others are straightforward consequences of the definitions. Firstly $S \subseteq S^{\uparrow\downarrow} \implies S^{\uparrow\downarrow\uparrow} \subseteq S^{\uparrow}$ by the antitone property. Given the relation $T \subseteq T^{\downarrow\uparrow}$ substitute $T = S^{\uparrow}$ to get the reverse inclusion. ■

Lemma 2.1.27
Let $(X, \leq)$ be a poset and S, T subsets of X. Then the intersections $S \cap S^{\uparrow}$ and $T \cap T^{\downarrow}$ contain at most one element. ***When they exist*** *write the elements as $\top_S$ and $\bot_T$ respectively, and are referred to as the* ***maximum*** *and* ***minimum*** *elements respectively.*

Proof. Given $x, y \in S \cap S^{\uparrow}$ then by definition $x \leq y$ and $y \leq x$. By anti-symmetry we have $x = y$ as required. ■

Definition 2.1.28 (Supremum and Infimum)
Let $(X, \leq)$ be a poset and $S \subseteq X$ a subset. We say a ***supremum*** *of S is the minimal upper bound, i.e. the unique element of*

$$S^{\uparrow} \cap S^{\uparrow\downarrow}$$

when it exists and write this as $\sup S$. Similarly an ***infimum*** *of S is the maximal lower bound, i.e. the unique element of*

$$S^{\downarrow} \cap S^{\downarrow\uparrow}$$

when it exists and write this as $\inf X$.

Lemma 2.1.29 (Maximum = Supremum)
Let $(X, \leq)$ be a poset and $S \subseteq X$ a subset. Then $\top_S$ exists if and only if $\sup S$ exists and is a member of S. In this case $\top_S = \sup S$.

Lemma 2.1.30
Let $(X, \leq)$ be a poset. Then $\{\sup S\}^{\uparrow} = S^{\uparrow}$ and $\{\inf T\}^{\downarrow} = T^{\downarrow}$ when these exist.

Lemma 2.1.31 (Sup is monotone and Inf is antitone)
Let $(X, \leq)$ be a poset and S, T subsets of X. Then $S \subseteq T \implies \sup S \leq \sup T$ and $\inf T \leq \inf S$ when these exist.

Proof. Note $S \subseteq T \implies T^{\uparrow} \subseteq S^{\uparrow}$ so $\sup T \in S^{\uparrow}$. By definition $\sup S \in S^{\uparrow\downarrow}$ therefore $\sup S \leq \sup T$.

Similarly $S \subseteq T \implies T^{\downarrow} \subseteq S^{\downarrow}$. By definition $\inf T \in T^{\downarrow} \implies \inf T \in S^{\downarrow}$. By definition $\inf S \in S^{\downarrow\uparrow}$ therefore $\inf T \leq \inf S$.
■

Remark 2.1.32
Note that $\emptyset^{\uparrow} = X$ and therefore $\sup \emptyset = \bot$ when it exists. Similarly $\inf \emptyset = \top$ when it exists.

When $\top$ exists $\sup X = \top$, otherwise it is not defined. Similarly when $\bot$ exists $\inf X = \bot$, otherwise it is not defined.

Definition 2.1.33 (Lattice)
A poset $(X, \leq)$ is a ***lattice*** *if every pair of elements $\{x, y\}$ admits both a supremum and infimum. In this case we write*

$$a \vee b := \sup\{a, b\}$$

and

$$a \wedge b := \inf\{a, b\}$$

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*These are called the **join** and **meet** operations. A subset Y is called a **sub-lattice** if*

$$a, b \in Y \implies a \wedge b \in Y \text{ and } a \vee b \in Y.$$

*Similarly it is a **complete lattice** if every subset S admits both a supremum and infimum. This is written*

$$\bigvee S := \sup S$$

and

$$\bigwedge S := \inf S$$

*Note a complete lattice has both a top and a bottom element (by considering $\sup \emptyset$ and $\inf \emptyset$), and a lattice admits **finite** joins and meets.*

Trivially

$$\bigwedge \{x\} = \bigvee \{x\} = x$$

Example 2.1.34 (Power Set)
*For a fixed set E the collection of subsets $\mathcal{P}(E)$ is a complete lattice under the union and intersection operator **with the convention that empty intersection is the whole set and empty union is the empty set***

In this case $\top = E$ and $\bot = \emptyset$.

Proposition 2.1.35 (Principal down-sets are lattices)
Let $(X, \leq)$ be a lattice and $x, y \in X$. Then the subsets $\{x\}^\uparrow$, $\{x\}^\downarrow$ and $\{x\}^\uparrow \cap \{y\}^\downarrow$ are sub-lattices.

Verifying a poset is a lattice is slightly easier than it may first appear.

Lemma 2.1.36 (Supremum is Infimum of upper bounds)
Let $(X, \leq)$ be a poset and S a subset of X. Then

$$\sup S = \inf S^\uparrow$$

when either exists. Dually

$$\inf S = \sup S^\downarrow$$

Proof. By definition $\sup S$ is the unique element of $S^\uparrow \cap S^{\uparrow\downarrow}$ and $\inf S^\uparrow$ is the unique element of $S^{\uparrow\downarrow} \cap S^{\uparrow\downarrow\uparrow}$. By (2.1.26) $S^{\uparrow\downarrow\uparrow} = S^\uparrow$ so they are equivalent. ∎

Proposition 2.1.37 (Criteria to be a Complete Lattice)
Let $(X, \leq)$ be a poset. Then the following are equivalent

a) X is a complete lattice
b) X admits arbitrary infimums (and in particular has $\top = \inf \emptyset$)
c) X admits arbitrary supremums (and in particular has $\bot = \sup \emptyset$)

In this case we have the relationships

$$\bigvee S = \bigwedge S^\uparrow$$

$$\bigwedge S = \bigvee S^\downarrow$$

Proof. a) $\implies$ b,c) is clear.

b,c) $\implies$ a) follows from the previous Lemma. ∎

Lemma 2.1.38
Let $(X, \leq)$ be a poset and $(Y, \leq)$ a sub-poset. Let $S \subseteq Y$ be a subset. Then $\inf_Y S$ exists if and only if $\inf_X S$ exists and belongs to Y. In this case they are equal.

Proof. Note in general that $T^{\downarrow,Y} = T^{\downarrow,X} \cap Y$ and $T^{\uparrow,Y} = T^{\uparrow,X} \cap Y$. Therefore

$$S^{\downarrow,Y} \cap S^{\downarrow\uparrow,Y} = S^{\downarrow,X} \cap S^{\downarrow\uparrow,X} \cap Y$$

Recall $\inf S$ is the unique element of $S^\downarrow \cap S^{\downarrow\uparrow}$ if it exists. Then the result follows easily. ∎

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Definition 2.1.39 (Moore Family)
*Let $(X, \leq)$ be a complete lattice. A sub-poset $(Y, \leq)$ is a **Moore family** over X if it satisfies the following property*

$$S \subseteq Y \implies \bigwedge_X S \in Y$$

In particular this includes the case $S = \emptyset$ and so $\top \in Y$.

Example 2.1.40 (Moore family of sets)
Given a fixed set E, then $\mathcal{P}(E)$ is a complete lattice ordered under inclusion. Then a family of subsets $\mathcal{F}$ is a Moore family precisely when

- $E \in \mathcal{F}$
- $A_{i \in I} \in \mathcal{F} \implies \bigcap_{i \in I} A_i \in \mathcal{F}$

Proposition 2.1.41 (Equivalent Formulations of Complete Sub-lattice)
Let $(X, \leq)$ be a complete lattice and $(Y, \leq)$ a sub-poset. Then the following are equivalent

a) $(Y, \leq)$ is a Moore family
b) $(Y, \leq)$ is a complete lattice
c) Y is the image of some closure operator $c : X \to X$

In this case the closure operator is given by

$$c(x) = \bigwedge_X \{y \in Y \mid x \leq y\}$$

For $S \subseteq Y$

$$\bigwedge_Y S = \bigwedge_X S$$

$$\bigvee_Y S = c\left(\bigvee_X S\right)$$

and for $S \subseteq X$ we have

$$c\left(\bigvee_X S\right) = \bigwedge_X (S^\uparrow \cap Y)$$

Proof. a) $\implies$ b) By (2.1.38) $S \subseteq Y \implies \bigwedge_Y S = \bigwedge_X S$. By (2.1.37) then Y is a complete lattice.

b) $\implies$ c) Suppose that $(Y, \leq)$ is a complete lattice then define the function $c : X \to X$ by $c(x) = \bigwedge_X \Gamma_x$ where $\Gamma_x = \{y \in Y \mid x \leq y\}$. We need to show that it is a closure operator. Evidently $x \in \Gamma_x^\downarrow$ and $c(x) \in \Gamma_x^{\downarrow\uparrow}$ by definition of infimum. Therefore $x \leq c(x)$ and c is extensive. Note $x \leq y \implies \Gamma_y \subseteq \Gamma_x$. By (2.1.31) we have $\inf \Gamma_x \leq \inf \Gamma_y$, whence $c(x) \leq c(y)$ and c is monotone.

Y is a complete lattice, so by (2.1.38) we have $c(x) \in Y$ so that $\mathrm{Im}(c) \subseteq Y$. We claim that $x \in Y \implies c(x) = x$. In this case $x \in \Gamma_x$ and $c(x) \in \Gamma_x^\downarrow$ whence $c(x) \leq x$ and therefore $x = c(x)$ as required. Therefore $Y \subseteq \mathrm{Fix}(c) \subseteq \mathrm{Im}(c) \subseteq Y$, whence $Y = \mathrm{Im}(c) = \mathrm{Fix}(c)$ and c is idempotent by (2.1.15). As c is extensive, monotone and idempotent it is by definition a closure operator.

c) $\implies$ a) In order for $Y \coloneqq \mathrm{Im}(c)$ to be a Moore family, we need to show $S \subseteq Y \implies \bigwedge_X S \in Y$. We claim that by properties of c we have

$$S \subseteq Y \implies c\left(S^\downarrow\right) \subseteq S^\downarrow$$

$$T \subseteq X \implies c(T^\uparrow) \subseteq T^\uparrow$$

Therefore c maps the singleton set $S^\downarrow \cap S^{\downarrow\uparrow} = \left\{\bigwedge_X S\right\}$ to itself. In otherwords $\bigwedge_X S \in \mathrm{Fix}(c) = \mathrm{Im}(c) = Y$ as required.

Define $\Gamma_x \coloneqq \{y \in Y \mid x \leq y\}$. We wish to show that $c(x) = \bigwedge_X \Gamma_x$. As $x \leq c(x)$ we have $c(x) \in \Gamma_x$. Furthermore $y \in \Gamma_x \implies c(x) \leq c(y) = y$. So $c(x) \in \Gamma_x^\downarrow$. Therefore $c(x) = \bot_{\Gamma_x} = \bigwedge_X \Gamma_x$ as required.

Finally by (2.1.30) $\left\{\bigvee_X S\right\}^\uparrow = S^\uparrow$ for any $S \subseteq X$. Therefore, as $c(x) = \bigwedge_X \Gamma_x$ we find

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$$c\left(\bigvee_X S\right) = \bigwedge_X \left(\left\{\bigvee_X S\right\}^{\uparrow} \cap Y\right) = \bigwedge_X (S^{\uparrow} \cap Y)$$

as required. In particular when $S \subseteq Y$ we find by (2.1.41)

$$\bigvee_Y S = \bigwedge_Y S^{\uparrow, Y} = \bigwedge_X S^{\uparrow, Y} = \bigwedge_X (S^{\uparrow} \cap Y) = c\left(\bigvee_X S\right)$$

■

Remark 2.1.42
For a given complete lattice $(X, \leq)$ we have established a correspondence between

$$\{\text{closure operators } \ c : X \to X\} \longleftrightarrow \{\text{complete sub-lattices } \ (Y, \leq)\}$$

Corollary 2.1.43 (Moore family admits a closure operator)
Let E be a fixed set and $\mathcal{F}$ a Moore family over $(\mathcal{P}(E), \subseteq)$. Then there exists a surjective closure operator $c : \mathcal{P}(E) \to \mathcal{F}$ given by

$$c(F) = \bigcap_{F \subseteq E_\alpha \in \mathcal{F}} E_\alpha$$

Any such closure operator $c : \mathcal{P}(E) \to \mathcal{P}(E)$ gives rise to a Moore family $\mathcal{F} := \mathrm{Im}(c)$.

Proposition 2.1.44 (Alternative expression for join)
Let $(X, \leq)$ be a complete lattice and $c : X \to X$ a closure operator with image Y. Then for any subset $S \subset X$

$$c\left(\bigvee_X S\right) = c\left(\bigvee_X c(S)\right) = \bigvee_Y c(S) = \bigwedge_X (S^{\uparrow} \cap Y)$$

i.e. it's the smallest "closed" set containing each element of S.

Proof. By (2.1.41) the expression for $c\left(\bigvee_X S\right)$ yields

$$c\left(\bigvee_X c(S)\right) = \bigwedge_X (c(S)^{\uparrow} \cap Y) = \bigwedge_X (S^{\uparrow} \cap Y) = c\left(\bigvee_X S\right)$$

where the middle equality follows because if $y \in Y$ then $c(s) \leq y \iff s \leq y$. Furthermore $c(S) \subseteq Y$ so the expression for $\bigvee_Y$ yields

$$c\left(\bigvee_X c(S)\right) = \bigvee_Y c(S).$$

as required. ■

2.1.5 Distributive Lattice

Proposition 2.1.45
Let $(X, \leq)$ be a lattice then the following relations hold

a) $x \wedge (y \vee z) \geq (x \wedge y) \vee (x \wedge z)$
b) $x \vee (y \wedge z) \leq (x \vee y) \wedge (x \vee z)$

Proof. a) By definition $x \wedge y \leq x$ and $x \wedge y \leq y \leq y \vee z$. Therefore $x \wedge y \leq x \wedge (y \vee z)$. By symmetry in y and z we have $x \wedge z \leq x \wedge (y \vee z)$. Whence $(x \wedge y) \vee (x \wedge z) \leq x \wedge (y \vee z)$ as required.

b) follows by duality ■

Definition 2.1.46 (Distributive Lattice)
*We say a lattice $(X, \leq)$ is **distributive** if it satisfies the following relations for all $x, y, z \in X$*

- $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$
- $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$

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Proposition 2.1.47
Let $(X, \leq)$ be a lattice then TFAE

a) $(X, \leq)$ *is a distributive lattice*
b) $x \wedge (y \vee z) \leq (x \wedge y) \vee (x \wedge z)$
c) $x \vee (y \wedge z) \geq (x \vee y) \wedge (x \vee z)$

Example 2.1.48
Any family of subsets closed under finite intersection and union is a distributive lattice.

2.1.6 Galois Connections

Definition 2.1.49 (Galois Connection)
Let $\left(X, \underset{X}{\leq}\right)$ and $\left(Y, \underset{Y}{\leq}\right)$ be posets. A pair of functions $(f_\star, f^\star)$

$$X \underset{f^\star}{\overset{f_\star}{\rightleftharpoons}} Y$$

*is called an **antitone Galois connection** if it satisfies the **adjoint property***

- $x \underset{X}{\leq} f^\star(y) \iff y \underset{Y}{\leq} f_\star(x) \quad \forall x \in X, y \in Y$

*We say it is a **monotone Galois connection** if instead*

- $x \underset{X}{\leq} f^\star(y) \iff f_\star(x) \underset{Y}{\leq} y \quad \forall x \in X, y \in Y$

We will assume that if not otherwise specified the connection is antitone.

Proposition 2.1.50 (Equivalent Condition for Galois Connection)
Let $\left(X, \underset{X}{\leq}\right)$ and $\left(Y, \underset{Y}{\leq}\right)$ be posets. Consider a pair of functions

$$X \underset{f^\star}{\overset{f_\star}{\rightleftharpoons}} Y$$

*Then this constitutes an **antitone Galois Connection** if and only if*

- $f_\star$ *and* $f^\star$ *are both antitone*
- $x \underset{X}{\leq} f^\star(f_\star(x))$ *and* $y \underset{Y}{\leq} f_\star(f^\star(y))$ *for all* $x \in X, y \in Y$ *(i.e.* $f^\star \circ f_\star$ *and* $f_\star \circ f^\star$ *are extensive)*

*Similarly it constitutes a **monotone Galois Connection** if and only if*

- $f_\star$ *and* $f^\star$ *are both monotone*
- $x \underset{X}{\leq} f^\star(f_\star(x))$ *and* $f_\star(f^\star(y)) \underset{Y}{\leq} y$ *for all* $x \in X, y \in Y$

Proof. We consider only the antitone case, as the monotone follows from duality (flip $\underset{Y}{\leq}$).

Suppose that $(f_\star, f^\star)$ satisfies the adjoint property

$$f_\star(x) = f_\star(x) \implies f_\star(x) \underset{Y}{\leq} f_\star(x) \implies x \underset{X}{\leq} f^\star(f_\star(x))$$

$$f^\star(y) = f^\star(y) \implies f^\star(y) \underset{Y}{\leq} f^\star(y) \implies y \underset{Y}{\leq} f_\star(f^\star(y))$$

whence the extensive property follows. Furthermore

$$x \underset{X}{\leq} x' \implies x \underset{X}{\leq} f^\star(f_\star(x')) \implies f_\star(x') \underset{Y}{\leq} f_\star(x)$$

$$y \underset{Y}{\leq} y' \implies y \underset{Y}{\leq} f_\star(f^\star(y')) \implies f^\star(y) \underset{X}{\leq} f^\star(y')$$

which shows that the functions $f_\star$ and $f^\star$ are antitone.

Conversely suppose they satisfy the given conditions. Then by the antitone and extensive properties in turn

$$x \underset{X}{\leq} f^\star(y) \implies f_\star(f^\star(y)) \underset{Y}{\leq} f_\star(x) \implies y \underset{Y}{\leq} f_\star(x)$$

and

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$$y \underset{Y}{\leq} f_\star(x) \implies f^\star(f_\star(x)) \underset{X}{\leq} f^\star(y) \implies x \underset{X}{\leq} f^\star(y).$$

which is the adjoint property as required. ∎

Definition 2.1.51 (Closed sets)
*Let $(f_\star, f^\star)$ be a Galois connection. Then define the **closed sets** to be*

$$X^\star := f^\star(Y)$$

$$Y^\star := f_\star(X)$$

Proposition 2.1.52 (Isomorphism on closed sets)
*Consider an antitone (resp. monotone) Galois connection $X \underset{f^\star}{\overset{f_\star}{\rightleftharpoons}} Y$. Then it restricts to a dual isomorphism (resp. order isomorphism) on **closed sets***

$$X^\star \underset{f^\star}{\overset{f_\star}{\rightleftharpoons}} Y^\star$$

Furthermore the following properties hold

- $f_\star \circ f^\star \circ f_\star = f_\star$
- $f^\star \circ f_\star \circ f^\star = f^\star$
- *In the antitone case $f^\star \circ f_\star$ and $f_\star \circ f^\star$ are closure operators.*
- *In the monotone case $f^\star \circ f_\star$ is a closure operator and $f_\star \circ f^\star$ is a kernel operator.*
- $X^\star = \mathrm{Fix}(f^\star \circ f_\star) = \mathrm{Im}(f^\star \circ f_\star)$
- $Y^\star = \mathrm{Fix}(f_\star \circ f^\star) = \mathrm{Im}(f_\star \circ f^\star)$

Proof. We detail the antitone case as the monotone case follows by duality. We first prove so-called triangular identities, for by the extensive property (2.1.50)

$$x \leq f^\star(f_\star(x)) \implies f_\star(f^\star(f_\star(x))) \leq f_\star(x)$$

and by the other extensive property

$$f_\star(x) \leq f_\star(f^\star(f_\star(x)))$$

whence they are equal. The other case is similar.

It's immediate that $f^\star \circ f_\star$ and $f_\star \circ f^\star$ are idempotent, and they are extensive by (2.1.50). And the composition of two antitone functions is monotone so $f^\star \circ f_\star$ and $f_\star \circ f^\star$ are closure operators.

Observe

$$\mathrm{Im}(f^\star \circ f_\star) \subseteq \mathrm{Im}(f^\star) \subseteq \mathrm{Fix}(f^\star \circ f_\star)$$

where the first inclusion is trivial and the second inclusion follows from the second triangular identity. However both sides are equal by (2.1.15) and the expression for $X^\star$ follows. The expression for $Y^\star$ follows similarly.

This shows that the maps are mutual inverses as required. ∎

In certain circumstances we may consider a smaller subset of X, by applying a suitable closure operator which is compatible with the Galois correspondence:

Proposition 2.1.53 (Subordinated Closure Operator)
Let $X \underset{f^\star}{\overset{f_\star}{\rightleftharpoons}} Y$ be a Galois connection and $c : X \to X$ be a closure operator with image X_c. Then

$$c(x) \leq (f^\star \circ f_\star)(x) \quad \forall x \in X \iff \mathrm{Im}(f^\star) \subseteq \mathrm{Fix}(c) \iff X^\star \subseteq X_c$$

In this case

$$f_\star(c(x)) = f_\star(x)$$

Proof. Suppose $c(x) \leq (f^\star \circ f_\star)(x)$. Substitute $x = f^\star(y)$ then, because c is extensive,

$$f^\star(y) \leq c(f^\star(y)) \leq (f^\star \circ f_\star \circ f^\star)(y) = f^\star(y).$$

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Therefore $c(f^\star(y)) = f^\star(y)$ and $\mathrm{Im}(f^\star) \subseteq \mathrm{Fix}(c)$ as required. Conversely suppose this holds, then by the monotone property of c and extensive property of $f^\star \circ f_\star$

$$x \leq (f^\star \circ f_\star)(x) \implies c(x) \leq c((f^\star \circ f_\star)(x)) = (f^\star \circ f_\star)(x).$$

as required. Finally by the extensive property of c

$$x \leq c(x) \leq (f^\star \circ f_\star)(x)$$

and by the antitone/monotone property of $f_\star$ and triangular identity

$$f_\star(x) \leq f_\star(c(x)) \leq f_\star(x)$$

whence $f_\star(c(x)) = f(x)$ as required. ∎

The meaning of the "adjoint" criterion can be explained by the following rather generic situation

Example 2.1.54 (Canonical example of an antitone Galois connection)
Suppose there is a predicate

$$\psi : X \times Y \to \{0, 1\}$$

Define a connection

$$\mathcal{P}(X) \underset{f^\star}{\overset{f_\star}{\rightleftarrows}} \mathcal{P}(Y)$$

by

$$f_\star(S) = \{y \in Y \mid \psi(x, y) = 1 \quad \forall x \in S\}$$

$$f^\star(T) = \{x \in X \mid \psi(x, y) = 1 \quad \forall y \in T\}$$

Then

$$S \subseteq f^\star(T) \iff \psi(s, t) = 1 \quad \forall s \in S \quad t \in T \iff T \subseteq f_\star(S)$$

Proposition 2.1.55 (Joins under Galois Correspondence)
Let $\left(X, \underset{X}{\leq}\right)$ *and* $\left(Y, \underset{Y}{\leq}\right)$ *be complete lattices with an antitone Galois connection* $X \underset{f^\star}{\overset{f_\star}{\rightleftarrows}} Y$*. Then for* $S \subseteq X$

$$f_\star\Big(\bigvee S\Big) = \bigwedge f_\star(S)$$

Similarly for $T \subseteq Y$ *we have*

$$f^\star\Big(\bigvee T\Big) = \bigwedge f^\star(T)$$

Proof. Let $a = \bigvee S$ and $b = \bigwedge f_\star(S)$. Then $s \leq a \implies f_\star(a) \leq f_\star(s)$ for all $s \in S$, which implies $f_\star(a) \leq b$. Similarly $b \leq f_\star(s) \implies s \leq f^\star(b)$ by the adjoint criterion. Therefore $a \leq f^\star(b)$ by definition of join, which implies $b \leq f_\star(a)$ by the adjoint criterion again. Whence $f_\star(a) = b$ as required.

The second statement follows from duality. ∎

2.1.7 Axiom of Choice

Theorem 2.1.56 (Axiom of choice)
There are a number of essentially equivalent forrmulations of the axiom of choice

a) *The Cartesian product of a non-empty family of sets is non-empty*
b) *For any set* X *of non-empty sets there exists a function* $f : X \to \bigcup X$ *such that* $A \in X \implies f(A) \in A$.
c) ***Zorn's Lemma*** *Suppose a partially ordered set* $(X, \leq)$ *is such that every chain in* X *has an upper bound in* X*. Then* X *contains at least one maximal element.*
d) *Every surjective function has a right inverse.*

Corollary 2.1.57 (Choose representatives)
Let $\pi : X \to Y$ *be a surjective function and* $T \subseteq Y$ *a subset. Then there exists a subset* $S \subseteq X$ *such that* $\pi|_S$ *is bijective.*

When T *is finite* $\#S = \#T$.

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Definition 2.1.58 (Finite Character)
*A family of sets $\mathcal{F}$ has **finite character** if it satisfies the following property*

$$A \in \mathcal{F} \iff (B \subseteq A \text{ and finite} \implies B \in \mathcal{F})$$

Corollary 2.1.59 (Tukey's Lemma)
Let $\mathcal{F}$ be a family of sets of finite character. Then it is chain-complete when ordered by inclusion.

In particular every set is contained in a maximal set.

2.1.8 Chain Conditions

Definition 2.1.60 (Totally ordered / chains)
*A poset $(\mathcal{F}, \leq)$ is **totally ordered** if $x \leq y$ or $y \leq x$ for all $x, y \in \mathcal{F}$.*

Definition 2.1.61 (Chain)
*A non-empty subset C of $\mathcal{F}$ is a **chain** if it is totally ordered under $\leq$.*

The length of the chain is simply $\ell(C) := |C| - 1$.

A chain C is

- ***saturated** if $x \leq z \leq y$ and $x, y \in C \implies z \in C$.*
- ***maximal** if it's not contained properly in another chain.*

Definition 2.1.62 (Chain-Complete)
*A poset $(\mathcal{F}, \leq)$ is **chain complete** if every chain C has a supremum in $\mathcal{F}$. It is **co-chain complete** if every chain C has an infimum in $\mathcal{F}$.*

Proposition 2.1.63 (Noetherian / Artinian Poset)
Let $(X, \leq)$ be a poset then the following conditions are equivalent

a) Any ascending chain

$$x_1 \leq x_2 \leq ... \leq x_n \leq ...$$

eventually stabilizes

b) Any non-empty subset $Y \subseteq X$ has a maximal element

*Such a poset is called **Noetherian**. If it satisfies the dual condition then it is called **Artinian**.*

Proof. a) $\implies$ b) If Y has no maximal elements then we may (by axiom of dependent choice) construct a strictly increasing sequence, which by definition does not stabilize.

b) $\implies$ a) Clear. ■

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2.2 Numbers

Informally we consider the set of integers

$$\mathbb{Z} = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$$

and the subset of natural numbers

$$\mathbb{N} = \{0, 1, 2, ..., \}$$

Although it's possible to construct the integers painstakingly from a small set of axioms (see ...) we instead for brevity simply state the most commonly used results as axioms.

2.2.1 Integers

We suppose the existence of a set $\mathbb{Z}$ with distinguished elements $0 \neq 1$ together with

- A binary operation

  $$+ : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$$

  and an involution

  $$(-) : \mathbb{Z} \to \mathbb{Z}$$

  satisfying

  - $-0 = 0$
  - $-(-x) = x$
  - $x = -x \iff x = 0$
  - $x + 0 = 0 = 0 + x$
  - $x + y = y + x$
  - $(x + y) + z = x + (y + z)$
  - $x + (-x) = 0 = (-x) + x$
  - $-(x + y) = (-x) + (-y)$

- A subset $\mathbb{N}$ such that

  - $0, 1 \in \mathbb{N}$
  - $x, y \in \mathbb{N} \implies x + y \in \mathbb{N}$
  - $x \in \mathbb{Z} \implies (x \in \mathbb{N}) \vee (-x \in \mathbb{N})$
  - $x \in \mathbb{N} \wedge -x \in \mathbb{N} \implies x = 0$

  which also satisfies the principle of induction

- Let $S \subseteq \mathbb{N}$ be a set such that
  - $0 \in S$
  - $x \in S \implies x + 1 \in S$

  then $S = \mathbb{N}$

It's possible to use these to show the existence of multiplication

Proposition 2.2.1 (Multiplication exists)
There exists a binary operation

$$\cdot : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$$

such that

- $x \cdot 0 = 0 = 0 \cdot x$
- $x \cdot 1 = x = 1 \cdot x$
- $xy = yx$
- $(xy)z = x(yz)$

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- $x(y+z) = xy + xz$
- $(y+z)x = yx + zx$
- $(-x)(y) = -(xy) = x(-y)$

We may also show the existence a partial ordering

Proposition 2.2.2 (Order exists)
There exists a relation $\leq$ on $\mathbb{Z}$ given by

$$x \leq y \iff y - x \in \mathbb{N}$$

which satisfies

$$x \leq y \vee y \leq x$$

$$x \leq y \wedge y \leq x \implies x = y$$

Define $x < y$ in the obvious way then it satisfies the usual trichotomy law, namely precisely one of the following holds

$$x < y, \quad x = y, \quad y < x$$

and further

- *$z > 0$ then $x < y \iff xz < yz$*
- *$z < 0$ then $x < y \iff yz < xz$*
- *$y > 1$ and $x > 0$ then $x < xy$*

Finally we can construct an absolute value function

Proposition 2.2.3
*There exists an **absolute value** function*

$$|\cdot| : \mathbb{Z} \to \mathbb{N}$$

such that

$$|x| = \begin{cases} x & 0 < x \\ 0 & x = 0 \\ -x & x < 0 \end{cases}$$

It satisfies

- $|x| = 0 \iff x = 0$
- $|x| = |y| \iff x = \pm y$
- $|xy| = |x||y|$
- $|x + y| \leq |x| + |y|$

In many cases it may be more convenient to use the following form of induction

Proposition 2.2.4 (Well-Ordering Principle)
Let $S \subset \mathbb{N}$ be a non-empty subset. Then it contains a minimal element d.

2.2.2 Arithmetic

Proposition 2.2.5 (Division Algorithm)
Let $x, y \in \mathbb{Z}$ be non-zero integers then there exists q, r such that

$$x = yq + r$$

and

$$0 \leq r < |y|$$

Furthermore (q, r) is the unique such pair.

Proof. Suppose first that $x, y > 0$. Let $S = \{x - yn \mid n \in \mathbb{Z}\} \cap \mathbb{N}$. Then $x \in S$ so it is non-empty. By the Well-Ordering principle it has a minimal element r. By assumption

$$x = yq + r$$

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for some $q \in \mathbb{Z}$ and $r \geq 0$. Suppose $r \geq y$, then $0 \leq x - y(q+1) < r$ contradicting minimality.

The case $x > 0, y < 0$ is then straightforward, as is the case $x < 0$.

For uniqueness suppose $yq' + r' = yq + r$ then $|y||q - q'| = |r' - r| < |y|$ from which it follows $|q - q'| = 0 \implies q = q' \implies r = r'$. ■

Corollary 2.2.6 (Ideals are Principal)
Let $S \subseteq \mathbb{Z}$ be a non-empty set such that

$$x, y \in S \implies x \pm y \in S$$

Then $S = d\mathbb{Z}$ for a unique $d \geq 0$.

Proof. First we claim that $0 \in S$. For if $x \in S$ then $0 = x - x \in S$ by assumption. Furthermore $x \in S \implies -x = 0 - x \in S$.

Consider the set $S' = (S \cap \mathbb{N}) \setminus \{0\}$. If it's empty then $S = \{0\}$ (for $x \in S \implies -x \in S$) and $d = 0$.

Otherwise it has a minimal element $d > 0$ by the well-ordering principle. Then by induction $d\mathbb{Z} \subseteq S$. Conversely suppose $y \in S$ then by the division algorithm $y = qd + r$ with $0 \leq r < d$. By assumption $r = y - qd \in S$ and by minimality must be equal to 0. Therefore $y \in d\mathbb{Z}$ and $d\mathbb{Z} = S$ as required. ■

Definition 2.2.7 (Divisibility)
Let $x, y \in \mathbb{Z}$ be two integers. We say that x divides y if there exists a such that $ax = y$. In this case we write

$$x \mid y$$

and

$$\frac{y}{x}$$

for the unique integer a such that $ax = y$.

Lemma 2.2.8
Let $x, y \in \mathbb{Z}$ be two integers then

$$x \mid y \implies |x| \leq |y|$$

In particular $x \mid y \wedge y \mid x \implies x = \pm y$.

Proposition 2.2.9 (Bezout's Theorem)
Let x, y be non-zero integers. Then there exists a unique positive integer d such that

- *d is a common divisor of x, y*
- *For any other common divisor e we have $e \mid d$.*

Further there exists integers a, b such that $ax + by = d$. We write this as (x, y).

Proof. Let $S = \{ax + by \mid a, b \in \mathbb{Z}\}$. Then by (2.2.6) we have $S = d\mathbb{Z}$ for a unique $d > 0$. As $x, y \in S$ by definition d is a common divisor, and by definition $d = d \cdot 1 = ax + by$ for some integers a, b. Suppose e is a common divisor then $d = ax + by = e(ap + bq)$ and $e \mid d$ as required.

Any two such common divisors have $d = \pm d'$ by the previous Lemma. Since they are positive and non-zero we have $d = d'$. ■

Proposition 2.2.10
Let a, x, y be non-zero integers then

$$|a|(x, y) = (ax, ay)$$

In particular

$$\left(\frac{x}{x, y}, \frac{y}{x, y}\right) = 1$$

Proof. This follows from the characterization of (x, y) as the minimal positive integer in the set $\{mx + ny\}$. ■

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2.2.3 Prime Factorization

Definition 2.2.11
Let $x \in \mathbb{Z}$ be a non-zero integer. We say that x

- *is a **unit** if it's equal to 1 or -1.*
- *is **prime** if it's not a unit and $x \mid p$ implies $x = \pm 1$ or $x = \pm p$*
- ***composite** otherwise*

Lemma 2.2.12
Let p be a positive prime and a a non-zero integer. Then precisely one of the following holds

- $(p, a) = 1$
- $(p, a) = p$ *and* $p \mid a$

Proof. Note that (p, a) is positive and divides both p and a so the result follows by definition of prime. ■

Proposition 2.2.13 (Euclid's Lemma)
Suppose $x \mid ab$ then $\frac{x}{x,a} \mid b$.

In particular if p is a prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. First suppose that $(x, a) = 1$. Then by assumption $zx = ab$ and by Bezout's Theorem $mx + na = 1$ for some integers m, n. Multiply by z to find that $abm + na = a(bm + n) = z$. Therefore $a(bm + n)x = ab$ and cancel a to find $x \mid b$ as required.

For the general case define $x' = x/(x, a)$ and $a' = a/(x, a)$. Then by (2.2.10) $(x', a') = 1$. Furthermore it's clear that $x' \mid a'b$ so we have $x' \mid b$ by the special case just proven.

Finally suppose $p \mid ab$. If $(p, a) = 1$ then $p \mid b$ by the first result. By (2.2.12) if this does not hold then $p \mid a$ as required.■

Using these results we may show that there exists a unique factorization into primes, unique up to multiplication by a unit.

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2.3 Matroids

The theory of bases of vector spaces (the Vector Spaces subsection) and transcendence bases of field extensions (the Transcendental Field Extensions subsection) have some formal similarities, as noted in [Wae91]. Here we use the theory of Matroids to formalise this precisely so that the proofs need not be repeated in each case.

Definition 2.3.1 (Matroid)
Consider a set X together with a closure operator $c : \mathcal{P}(X) \to \mathcal{P}(X)$ ("span" operator) such that $c(\emptyset) = \emptyset$. We say

- *$S \subseteq X$ is **independent** if $x \in S \implies x \notin c(S \setminus \{x\})$*
- *$\Gamma \subseteq X$ is **spanning** if $c(\Gamma) = X$.*

Note by definition X is spanning and $\emptyset$ is independent. Moreover all singletons $\{x\}$ are independent.

*We call the pair (X, c) a **matroid** if it also satisfies the following properties*

- ***Finitary** $x \in c(\Gamma) \implies x \in c(\Gamma')$ for some finite subset Γ' of Γ.*
- ***Exchange Property** For all $x, y \in X$ and $Y \subseteq X$ we have*

$$x \in c(Y \cup \{y\}) \setminus c(Y) \implies y \in c(Y \cup \{x\})$$

*We say (X, c) has **finite rank** if it has a finite spanning set.*

*Finally we say $\mathcal{B} \subseteq X$ is a **basis** if it is both **spanning** and **independent**.*

We begin with some elementary characterizations of independent sets

Lemma 2.3.2
Suppose $S \subseteq X$ is a subset

a) $A \subseteq c(S) \implies c(S \cup A) = c(S)$
b) S is independent if and only if no proper subset has the same span.

Proof. We prove each in turn

a) By monotonicity

$$c(S) \subseteq c(S \cup A) \subseteq c(c(S) \cup A) = c(c(S)) = c(S)$$

b) Suppose S is independent and $S' \subsetneq S$ is a proper subset such that $c(S') = c(S)$. Choose $x \in S \setminus S'$ then by definition $x \in S \implies x \in c(S) = c(S') \subseteq c(S \setminus \{x\})$ contradicting independence.

Conversely suppose for some $x \in S$ we have $x \in c(S \setminus \{x\})$. Define $S' \coloneqq S \setminus \{x\}$. Then $x \in c(S')$ implies $c(S) = c(S')$ by a). As S' is a proper subset this contradicts the hypothesis.

■

Lemma 2.3.3
Every subset of an independent set is independent. Furthermore the family of independent sets has finite character.

Proof. The first statement is straightforward. Suppose S a dependent set such that every finite subset is independent. Then there exists $x \in S$ such that $x \in c(S \setminus \{x\})$. Then by the finitary property there exists a finite subset $S' \subseteq S \setminus \{x\}$ such that $x \in c(S')$. Therefore by definition $S' \cup \{x\}$ is not independent, a contradiction. ■

The finitary condition ensures that $\mathcal{E}$ is "inductively ordered"

Corollary 2.3.4
Let $\{S_i\}_{i \in I}$ be a chain of independent subsets. Then $S = \bigcup_{i \in I} S_i$ is also independent.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $F \subseteq S$ be a finite subset. For each $x \in F$ choose $i_x \in I$ with $x \in S_{i_x}$. As $\{S_i\}_{i \in I}$ is a chain and F is finite, the sets $\left\{S_{i_x}\right\}_{x \in F}$ are totally ordered by inclusion and so admit a maximum S_{i_0}, whence $F \subseteq S_{i_0}$. By (2.3.3) F is independent, being a subset of the independent set S_{i_0}. As $F \subseteq S$ was an arbitrary finite subset, S is independent by (2.3.3). ■

Lemma 2.3.5 (Extension Property)
Suppose S is an independent set and $x \notin c(S)$, then $S \cup \{x\}$ is independent.

Proof. We require to prove that for all $y \in S$ we have $y \notin c(S \cup \{x\} \setminus \{y\})$. By independence of S we have $y \notin c(S \setminus \{y\})$, so by the Exchange Property $y \in c(S \cup \{x\} \setminus \{y\})$ would imply $x \in c(S)$, contradicting the hypothesis. ■

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Proposition 2.3.6
Let $\mathcal{F}$ be a family of independent sets which satisfies the following properties

- $\emptyset \in \mathcal{F}$
- ***extension property*** $S \in \mathcal{F}$ *and* $x \notin c(S) \implies S \cup \{x\} \in \mathcal{F}$
- $\mathcal{F}$ *has finite character*

then $\mathcal{F}$ consists of all independent sets.

Proof. It's enough to show that $\mathcal{F}$ contains all finite independent sets, which follows by induction on $\#S$. For given an independent set S and $x \in S$, then by definition $x \notin c(S \setminus \{x\})$. By the induction hypothesis $S \setminus \{x\} \in \mathcal{F}$ whence by the extension property $S \in \mathcal{F}$ as required. ∎

Proposition 2.3.7 (Basis exists)
Let (X, c) be a matroid, S independent and Γ a subset such that $S \subseteq \Gamma$. Then there exists an independent set $\mathcal{B}$ such that $S \subseteq \mathcal{B} \subseteq \Gamma$ and $c(\mathcal{B}) = c(\Gamma)$.

In particular if Γ is spanning then $\mathcal{B}$ is a basis.

Proof. Consider the collection

$$\mathcal{I} = \{T \text{ independent} \mid S \subseteq T \subseteq \Gamma\}$$

By (2.3.4), $\mathcal{I}$ is chain-complete. Therefore it has a maximal element $\mathcal{B}$ by Zorn's Lemma. Suppose $x \in \Gamma \setminus c(\mathcal{B})$ then $\mathcal{B} \cup \{x\}$ is independent by (2.3.5), contradicting maximality. Therefore $\Gamma \subseteq c(\mathcal{B}) \implies c(\Gamma) \subseteq c(\mathcal{B})$. The reverse inequality is clear so that $c(\Gamma) = c(\mathcal{B})$. ∎

Corollary 2.3.8 (Criteria for bases)
Let (X, c) be a matroid. Then the following are equivalent

a) $\mathcal{B}$ is a basis
b) $\mathcal{B}$ is a minimal spanning set
c) $\mathcal{B}$ is maximally independent (possibly in some spanning set Γ)

Proof. a) $\implies$ b). Let $\mathcal{B}$ be a basis and $\Gamma \subseteq \mathcal{B}$ a spanning set. Then by (2.3.2) b) $\Gamma = \mathcal{B}$.

b) $\implies$ a). Let $\mathcal{B}$ be a minimal spanning set, then by (2.3.7) there exists a subset $\mathcal{B}'$ which is a basis, and in particular spanning. By minimality $\mathcal{B} = \mathcal{B}'$.

c) $\implies$ a). By (2.3.7) there exists a basis $\mathcal{B}'$ containing $\mathcal{B}$. By maximality $\mathcal{B}' = \mathcal{B}$.

a) $\implies$ c). Suppose $\mathcal{B} \subseteq S$ where S is independent. Then S is spanning too, and so by (2.3.2) b) $\mathcal{B} = S$. ∎

Definition 2.3.9 (Sub-matroid)
*A subset $Y \subseteq X$ is a **sub-matroid** if $c(Y) = Y$. In this case $S \subseteq Y \implies c(S) \subseteq Y$ and so we have an induced matroid structure (Y, c).*

Lemma 2.3.10 (Mini Exchange Lemma)
Let $S \subseteq \Gamma$ be finite sets and $x \in X \setminus S$ such that

- *S is independent*
- *$x \in c(\Gamma) \setminus c(S)$*

Then there exists $y \in \Gamma \setminus S$ such that $c(\Gamma \setminus \{y\} \cup \{x\}) = c(\Gamma)$.

Proof. We may assume without loss of generality that $x \notin \Gamma$.

Consider $\tilde{\Gamma} \subseteq \Gamma$ minimal subject to $S \subseteq \tilde{\Gamma}$ and $x \in c(\tilde{\Gamma})$. If $S = \tilde{\Gamma}$ then $x \in c(S)$ a contradiction. Therefore we may choose $y \in \tilde{\Gamma} \setminus S$. By minimality we have $x \in c(\tilde{\Gamma}) \setminus c(\tilde{\Gamma} \setminus \{y\})$. Therefore by the Exchange Property we have $y \in c(\tilde{\Gamma} \setminus \{y\} \cup \{x\})$. Then by (2.3.2) applied twice

$$c(\Gamma \setminus \{y\} \cup \{x\}) = c(\Gamma \cup \{x\}) = c(\Gamma)$$

as required. ∎

Proposition 2.3.11 (Exchange Lemma)
Let S be an independent set and Γ be a finite set such that $S \subseteq c(\Gamma)$. Then there exists a subset $T \subseteq \Gamma$ such that

- *$\#T = \#S$*

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- $c(\Gamma \setminus T \cup S) = c(\Gamma)$.

In particular $\#S \leq \#\Gamma$.

Proof. By considering the sub-matroid $(c(\Gamma), c)$ we may assume without loss of generality that $X = c(\Gamma)$.

Let $n = \#\Gamma$ and consider the set of pairs

$$\mathcal{F} \coloneqq \{(A, B) \mid A \subseteq S, \quad B \subseteq \Gamma, \quad \#A = \#B, \quad c(\Gamma \setminus B \cup A) = X\}$$

Essentially $\mathcal{F}$ is the set of swaps we may perform from S to Γ whilst preserving the span. It is non-empty because $(\emptyset, \emptyset) \in \mathcal{F}$ and we wish to show that $(S, B) \in \mathcal{F}$ for some B.

Observe that for all $(A, B) \in \mathcal{F}$ we have $\#B \leq n$ so choose a pair such that $\#B$ is maximal. We wish to show that in this case $A = S$, so we suppose to the contrary that $A \subsetneq S$.

We claim that $B \subsetneq \Gamma$, for $B = \Gamma$ implies by construction $c(A) = X$ which would imply $c(A) = c(S)$, contradicting the criteria for independence of S given by (2.3.2) b).

Define $\Gamma' \coloneqq \Gamma \setminus B \cup A$. Then by assumption $c(\Gamma') = X$ and $A \subseteq \Gamma'$ is independent. Choose $x \in S \setminus A$ then by definition of independence $x \notin c(A)$. By (2.3.10) there exists $y \in \Gamma' \setminus A$ such that

$$c(\Gamma' \setminus \{y\} \cup \{x\}) = c(\Gamma') = X$$

Note by construction that $y \notin B$, so we see that $(A \cup \{x\}, B \cup \{y\}) \in \mathcal{F}$ has greater length, which contradicts maximality. ∎

Corollary 2.3.12 (Bases have the same cardinality)
Every base of a finite rank matroid is finite and of the same size. Denote this by $r(X)$.

Proof. There is a finite basis by (2.3.7). Then apply (2.3.11). ∎

Corollary 2.3.13
Let (X, c) *be a finite-rank matroid and* $S \subseteq X$ *an independent subset, then* $\#S \leq r(X)$. *Similarly a spanning subset* Γ *satisfies* $\#\Gamma \geq r(X)$.

Proof. Apply (2.3.7) and (2.3.12). ∎

Corollary 2.3.14 (Basis Criteria)
Let $\mathcal{B}$ *be a subset of a finite-rank matroid* (X, c). *Then the following are equivalent*

a) $\mathcal{B}$ *is a basis*
b) $\mathcal{B}$ *is independent and* $\#\mathcal{B} \geq r(X)$
c) $\mathcal{B}$ *is spanning and* $\#\mathcal{B} \leq r(X)$

Proof. Apply (2.3.7) and (2.3.12). ∎

Proposition 2.3.15
Let $Y \subseteq X$ *be a sub-matroid of a finite-rank matroid. Then* $Y = X \iff r(Y) = r(X)$.

Proof. Let $\mathcal{B}$ be a basis for Y then $\#\mathcal{B} = r(Y) = r(X)$ and is a-fortiori independent in X. Therefore by (2.3.14) $\mathcal{B}$ is a basis for X and hence $X = c(\mathcal{B}) = Y$. ∎

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2.4 Decomposition in Noetherian and Distributive Lattices

An analogue of irreducible factorization in rings (see the Factorisation section) applies to Noetherian Lattices. Furthermore uniqueness holds when the lattice is distributive. For a canonical reference see [Bir40].

Definition 2.4.1 (Meet-Prime and Meet-Irreducible)
Let $(X, \leq)$ be a lattice and $x \in X$. Then we say that x is

- ***meet-irreducible*** *if $y \wedge z = x \implies y = x$ or $z = x$*
- ***meet-prime*** *if $y \wedge z \leq x \implies y \leq x$ or $z \leq x$*
- ***join-irreducible*** *if $y \vee z = x \implies y = x$ or $y = z$*
- ***join-prime*** *if $x \leq y \vee z \implies x \leq y$ or $x \leq z$*

The following result is proven in [Bir40] Ch. IX Lemma 4.1.

Proposition 2.4.2 (Prime = Irreducible)
Let $(X, \leq)$ be a lattice. In general meet-prime $\implies$ meet-irreducible and join-prime $\implies$ join-irreducible. If X is a distributive lattice, then the converse holds.

In the distributive case we denote by $\mathcal{M}(X)$ and $\mathcal{J}(X)$ the sub-poset of meet-prime and join-prime elements respectively.

Proof. The first statement is straightforward. Conversely suppose X is a distributive lattice and x is meet-irreducible. If $y \wedge z \leq x$ then $x = x \vee (y \wedge z) = (y \vee x) \wedge (z \vee x) \implies x = y \vee x$ or $x = z \vee x$, whence the result follows. ∎

Proposition 2.4.3
Let $(X, \leq)$ be a distributive lattice and Y a sub-lattice, then

$$\mathcal{M}(Y) = \mathcal{M}(X) \cap Y$$

$$\mathcal{J}(Y) = \mathcal{J}(X) \cap Y$$

In particular this holds when $Y = \{x\}^{\uparrow}, \{y\}^{\downarrow}, \{x\}^{\uparrow} \cap \{y\}^{\downarrow}$.

Proposition 2.4.4
Let $(X, \leq)$ be a distributive lattice. If it is chain-complete (resp. co-chain-complete) then so is $\mathcal{J}(X)$ (resp. $\mathcal{M}(X)$).

Every join-prime element is bounded above by a maximal join-prime element, and every meet-prime element is bounded below by a minimal meet-prime element

Proof. Let C be a chain of join-prime elements and $x := \bigvee C$. Suppose that $y \vee z \geq x$ then $y \vee z \geq w$ for all $w \in C$. Then $y \geq w$ or $z \geq w$ for all $w \in C$. Let $C_1 := \{w \in C \mid y \geq w\}$. If $C_1 = C$ then we are done as $x \leq y$. Otherwise suppose $w_0 \notin C_1$ then by prime-ness $z \geq w_0$. Clearly $w \leq w_0 \implies w \leq z$. Further $w \geq w_0 \implies w \not\leq y$ (as otherwise $w_0 \leq y$) whence $w \leq z$. Therefore $x \leq z$.

The last statement follows from Zorn's Lemma by considering the sub-lattices $\{x\}^{\uparrow}$ and $\{y\}^{\downarrow}$ which inherit the chain complete properties. ∎

Definition 2.4.5
Let $(X, \leq)$ be a lattice and $Y \subseteq X$ a finite subset. Then we say that Y is

- ***(meet-)irredundant*** *if no proper subset has the same meet*
- ***incomparable*** *(or an **antichain**) if no two elements are comparable*

Lemma 2.4.6 (incomparable $\iff$ irredundant)
Let $(X, \leq)$ be a lattice and $Y \subseteq X$ then Y meet-irredundant $\implies$ Y incomparable. Conversely if Y is a finite incomparable subset of meet-prime elements then Y is meet-irredundant.

Proof. The first part is straightforward, for if $y_1 \lneq y_2$ are elements of Y then $\bigwedge Y = \bigwedge Y \setminus \{y_2\}$.

Conversely suppose $Y' \subsetneq Y$ is such that $\bigwedge Y' = \bigwedge Y$. Choose $y_2 \in Y \setminus Y'$, then $\bigwedge Y' \leq y_2$, whence by definition of meet-prime (and induction) $y_1 \leq y_2$ for some $y_1 \in Y'$. ∎

The following is [Bir40] Chapter IX Theorem 9

Proposition 2.4.7 (Decomposition in Noetherian Lattice)
Let $(X, \leq)$ be a Noetherian distributive lattice. Then every element $x \in X$ has a unique decomposition

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$$x = x_1 \wedge ... \wedge x_n$$

where x_i are meet-prime and meet-irredundant (equivalently incomparable). These are precisely the meet-primes minimal over x.

Dually, if $(X, \leq)$ is an Artinian distributive lattice, then every element $x \in X$ has a unique decomposition

$$x = x_1 \vee ... \vee x_n$$

where x_i are join-prime and join-irredundant (equivalently incomparable). These are precisely the join-primes maximal below x.

Proof. Let Y be the subset of elements which are not finite meets of meet-prime elements, and suppose it is non-empty. Then by (2.1.63) Y has a maximal element x_0. It cannot be meet-prime, and therefore not meet-irreducible (2.4.2), so there must exist elements $y_0, z_0 \in X$ such that $x_0 = y_0 \wedge z_0$ but $x_0 \lneq y_0$ and $x_0 \lneq z_0$. By maximality y_0, z_0 are finite meets of prime elements, and therefore so is x_0 a contradiction.

Therefore we have a decomposition into distinct primes

$$x = x_1 \wedge ... \wedge x_n$$

Consider the family of subsets of $\{x_1, ..., x_n\}$ which have the same meet. Then there exists a minimal subset which by definition is meet-irredundant and by (2.4.6) incomparable. Suppose there is another such decomposition $x = x'_1 \wedge ... \wedge x'_m$. Then for every $i = 1...n$ we have $x'_{\sigma(i)} \leq x_i$ and for every $j = 1...m$ we have $x_{\tau(j)} \leq x'_j$ whence $x_{\tau(\sigma(i))} \leq x'_{\sigma(i)} \leq x_i$. As the decomposition is incomparable we have $\tau(\sigma(i)) = i$ and $x'_{\sigma(i)} = x_i$. Therefore σ is injective and $n \leq m$. By symmetry $m \leq n$ and σ is a bijection. In otherwords the decomposition is unique.

Note $x \leq z$ and z meet-prime implies $x_j \leq z$ for some j. Therefore if z is a minimal prime then $x_j = z$ and all the meet-primes minimal over x must occur in the decomposition. Similarly if $z \leq x_i$ then by incomparability $x_j = z = x_i$. Therefore each x_i is also minimal. ∎

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2.5 Krull Dimension

The purpose of this section is to abstract the notions of Krull Dimension in commutative ring theory and topology. A more standard approach (eg EGA IV) would be to develop the topological notion first, and then link to commutative ring case using the prime spectrum. Generally the concept is not well-behaved, so stronger conditions are defined which generally hold in geometric cases. Principle references are (EGA0 IV 14.3, Heinrich).

Definition 2.5.1 (Finite-Dimensional Poset)
*Let $(\mathcal{G}, \leq)$ be a poset, we say that it is **finite-dimensional** if*

$$\dim(\mathcal{G}) := \sup\{\ell(C) \mid C \subseteq \mathcal{G} \text{ a chain}\} < \infty$$

In this case we define

$$\begin{aligned}\dim(x) &:= \dim\big(\{x\}^{\downarrow}\big)\\ \operatorname{codim}(y) &:= \dim\big(\{y\}^{\uparrow}\big)\\ \operatorname{codim}(y, x) &:= \dim\big(\{x\}^{\downarrow} \cap \{y\}^{\uparrow}\big)\end{aligned}$$

Note $\mathcal{G}$ is both Noetherian and Artinian, but finite-dimensionality is a stronger condition. Note also that $\{x\}^{\downarrow}, \{y\}^{\uparrow}$ and $\{x\}^{\downarrow} \cap \{y\}^{\uparrow}$ are finite-dimensional posets

Definition 2.5.2 (Krull Lattice)
*Let $(\mathcal{F}, \leq)$ be an **Artinian** distributive lattice. We say it is a **Krull Lattice** if the poset of join-prime elements $\mathcal{J}(\mathcal{F})$ is finite-dimensional and define*

$$\dim(\mathcal{F}) := \dim(\mathcal{J}(\mathcal{F}))$$

By (2.4.3) we have $\mathcal{H} = \{x\}^{\downarrow}, \{y\}^{\uparrow}, \{x\}^{\downarrow} \cap \{y\}^{\uparrow}$ are Krull Lattices such that

$$\mathcal{J}(\mathcal{H}) = \mathcal{J}(\mathcal{F}) \cap \mathcal{H}$$

For $x, y \in \mathcal{F}$ we have a unique decomposition into maximal join-prime elements $x_i, y_j \in \mathcal{J}(\mathcal{F})$ (2.4.7)

$$x = x_1 \vee \ldots \vee x_n$$

$$y = y_1 \vee \ldots \vee y_m$$

Note that $y \leq x \iff$ for all j we have $y_j \leq x_i$ for some i and we may define

$$\begin{aligned}\dim(x) &:= \max_i \dim(x_i) = \dim\big(\{x\}^{\downarrow}\big)\\ \operatorname{codim}(y) &:= \min_j \operatorname{codim}\big(y_j\big)\\ \operatorname{codim}(y, x) &:= \min_j \max_i \big\{\operatorname{codim}\big(y_j, x_i\big) \mid y_j \leq x_i\big\}\\ &= \min_j \operatorname{codim}\big(y_j, x\big)\end{aligned}$$

note it's required to be careful in definition of co-dimension in order to have a sensible co-dimension formula. Note also that

$$\dim\big(y; \{x\}^{\downarrow}\big) = \dim(y)$$

Remark 2.5.3
For the topological case, we would define $\mathcal{F}$ to be the closed subsets of X and $\mathcal{J}(\mathcal{F})$ would be the collection of irreducible closed subsets.

Proposition 2.5.4 (Extending chains)
Let $(\mathcal{G}, \leq)$ be a finite-dimensional poset

- *Every chain is contained in a saturated chain*
- *Every chain is contained in a maximal chain*
- *Every maximal chain is of the form*

  $$x_0 \lneq x_1 \ldots \lneq x_n$$

  for x_0 minimal and x_n maximal in $\mathcal{G}$.

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Definition 2.5.5 (Properties)
Let $\mathcal{G}$ be a finite-dimensional poset. Then we say it is

- ***Irreducible*** *if it has a top element*
- ***Equidimensional*** *if every maximal element has the same dimension*
- ***Equicodimensional*** *if every minimal element has the same codimension*
- ***Biequidimensional*** *if every maximal chain has the same length*
- ***Quasi-Biequidimensional*** *if $\{x\}^{\downarrow}$ is biequidimensional for all x (equivalently all maximal x)*
- ***Catenary*** *if for every pair $y \leq x$, every saturated chain in $[y,x] := \{y\}^{\uparrow} \cap \{x\}^{\downarrow}$ has the same length, namely* $\operatorname{codim}(y,x)$.

If $\mathcal{F}$ is a Krull Lattice then we say it inherits these properties from $\mathcal{J}(\mathcal{F})$. Note if $\mathcal{F}$ is irreducible then it also has (the same) top element.

Trivially irreducible implies equidimensional. Similarly biequidimensional implies both equidimensional and equicodimensional, but not conversely.

Finally ***quasi-biequidimensional + equidimensional $\iff$ biequidimensional.***

Proposition 2.5.6 (Simple Properties)
Let $\mathcal{G}$ be a finite-dimensional poset then

a) If x is maximal then $\operatorname{codim}(x) = 0$
b) If x is minimal then $\dim(x) = 0$
c) $\dim(\mathcal{G}) = \sup\{\dim(x) \mid x \text{ maximal}\} = \sup\{\operatorname{codim}(x) \mid x \text{ minimal}\}$
d) For all $z \leq y \leq x$ we have $\operatorname{codim}(z,y) + \operatorname{codim}(y,x) \leq \operatorname{codim}(z,x)$

If $\mathcal{F}$ is a Krull Lattice then

e) For all $y \leq x$ we have $\dim(y) + \operatorname{codim}(y,x) \leq \dim(x)$
f) For all $y \leq x$ we have $\operatorname{codim}(y,x) = 0 \iff y_j = x_i$ *some i,j*

Alternatively $\operatorname{codim}(y,x) > 0$ *if and only if* $\big(y_j \leq x_i \implies y_j \neq x_i\big)$.

Proof. e) The case of a finite-dimensional poset is (relatively) clear. In the general case then we have

$$\dim\big(y_j\big) + \operatorname{codim}(y,x) \leq \dim\big(y_j\big) + \operatorname{codim}\big(y_j,x\big) = \max_i\big(\dim\big(y_j\big) + \operatorname{codim}\big(y_j,x_i\big)\big) \leq \max_i \dim(x_i) = \dim(x)$$

and taking max over j yields the result.

f) The case $x,y \in \mathcal{J}(\mathcal{F})$ is clear by (2.5.4). For the general case $\operatorname{codim}(y,x) = 0 \iff \operatorname{codim}\big(y_j,x\big) = 0$ for some $j \iff \big(y_j \leq x_i \implies \operatorname{codim}\big(y_j,x_i\big) = 0\big) \iff y_j = x_i$ for some i,j. ∎

Corollary 2.5.7 (Codimension 1 formula)
Let $\mathcal{F}$ be a Krull Lattice with $y \in \mathcal{F}$, $x \in \mathcal{J}(\mathcal{F})$ and $y \leq x$. Then

$$\dim(y) = \dim(x) - 1 \implies \operatorname{codim}(y,x) = 1$$

Suppose further that $\mathcal{F}$ is irreducible then

$$\dim(y) = \dim(\mathcal{F}) - 1 \implies \operatorname{codim}(y) = 1$$

Proof. By (2.5.6) e) $\operatorname{codim}(y,x) \leq 1$. If $\operatorname{codim}(y,x) = 0$ then by f) we see that $y_j = x$ for some j, whence $y = x$ which contradicts $\dim(y) = \dim(x) - 1$. Therefore $\operatorname{codim}(y,x) = 1$ as required.

If $\mathcal{F}$ is irreducible the result follows with $x = \top$. ∎

Remark 2.5.8 (Duality)
We note that the concepts of ***dimension*** *(of a poset).* ***biequidimensional*** *and* ***catenary*** *are* self-dual, *in the sense that they are preserved when considering the dual poset* $\left(\mathcal{G}, \overset{d}{\leq}\right)$.

Similarly the concepts of ***equidimensional*** *and* ***equicodimensional*** *are dual to each other.*

Proposition 2.5.9
Let $\mathcal{G}$ be a finite-dimensional poset. Then the following are equivalent

- *$\mathcal{G}$ is catenary*

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- *For every triplet $z < y < x$ in $\mathcal{G}$ we have*

$$\operatorname{codim}(z, x) = \operatorname{codim}(z, y) + \operatorname{codim}(y, x)$$

When $\mathcal{G}$ is irreducible this is equivalent to

$$\operatorname{codim}(z) = \operatorname{codim}(z, y) + \operatorname{codim}(y)$$

for all $z < y$.

Proof. Suppose $\mathcal{G}$ is catenary. Choose saturated chains C_1 in $[z, y]$ and C_2 in $[y, x]$. One may show that $C_1 \cap C_2 = \{y\}$ and $C_1 \cup C_2$ is a saturated chain in $[x, z]$. The result follows by definition of catenary.

Conversely, consider a saturated chain between x and y of length n

$$x = x_0 \lneq x_1 \lneq ... \lneq x_n = y$$

Then clearly $\operatorname{codim}(x_i, x_{i+1}) = 1$ for all $i = 0...n-1$. Therefore by repeatedly applying the relation we may show that

$$\operatorname{codim}(x, y) = n$$

and therefore every saturated chain between x and y has the same length.

When $\mathcal{G}$ is irreducible we may deduce the second formula by setting $x = \top$. Conversely we may see that

$$\operatorname{codim}(z, x) = \operatorname{codim}(z) - \operatorname{codim}(x) = \operatorname{codim}(z) - \operatorname{codim}(y) + \operatorname{codim}(y) - \operatorname{codim}(x) = \operatorname{codim}(z, y) + \operatorname{codim}(y, x)$$

■

Lemma 2.5.10
Let $\mathcal{G}$ be a biequidimensional finite-dimensional poset. Then for $x \in \mathcal{G}$ we have

a) $\{x\}^{\downarrow}$ and $\{x\}^{\uparrow}$ are biequidimensional
b) $\dim(\mathcal{G}) = \dim(x) + \operatorname{codim}(x)$
c) If x is maximal then $\dim(x) = \dim(\mathcal{G})$ and in particular $\mathcal{G}$ is equidimensional
d) If x is minimal then $\operatorname{codim}(x) = \dim(\mathcal{G})$ and in particular $\mathcal{G}$ is equicodimensional

In particular $\mathcal{G}$ is quasi-biequidimensional.

Proof. Consider a fixed maximal chain C of $\{x\}^{\uparrow}$, necessarily containing x. Any maximal chain C' of $\{x\}^{\downarrow}$ also contains x and combines with C to yield a maximal chain of $\mathcal{F}$. Whence $\ell(C') + \ell(C) = \dim(\mathcal{F})$ and $\{x\}^{\downarrow}$ is biequidimensional. By duality $\{x\}^{\uparrow}$ is biequidimensional and $\ell(C) = \operatorname{codim}(x)$ from which the formula follows.

If x is maximal then clearly $\operatorname{codim}(x) = 0$, and similarly if x is minimal then $\dim(x) = 0$, so the last two statements follow immediately. ■

Proposition 2.5.11 (Equivalent Characterizations of Biequidimensionality)
Let $\mathcal{G}$ be a finite-dimensional poset. Then the following are equivalent

a) $\mathcal{G}$ is quasi-biequidimensional
b) $\mathcal{G}$ is catenary and for every maximal x we have $\{x\}^{\downarrow}$ is equicodimensional
c) $\mathcal{G}$ satisfies $\dim(x) = \dim(y) + \operatorname{codim}(y, x)$ for $y \leq x$
d) $\mathcal{G}$ satisfies c) whenever $\operatorname{codim}(y, x) = 1$

Furthermore the following relationship holds

$$\operatorname{codim}(y) = \operatorname{codim}(y, x) + \operatorname{codim}(x) \quad y \leq x$$

Proof. a) $\implies$ c). By assumption $\{x\}^{\downarrow}$ is biequidimensional. Then

$$\dim(x) = \dim\big(\{x\}^{\downarrow}\big) \overset{(2.5.10)}{=} \dim\big(y; \{x\}^{\downarrow}\big) + \operatorname{codim}\big(y; \{x\}^{\downarrow}\big) = \dim(y) + \operatorname{codim}(y, x)$$

For c) $\implies$ b). Suppose $z < y < x$ in $\mathcal{G}$ then by the codimension formula applied twice

$$\operatorname{codim}(z, x) = \dim(x) - \dim(z) = \dim(x) - \dim(y) + \dim(y) - \dim(z) = \operatorname{codim}(y, x) + \operatorname{codim}(z, y)$$

so by (2.5.9) $\mathcal{G}$ is catenary. Let x be a maximal element and z a minimal element of $\{x\}^{\downarrow}$, then by the codimension formula

$$\operatorname{codim}(z, x) = \dim(x) - \dim(z) = \dim(x)$$

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whence $\{x\}^{\downarrow}$ is equicodimensional.

b) $\implies$ a). Let x be a maximal element and C a maximal chain in $\{x\}^{\downarrow}$ with minimum element y then, as $\mathcal{G}$ is catenary, we have $\ell(C) = \operatorname{codim}(y, x)$. As $\{x\}^{\downarrow}$ is equicodimensional we have $\operatorname{codim}(y, x) = \dim(x)$ which shows that $\{x\}^{\downarrow}$ is biequidimensional.

Clearly c) $\implies$ d). Conversely for d) $\implies$ a) consider a maximal chain in $\{x\}^{\downarrow}$

$$x_0 \lneq \ldots \lneq x_n = x$$

Then clearly $\operatorname{codim}(x_i, x_{i+1}) = 1$ whence by induction $\ell(C) = \dim(x)$.

The final statement follows from (2.5.9) because $\{x\}^{\downarrow}$ is irreducible and catenary. ∎

Remark 2.5.12
This is a corrected version of EGA IV 14.3.3, as noted by Heinrich.

Corollary 2.5.13
When $\mathcal{F}$ is a quasi-biequidimensional Krull Lattice then we have the following codimension formulas for all $x \in \mathcal{J}(\mathcal{F})$ and $y \in \mathcal{F}$ such that $y \leq x$

$$\dim(x) = \dim(y) + \operatorname{codim}(y, x)$$
$$\operatorname{codim}(y) = \operatorname{codim}(y, x) + \operatorname{codim}(x)$$

Furthermore if $\mathcal{F}$ is biequidimensional then for all $y \in \mathcal{F}$

$$\dim(\mathcal{F}) = \dim(y) + \operatorname{codim}(y)$$

Proof. The first two relations hold when $x, y \in \mathcal{J}(\mathcal{F})$ by (2.5.11). We may generalise this as follows.

For the first relation and fixed $x \in \mathcal{J}(\mathcal{F})$ we see $\dim(y_j)$ is maximal precisely when $\operatorname{codim}(y_j, x)$ is minimal. Therefore it holds for all $y \in \mathcal{F}$.

Similarly for the second relation we see that $\operatorname{codim}(y_j, x)$ is minimal precisely when $\operatorname{codim}(y_j)$ is minimal.

The final statement follows from the first relation by taking the supremum over all maximal elements x. ∎

Corollary 2.5.14
Let $\mathcal{G}$ be an irreducible finite-dimensional poset. Then the following are equivalent

a) $\mathcal{G}$ is biequidimensional
b) $\mathcal{G}$ is quasi-biequidimensional
c) $\mathcal{G}$ is catenary and equicodimensional
d) $\dim(x) = \dim(y) + \operatorname{codim}(y, x) \quad \forall y \leq x$
e) $\mathcal{G}$ satisfies d) when $\operatorname{codim}(y, x) = 1$

Proof. The equivalence follows from (2.5.11) by noting that an irreducible poset has only one maximal element and is in particular equidimensional. ∎

Proposition 2.5.15
Let $\mathcal{F}$ be a quasi-biequidimensional Krull Lattice. Then for all $x \in \mathcal{F}$ we have $\{x\}^{\downarrow}$ is a quasi-biequidimensional Krull Lattice.

Proof. Consider $y \in \mathcal{J}(\{x\}^{\downarrow}) \overset{(2.4.3)}{=} \mathcal{J}(\mathcal{F}) \cap \{x\}^{\downarrow}$. Then by definition $\{y\}^{\downarrow}$ is biequidimensional, and so $\{x\}^{\downarrow}$ is quasi-biequidimensional. ∎

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2.6 Category Theory

2.6.1 Categories

Definition 2.6.1 (Category)
A (locally small) ***category*** $\mathcal{C}$ *consists of*

- *a class* $\mathrm{ob}(\mathcal{C})$ *of objects*
- *for every pair of objects* $X, Y \in \mathrm{ob}(\mathcal{C})$ *a* set *of* ***morphisms*** $\mathrm{Mor}(X, Y)$
- *for every three objects* X, Y, Z *a* ***law of composition***

$$\begin{aligned} \mathrm{Mor}(X,Y) \times \mathrm{Mor}(Y,Z) &\to \mathrm{Mor}(X,Z) \\ (g,f) &\to g \circ f \end{aligned}$$

such that the following conditions hold

- $h \circ (g \circ f) = (h \circ g) \circ f$ *(**associativity**)*
- *There exists* $1_X \in \mathrm{Mor}(X,X)$ *such that* $1_X \circ f = f$ *and* $g \circ 1_X = g$.

Example 2.6.2
The category of sets is **Set** *with maps in the usual way. Note associativity is automatically satisfied.*

Example 2.6.3 (n-pointed category)
Given a category $\mathcal{C}$ *where objects are sets, we may consider the pointed category* $(\mathcal{C}, \star^n)$ *consisting of pairs* (A, a) *where* $A \in \mathrm{ob}(\mathcal{C})$ *and* $a \in A^n$. *We consider only morphisms* $f : A \to B$ *such that* $f(a_i) = b_i$.

Definition 2.6.4 (Opposite Category)
Given a category $\mathcal{C}$ *the* ***opposite category*** *is denoted* $\mathcal{C}^{op}$ *and given by*

- *The same class of objects* $\mathrm{ob}(\mathcal{C}^{op}) = \mathrm{ob}(\mathcal{C})$
- *For every pair of objects* $X, Y \in \mathrm{ob}(\mathcal{C})$ *the morphisms are reversed*

$$\mathrm{Mor}^{op}(X,Y) := \mathrm{Mor}(Y,X)$$

- *The law of composition is reversed*

$$\begin{aligned} \mathrm{Mor}^{op}(X,Y) \times \mathrm{Mor}^{op}(Y,Z) &\to \mathrm{Mor}^{op}(X,Z) \\ (g,f) &\to f \circ g \end{aligned}$$

Example 2.6.5
The polynomial ring $A[X]$ *is an initial object in the category of pointed* A*-algebras.*

Definition 2.6.6 (Isomorphism)
A morphism $f : X \to Y$ *is an* ***isomorphism*** *if there exists* $g : Y \to X$ *such that*

$$g \circ f = 1_X \quad \text{and} \quad f \circ g = 1_Y.$$

Definition 2.6.7 (Functor)
A ***covariant functor*** $F : \mathcal{C} \to \mathcal{D}$ *consists of a mapping of objects*

$$F : \mathrm{ob}(\mathcal{C}) \to \mathrm{ob}(\mathcal{D})$$

together with a mapping of morphisms

$$F(-) : \mathrm{Mor}(X,Y) \to \mathrm{Mor}(F(X), F(Y))$$

which satisfies

- $F(1_X) = 1_{F(X)}$
- $F(f \circ g) = F(f) \circ F(g)$

A ***contravariant functor*** $\mathcal{C} \to \mathcal{D}$ *is equivalent to a covariant functor on the opposite category* $\mathcal{C}^{op} \to \mathcal{D}$, *namely where arrows are reversed.*

Definition 2.6.8 (Full and faithful)
A functor F *is said to be*

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- ***Faithful*** *if* $F(-)$ *is injective.*
- ***Full*** *if* $F(-)$ *is surjective.*

Definition 2.6.9 (Concrete Category)
A ***concrete category*** *is a pair* $(\mathcal{C}, U)$ *where* $\mathcal{C}$ *and a "****forgetful functor****"* $U : \mathcal{C} \to \mathbf{Set}$ *which is faithful.*

Example 2.6.10 (Forgetful Functor)
The category of groups (resp. rings, modules, ...) is a concrete category in the obvious way.

Definition 2.6.11 (Mor functor)
For any objects $X, Y, Z \in \mathrm{ob}(\mathcal{C})$*, there is a canonical covariant functor*

$$\begin{aligned} \mathrm{Mor}(X, -) : \mathcal{C} &\longrightarrow \mathbf{Set} \\ Y &\longrightarrow \mathrm{Mor}(X, Y) \end{aligned}$$

which acts on a morphism $f : Y \to Z$ *by*

$$\begin{aligned} \mathrm{Mor}(X, f) : \mathrm{Mor}(X, Y) &\to \mathrm{Mor}(X, Z) \\ g &\to f \circ g \end{aligned}$$

It's a functor precisely because composition of functions is associative. Similarly there's a canonical contravariant functor $\mathrm{Mor}(-, Y)$*.*

Definition 2.6.12 (Natural Transformation)
Let $F, G : \mathcal{C} \to \mathcal{D}$ *be covariant functors. A* ***natural transformation*** $\eta : F \Rightarrow G$ *consists of a family of morphisms*

$$\eta_Z : F(Z) \to G(Z) \quad Z \in \mathrm{ob}(\mathcal{C})$$

such that the following diagram commutes for all $f : Z \to Z'$

$$\begin{array}{ccc} F(Z) & \xrightarrow{\eta_Z} & G(Z) \\ \Big\downarrow{\scriptstyle F(f)} & & \Big\downarrow{\scriptstyle G(f)} \\ F(Z') & \xrightarrow{\eta_{Z'}} & G(Z') \end{array}$$

Definition 2.6.13 (Natural isomorphism)
A natural transformation $\eta : F \Rightarrow G$ *is a natural isomorphism if* η_Z *is an isomorphism for all* $Z \in \mathcal{C}$*.*

Definition 2.6.14 (Functor Category)
Let I *be a small category and* $\mathcal{D}$ *a category. The* ***functor category*** $\mathcal{D}^I$ *has*

- *objects: functors* $F : I \to \mathcal{D}$*;*
- *morphisms* $F \to G$*: natural transformations* $\eta : F \Rightarrow G$ *(2.6.12);*
- *composition: for* $\eta : F \Rightarrow G$ *and* $\theta : G \Rightarrow H$*,* $(\theta \circ \eta)_Z := \theta_Z \circ \eta_Z$ *for* $Z \in \mathrm{ob}(I)$ *– again a natural transformation, since for* $f : Z \to Z'$ *in* I*, using naturality of* θ *then of* η*,*

$$H(f) \circ \theta_Z \circ \eta_Z = \theta_{Z'} \circ G(f) \circ \eta_Z = \theta_{Z'} \circ \eta_{Z'} \circ F(f);$$

- *identities:* 1_F*, with* $(1_F)_Z := 1_{F(Z)}$ *(trivially natural).*

Associativity and the identity laws hold pointwise at each $Z \in \mathrm{ob}(I)$*, being inherited from those of* $\mathcal{D}$*; so* $\mathcal{D}^I$ *is indeed a category.*

Definition 2.6.15
We say $\mathcal{C}' \subset \mathcal{C}$ *is a* ***subcategory*** *if*

- $\mathrm{ob}(\mathcal{C}') \subseteq \mathrm{ob}(\mathcal{C})$
- $\mathrm{Mor}_{\mathcal{C}'}(Z, W) \subseteq \mathrm{Mor}_{\mathcal{C}}(Z, W)$
- *Composition agrees when it is well-defined*

We say $\mathcal{C}'$ *is a* ***full subcategory*** *if additionally* $\mathrm{Mor}_{\mathcal{C}'}(Z, W) = \mathrm{Mor}_{\mathcal{C}}(Z, W)$*.*

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2.6.2 Comma and Slice Categories

Definition 2.6.16 (Comma Category)
*Let $F : \mathcal{C} \to \mathcal{E}$ and $G : \mathcal{D} \to \mathcal{E}$ be functors. The **comma category** $(F \downarrow G)$ is the category with*

- *objects: triples (C, D, f) where $C \in \mathcal{C}$, $D \in \mathcal{D}$, and $f : F(C) \to G(D)$ is a morphism in $\mathcal{E}$*
- *morphisms: a morphism $(C, D, f) \to (C', D', f')$ is a pair (g, h) where $g : C \to C'$ is a morphism in $\mathcal{C}$ and $h : D \to D'$ is a morphism in $\mathcal{D}$, such that*

$$G(h) \circ f = f' \circ F(g)$$

with composition $(g', h') \circ (g, h) := (g' \circ g, h' \circ h)$ and identities $1_{(C,D,f)} := (1_C, 1_D)$, inherited from $\mathcal{C}$ and $\mathcal{D}$.

Definition 2.6.17 (Slice Category)
*Let $\mathcal{C}$ be a category and $X \in \mathcal{C}$, and let $\underline{X} : \mathbf{1} \to \mathcal{C}$ be the functor with $\underline{X}(\star) = X$. The **slice category** $\mathcal{C}/X$ (also denoted $\mathcal{C} \downarrow X$) is the comma category $(\mathrm{id}_{\mathcal{C}} \downarrow \underline{X})$ (2.6.16).*

Explicitly its objects are pairs (Y, f) where $Y \in \mathrm{ob}(\mathcal{C})$ and $f : Y \to X$ is a morphism in $\mathcal{C}$, and a morphism $(Y, f) \to (Y', f')$ is a morphism $g : Y \to Y'$ in $\mathcal{C}$ such that $f' \circ g = f$.

*Dually the **coslice category** $X/\mathcal{C}$ (also denoted $X \downarrow \mathcal{C}$) is the comma category $(\underline{X} \downarrow \mathrm{id}_{\mathcal{C}})$, with objects pairs (Y, f) where $Y \in \mathrm{ob}(\mathcal{C})$ and $f : X \to Y$ is a morphism in $\mathcal{C}$, and a morphism $(Y, f) \to (Y', f')$ is a morphism $g : Y \to Y'$ in $\mathcal{C}$ such that $g \circ f = f'$.*

2.6.3 Product Categories and Bifunctors

Definition 2.6.18 (Product Category)
Given two categories $\mathcal{C}$, $\mathcal{D}$ we may construct the product category $\mathcal{C} \times \mathcal{D}$ as follows

- *The objects are given by pairs*

$$\mathrm{ob}(\mathcal{C} \times \mathcal{D}) := \mathrm{ob}(\mathcal{C}) \times \mathrm{ob}(\mathcal{D})$$

- *The morphisms are given by pairs*

$$\mathrm{Mor}((C, C'), (D, D')) := \mathrm{Mor}(C, C') \times \mathrm{Mor}(D, D')$$

  Concretely given $f : C \to C'$ and $g : D \to D'$ write $(f, g) : (C, D) \to (C', D')$
- *The law of composition is determined by*

$$(f, g) \circ (h, k) := (f \circ h, g \circ k)$$

It's clear that the law of composition is associative and the identity morphism for (C, D) is $(1_C, 1_D)$. Furthermore we observe the following property

$$(f, 1_D) \circ (1_C, g) = (f, g) = (1_C, g) \circ (f, 1_D) \tag{*}$$

Definition 2.6.19 (Bifunctor)
*A functor on a product category, $F : \mathcal{C} \times \mathcal{D} \to \mathcal{E}$ is termed a **bifunctor**. F induces a family of functors*

- *$F(C, -) : \mathcal{D} \to \mathcal{E}$ given by*

$$F(C, g) := F(1_C, g)$$

- *$F(-, D) : \mathcal{C} \to \mathcal{E}$ given by*

$$F(f, D) := F(f, 1_D)$$

which satisfy the compatibility conditions

$$\begin{array}{ccc} F(C,D) & \xrightarrow{F(f,D)} & F(C',D) \\ {\scriptstyle F(C,g)}\downarrow & \searrow^{F(f,g)} & \downarrow{\scriptstyle F(C',g)} \\ F(C,D') & \xrightarrow[F(f,D')]{} & F(C',D') \end{array}$$

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by applying F to (*).

Proposition 2.6.20 (Reconstruct Bifunctor)
Consider a family of functors $\{F_L(C) : \mathcal{D} \to \mathcal{E}\}_{C \in \mathrm{ob}(\mathcal{C})}$ and $\{F_R(D) : \mathcal{C} \to \mathcal{E}\}_{D \in \mathrm{ob}(\mathcal{D})}$. Suppose the following conditions hold

- $F_L(C)(D) = F_R(D)(C)$
- $F_L(C')(g) \circ F_R(D)(f) = F_R(D')(f) \circ F_L(C)(g)$

then these determine a well-defined bifunctor given by

$$F(f \times g) \coloneqq F_L(C')(g) \circ F_R(D)(f)$$

Proposition 2.6.21
Let $F : \mathcal{C} \times \mathcal{D} \to \mathcal{E}$ be a bifunctor. For $f : C \to C'$ then there is a natural transformation

$$F(f, -) : F(C, -) \Rightarrow F(C', -)$$

given by $F(f, -)_D \coloneqq F(f, 1_D)$. Similarly for $g : D \to D'$ then there is a natural transformation

$$F(-, g) : F(-, D) \Rightarrow F(-, D')$$

Proof. The naturality condition is immediate from (*) and the commutative diagram in (2.6.19). ∎

Proposition 2.6.22 (Criteria for Natural Transformation)
Let $F, G : \mathcal{C} \times \mathcal{D} \to \mathcal{E}$ be two bifunctors and suppose we have a family of natural transformations

$$\eta_C : F(C, -) \Rightarrow G(C, -)$$

for all $C \in \mathcal{C}$. Then the following are equivalent

a) $\eta : F(-, -) \Rightarrow G(-, -)$ is a natural transformation
b) The following diagram commutes for all $D \in \mathcal{D}$ and $f : C \to C'$

$$\begin{array}{ccc} F(C, D) & \xrightarrow{\eta_{C,D}} & G(C, D) \\ {\scriptstyle F(f, 1_D)}\downarrow & & \downarrow{\scriptstyle G(f, 1_D)} \\ F(C', D) & \xrightarrow{\eta_{C',D}} & G(C', D) \end{array}$$

Proof. a) $\implies$ b) This diagram is a special case of the naturality condition.

b) $\implies$ a) Suppose $f : C \to C'$ and $g : D \to D'$ then we require to show that

$$G(f, g)\big(\eta_{C,D}(\phi)\big) = \eta_{C',D'}(F(f, g)(\phi))$$

However

$$\begin{aligned} G(f, g)\big(\eta_{C,D}(\phi)\big) &= G(f, 1_{D'})\big(G(1_C, g)\big(\eta_{C,D}(\phi)\big)\big) \\ &= G(f, 1_{D'})\eta_{C,D'}(F(1_C, g)(\phi)) \\ &= \eta_{C',D'}(F(f, 1_{D'})(F(1_C, g)(\phi))) \\ &= \eta_{C',D'}(F(f, g)(\phi)) \end{aligned}$$

where we have used that $\eta_{C,D}$ is natural in D and C individually. ∎

Example 2.6.23 (Mor is a bifunctor)
The canonical example is the following, given any locally small category $\mathcal{C}$, we have a bifunctor

$$\mathrm{Mor} : \mathcal{C}^{op} \times \mathcal{C} \to \mathbf{Set}$$

given by the set of morphisms. The action on morphisms is given by

$$\mathrm{Mor}(f, g)(\phi) = g \circ \phi \circ f$$

and we verify by associativity of $\mathcal{C}$ that

$$\mathrm{Mor}((f, g) \circ (h, k)) = \mathrm{Mor}(f, g) \circ \mathrm{Mor}(h, k)$$

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and it's clear that the commutativity condition in (*) is satisfied.

Example 2.6.24 (Concrete bifunctors)
Let $\mathcal{C}$, $\mathcal{D}$ be concrete categories and consider the following bifunctor

$$F : \mathcal{D}^{op} \times \mathcal{C} \to \mathbf{Set}$$
$$F(D, C) := \{\phi : D \to C \mid P(\phi)\}$$

where $P(-)$ is some predicate such that $P(\phi) \implies P(g \circ \phi \circ f)$.

2.6.4 Equivalence of categories

Definition 2.6.25 (Equivalence of categories)
*Let $\mathcal{C}$, $\mathcal{D}$ be categories. An **equivalence of categories** consists of a pair of functors (either both covariant or both contravariant)*

$$\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$$

together with natural isomorphisms

$$\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow GF$$

$$\epsilon : FG \Rightarrow \mathrm{id}_{\mathcal{D}}$$

We say F is an equivalence of categories if there exists some G satisfying these conditions.

Definition 2.6.26
*We say a functor $F : \mathcal{C} \to \mathcal{D}$ is **essentially surjective** if for all $D \in \mathcal{D}$ there exists $C \in \mathcal{C}$ such that $F(C)$ is isomorphic to D.*

Lemma 2.6.27
Let $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{C}$ be functors.

If there exists a natural isomorphism $\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow GF$ then F is faithful. Explicitly $F(-)$ has a left-inverse given by

$$g \to \eta_{C'}^{-1} \circ G(g) \circ \eta_C$$

Furthermore $GF(\eta_C) = \eta_{GF(C)}$.

Proof. Consider the sequence of maps

$$\mathrm{Mor}(C, C') \overset{F(-)}{\longrightarrow} \mathrm{Mor}(F(C), F(C')) \overset{G(-)}{\longrightarrow} \mathrm{Mor}(GF(C), GF(C')) \overset{\mathrm{Mor}\left(\eta_C, \eta_{C'}^{-1}\right)}{\longrightarrow} \mathrm{Mor}(C, C')$$

Note that the composite of this map is given by

$$f \to \eta_{C'}^{-1} \circ GF(f) \circ \eta_C = \eta_{C'}^{-1} \circ \eta_{C'} \circ f = f$$

in other words $F(-)$ has a left inverse and therefore F is faithful.

Note by naturality we have $GF(\eta_C) \circ \eta_C = \eta_{GF(C)} \circ \eta_C$. Since η_C is an isomorphism we may cancel to find $GF(\eta_C) = \eta_{GF(C)}$. ∎

Proposition 2.6.28 (Equivalence is full and faithful)
Let $F : \mathcal{C} \to \mathcal{D}$ be a functor then the following are equivalent

- *F is full, faithful and essentially surjective (2.6.26)*
- *F is an equivalence of categories (2.6.25)*

In other words $F(-)$ is bijective and hence has a two-sided inverse. Explicitly it is given by

$$\mathrm{Mor}(C, C') \longleftrightarrow \mathrm{Mor}(F(C), F(C'))$$
$$f \to F(f)$$
$$\eta_{C'}^{-1} \circ G(g) \circ \eta_C \leftarrow g$$

Proof. We prove only the second implies the first. By assumption there is an equivalence with G and by the previous Lemma both F and G are faithful by considering η and ϵ^{-1} in turn. Further the given map is already shown to be a left inverse. We claim it's also a right inverse, for consider

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$$g' := F\left(\eta_{C'}^{-1}\right) \circ FG(g) \circ F(\eta_C).$$

We claim that $G(g') = G(g)$. As G is faithful this would imply $g' = g$ and the given map is a right inverse as required. Observe

$$G(g') = GF\left(\eta_{C'}^{-1}\right) \circ GFG(g) \circ GF(\eta_C) = \eta_{GF(C')}^{-1} \circ GFG(g) \circ \eta_{GF(C)} = \eta_{GF(C')}^{-1} \circ \eta_{GF(C')} \circ G(g) = G(g)$$

where we have used the result that $GF(\eta_C) = \eta_{GF(C)}$. Since the maps are mutual inverses we see that F is full and faithful as required.

Given $D \in \mathcal{D}$ then $F(G(D))$ is isomorphic to D via ϵ so F is essentially surjective. ■

Proposition 2.6.29 (Duality)
Let $(-)^\star : \mathcal{C} \to \mathcal{C}$ be a contravariant functor such that there is a natural isomorphism

$$\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow (-)^{\star\star}$$

then $(-)^\star$ is an equivalence of categories and in particular full and faithful.

Proof. Define $\epsilon = \eta^{-1}$ to determine the equivalence of categories. By the previous result then $(-)^\star$ is full and faithful.■

2.6.5 Properties of Morphisms

Definition 2.6.30 (Injective, Surjective and Bijective)
Let $(\mathcal{C}, U)$ be a concrete category (2.6.9) and $f : X \to Y$ a morphism. Then we say

- *f is **injective** if $U(f)$ is injective*
- *f is **surjective** if $U(f)$ is surjective*

Remark 2.6.31
Note if f is both surjective and injective it need not be an isomorphism.

The concepts of monic/split-monic, epic/split-epic, iso generalize the notion of injective, surjective and bijective to general categories as we shall see.

Definition 2.6.32 (Monomorphism / Epimorphism)
*A morphism $f : X \to Y$ is said to be a **monomorphism** (or **monic**) if*

$$f \circ g_1 = f \circ g_2 \Longrightarrow g_1 = g_2$$

for all $g_1, g_2 : Z \to X$.

*Dually, f is said to be an **epimorphism** (or **epic**) if*

$$g_1 \circ f = g_2 \circ f \Longrightarrow g_1 = g_2$$

for all $g_1, g_2 : Y \to Z$.

Remark 2.6.33 (Monic/Epic Duality)
A morphism $f : X \to Y$ of $\mathcal{C}$ is monic if and only if $f^{op} : Y \to X$ is epic in $\mathcal{C}^{op}$, and dually f is epic if and only if f^{op} is monic in $\mathcal{C}^{op}$ (2.6.4): composition in $\mathcal{C}^{op}$ reverses the order of composition in $\mathcal{C}$, so the defining cancellation property of monic in $\mathcal{C}$ is precisely the defining cancellation property of epic in $\mathcal{C}^{op}$, and vice versa.

Definition 2.6.34 (Split-monic / Split-epic)
*A morphism $f : X \to Y$ is **split-monic** if it has a left inverse, $g : Y \to X$*

$$g \circ f = 1_X$$

*We say g is a **section** of f.*

*Dually, f is **split-epic** if it has a right inverse, $g : Y \to X$*

$$f \circ g = 1_Y$$

*We say g is a **retraction** of f.*

Proposition 2.6.35 (Split Monic⇒Monic)
For a general category $\mathcal{C}$ we have

- *split-monic $\Longrightarrow$ monic*

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- *split-epic* $\implies$ *epic*

We say an **isomorphism** is a morphism with a two-sided inverse. We can refine the criteria for f to be an isomorphism using the notions just defined

Proposition 2.6.36 (Isomorphism Criteria)
Let $f : X \to Y$ be a morphism. Then the following are equivalent

*a) f is an **isomorphism***
*b) f is both **split-epic** and **split-monic***
*c) f is **split-epic** and **monic***
*d) f is **split-monic** and **epic***

In this case a morphism g is a retraction if and only if it is a section. And such a g is unique, so we denote it by f^{-1}

Proof. This is mostly formal

a) $\implies$ b) Clear.

b) $\implies$ c,d) This follows from (2.6.35).

c) $\implies$ b) Suppose g is a retraction of f, that is $fg = 1_Y$. Then $f(gf) = (fg)f = 1_Y \circ f = f = f \circ 1_X$. As f is monic we conclude that $gf = 1_X$ and g is a section of f.

d) $\implies$ b) Analogous

b) $\implies$ a) We've shown that any retraction is a section and vice-versa. Furthermore by monic/epic-ness a retraction or section is unique. ■

Lemma 2.6.37 (Factors of an Isomorphism)
Let $f : X \to Y$ be an isomorphism.

a) If $f = m \circ g$ for some monomorphism m, then m and g are both isomorphisms.
b) If $f = h \circ e$ for some epimorphism e, then h and e are both isomorphisms.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

a) Since f is an isomorphism, $m \circ (g \circ f^{-1}) = f \circ f^{-1} = 1_Y$, so m is split-epic. As m is also monic, (2.6.36) gives that m is an isomorphism, and so $g = m^{-1} \circ f$ is also an isomorphism, being a composite of isomorphisms.

b) Dually, since f is an isomorphism, $(f^{-1} \circ h) \circ e = f^{-1} \circ f = 1_X$, so e is split-monic. As e is also epic, (2.6.36) gives that e is an isomorphism, and so $h = f \circ e^{-1}$ is also an isomorphism. ■

Proposition 2.6.38
For the category **Set** *we have*

- *split-monic* $\iff$ *monic* $\iff$ *injective*
- *split-epic* $\iff$ *epic* $\iff$ *surjective*
- *isomorphism* $\iff$ *bijective*

Definition 2.6.39 (Balanced Category)
*A category $\mathcal{C}$ is **balanced** if every morphism which is both monic and epic is an isomorphism.*

Example 2.6.40
Set *is balanced: by (2.6.38), the monic and epic morphisms are precisely the injective and surjective ones, so a morphism which is both is bijective, hence an isomorphism. The category of groups is balanced for the same reason, a bijective group homomorphism has a group homomorphism as its (set-theoretic) inverse.*

Not every category is balanced. In **Top** *the map $[0, 2\pi) \to S^1$, $t \mapsto e^{it}$ is continuous and bijective, hence monic and epic, but its inverse is not continuous, so it is not an isomorphism. Likewise in* **Rng** *the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is monic and epic (any two ring homomorphisms out of $\mathbb{Q}$ agreeing on $\mathbb{Z}$ agree everywhere) but is not an isomorphism.*

Definition 2.6.41 (Preserves/Reflects)
Let $\mathcal{P}$ be a property of morphisms and $F : \mathcal{C} \to \mathcal{D}$ be a functor then we say

- *F **preserves** $\mathcal{P}$ if (f satisfies $\mathcal{P} \implies F(f)$ satisfies $\mathcal{P}$)*
- *F **reflects** $\mathcal{P}$ if ($F(f)$ satisfies $\mathcal{P} \implies f$ satisfies $\mathcal{P}$)*

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Proposition 2.6.42
Let $F : \mathcal{C} \to \mathcal{D}$ be a covariant functor then

- *F preserves split-monic, split-epic and iso morphisms.*

If in addition F is faithful then

- *F reflects monic and epic morphisms.*

and if F is full and faithful then

- *F reflects split-epic, split-monic and isomorphisms.*

Similar statements apply when F is contravariant.

Proof. The first statement is easy, for example if $gf = 1_X$ then $F(g) \circ F(f) = 1_{F(X)}$.

Suppose F is faithful, $F(f)$ is monic and $fg_1 = fg_2$. Then $F(f)F(g_1) = F(f)F(g_2) \implies F(g_1) = F(g_2)$ by assumption. As F is faithful $g_1 = g_2$ as required. The other statement is similar.

Suppose F is full and faithful and $F(f)$ is split-monic. Then $hF(f) = 1_{F(X)}$. As F is full $h = F(g)$ and $1_{F(X)} = F(gf)$. As F is faithful then $gf = 1_X$. the other statements are similar. ∎

Proposition 2.6.43
Let $(\mathcal{C}, U)$ be a concrete category (2.6.9) then

- *f split-monic $\implies$ f injective $\implies$ f monic*
- *f split-epic $\implies$ f surjective $\implies$ f epic*
- *f isomorphism $\implies$ f bijective*

Proof. Suppose f is split-monic, then $U(f)$ is split-monic by (2.6.42) and so by (2.6.38) $U(f)$ is injective.

Suppose $U(f)$ injective, then by (2.6.38) $U(f)$ is monic. By (2.6.42) U reflects monics and so f is monic.

The other statements are similar. ∎

We can restate the criteria for split-epic/split-monic

Proposition 2.6.44
Let $f : X \to Y$ be a morphism then

- *f is split-monic if and only if $\mathrm{Mor}(f, Z)$ is surjective for all $Z \in \mathcal{C}$*
- *f is epic if and only if $\mathrm{Mor}(f, Z)$ is injective for all $Z \in \mathcal{C}$*

dually

- *f is split-epic if and only if $\mathrm{Mor}(Z, f)$ is surjective for all $Z \in \mathcal{C}$*
- *f is monic if and only if $\mathrm{Mor}(Z, f)$ is injective for all $Z \in \mathcal{C}$*

Proof. f is epic (resp. monic) iff $\mathrm{Mor}(f, Z)$ (resp. $\mathrm{Mor}(Z, f)$) is injective precisely by the definitions.

Suppose f is split-monic, then $gf = 1_X \implies (hg)f = h$ for any h. That is $\mathrm{Mor}(f, Z)$ is surjective. Conversely if it's surjective then let $Z = X$ and choose g such that $gf = \mathrm{Mor}(f, X)(g) = 1_X$.

A similar statement follows dually for f split-epic, and f monic. ∎

Corollary 2.6.45 (Isomorphism Criteria)
Let $f : X \to Y$ be a morphism then TFAE

- *f is an isomorphism*
- *$\mathrm{Mor}(f, Z)$ is bijective for all $Z \in \mathcal{C}$*
- *$\mathrm{Mor}(Z, f)$ is bijective for all $Z \in \mathcal{C}$*

Proof. This follows from combining (2.6.44) with (2.6.36). ∎

Definition 2.6.46 (Conservative)
*We say a functor $F : \mathcal{C} \to \mathcal{D}$ is **conservative** if it reflects isomorphisms.*

*We say a concrete category $(\mathcal{C}, U)$ is **conservative** if the functor $U : \mathcal{C} \to \mathbf{Set}$ is conservative.*

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Proposition 2.6.47
Let $(\mathcal{C}, U)$ be a conservative concrete category which admits directed limits. Then the following are equivalent

a) U preserves directed colimits
b) U reflects directed colimits

2.6.6 Pointed Categories

Definition 2.6.48 (Initial object)
*An **initial object** of a category $\mathcal{C}$ is an object $\emptyset$ such that for all objects X*

$$\mathrm{Mor}(\emptyset, X) = \{\eta_\emptyset^X\}$$

consists of a single element.

In this case an initial object is unique up to unique isomorphism (2.6.49).

Proposition 2.6.49 (Criteria to be Initial)
Let $\emptyset$ be an initial object and $X \in \mathcal{C}$. Then the following are equivalent

a) X is an initial object
b) there is an isomorphism $\phi : X \xrightarrow{\sim} \emptyset$
c) there exists a morphism $\phi : X \to \emptyset$ and only one endomorphism $1_X : X \to X$

In this case the isomorphism is unique.

Proof. a) $\implies$ c) by definition

c) $\implies$ b) By assumption $\eta_\emptyset^X \circ \phi = 1_X$, and by definition of initial $\phi \circ \eta_\emptyset^X = 1_\emptyset$.

b) $\implies$ a) We have a bijection

$$\begin{aligned} \mathrm{Mor}(\emptyset, Y) &\longleftrightarrow \mathrm{Mor}(X, Y) \\ f &\to f \circ \phi \\ g \circ \phi^{-1} &\leftarrow g \end{aligned}$$

so as $\emptyset$ is initial, $\mathrm{Mor}(\emptyset, Y)$ is a singleton for every Y, hence so is $\mathrm{Mor}(X, Y)$, that is X is initial. This closes the cycle a) $\implies$ c) $\implies$ b) $\implies$ a), so all three are equivalent.

Finally, since X is then initial, $\mathrm{Mor}(X, \emptyset)$ is a singleton, so there is at most one morphism $X \to \emptyset$; in particular the isomorphism ϕ is unique. ∎

Definition 2.6.50 (Terminal Object)
*A **terminal object** of a category $\mathcal{C}$ is an object $\star$ such that for all objects $X \in \mathcal{C}$ we have*

$$\mathrm{Mor}(X, \star) = \{!_X\}$$

is a singleton.

In this case a terminal object is unique up to unique isomorphism (2.6.51).

Proposition 2.6.51 (Criteria to be Terminal)
Let $\star$ be a terminal object and $X \in \mathcal{C}$. Then the following are equivalent

a) X is a terminal object
b) there is an isomorphism $\phi : \star \xrightarrow{\sim} X$
c) there exists a morphism $\phi : \star \to X$ and only one endomorphism $1_X : X \to X$

In this case the isomorphism is unique.

Proof. By (2.6.4), $\mathrm{Mor}_{\mathcal{C}^{\mathrm{op}}}(\emptyset, X) = \mathrm{Mor}_{\mathcal{C}}(X, \emptyset)$ for all X, so $\star$ is a terminal object of $\mathcal{C}$ if and only if $\star$ is an initial object of $\mathcal{C}^{\mathrm{op}}$; likewise a morphism $\star \to X$ (resp. isomorphism, resp. endomorphism of X) in $\mathcal{C}$ is exactly a morphism $X \to \star$ (resp. isomorphism, resp. endomorphism of X) in $\mathcal{C}^{\mathrm{op}}$. So applying (2.6.49) in $\mathcal{C}^{\mathrm{op}}$, with $\emptyset := \star$, gives exactly the stated equivalence of a), b), c), together with the uniqueness of the isomorphism. ∎

Lemma 2.6.52 (Equivalences of Categories Preserve Terminal Objects)
Let $F : \mathcal{C} \to \mathcal{D}$ be an equivalence of categories, with G and natural isomorphisms $\eta : \mathrm{id}_\mathcal{C} \Rightarrow GF$, $\epsilon : FG \Rightarrow \mathrm{id}_\mathcal{D}$ as in (2.6.25). Then $T \in \mathcal{C}$ is a terminal object of $\mathcal{C}$ if and only if $F(T)$ is a terminal object of $\mathcal{D}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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Suppose T is terminal. By (2.6.28), F is full and faithful, so for every $C \in \mathcal{C}$ the map $\mathrm{Mor}(C,T) \to \mathrm{Mor}(F(C),F(T))$, $f \mapsto F(f)$, is a bijection; as T is terminal $\mathrm{Mor}(C,T)$ is a singleton, so $\mathrm{Mor}(F(C),F(T))$ is too. Now let $Y \in \mathcal{D}$ be arbitrary and set $C \coloneqq G(Y)$; the isomorphism $\epsilon_Y : F(C) \to Y$ gives a bijection $\mathrm{Mor}(F(C),F(T)) \to \mathrm{Mor}(Y,F(T))$, $g \mapsto g \circ \epsilon_Y^{-1}$, so combining with the above, $\mathrm{Mor}(Y,F(T))$ is a singleton. As $Y \in \mathcal{D}$ was arbitrary, $F(T)$ is terminal.

Conversely, G is itself an equivalence of categories (with F, $\epsilon^{-1} : \mathrm{id}_{\mathcal{D}} \Rightarrow FG$ and $\eta^{-1} : GF \Rightarrow \mathrm{id}_{\mathcal{C}}$ playing the roles of F, η, ϵ in (2.6.25)), so if $F(T)$ is terminal then applying the above (with F,G,T replaced by $G,F,F(T)$) shows $GF(T)$ is terminal; as $\eta_T : T \to GF(T)$ is an isomorphism, T is then terminal too by (2.6.51). ∎

Definition 2.6.53 (Terminal Category)
*The **terminal category** $\mathbf{1}$ is the category with a single object $\star$ and a single morphism $1_\star$.*

For any category $\mathcal{C}$ and object $X \in \mathcal{C}$, we write $\underline{X} : \mathbf{1} \to \mathcal{C}$ for the functor with $\underline{X}(\star) = X$.

Definition 2.6.54 (Zero Object / Pointed Category)
*A **zero object** of a category $\mathcal{C}$ is both an initial and terminal object. We denote this by $\mathbf{0}$ or $\mathbf{0}_{\mathcal{C}}$ when we need to emphasise the category $\mathcal{C}$*

For two objects $X,Y \in \mathcal{C}$ we denote by $0_{X,Y}$ the composite morphism $X \to 0 \to Y$. This is independent of the choice of zero object.

*A category with a zero object is **pointed**. A category $\mathcal{C}$ is pointed iff $\mathcal{C}^{op}$ is pointed. Further $\mathbf{0}_{\mathcal{C}} = \mathbf{0}_{\mathcal{C}^{op}}$.*

Proposition 2.6.55
Let $\mathcal{C}$ be a pointed category and $X \in \mathcal{C}$. Then the following are equivalent

a) X is an initial object
b) X is a terminal object
c) X is a zero object

Proposition 2.6.56
Let $\mathcal{C}$ be a pointed category. Then the morphism $0 \to X$ is a monomorphism and the morphism $Y \to 0$ is an epimorphism.

Proposition 2.6.57 (Functors Preserve Zeros)
Let $F : \mathcal{C} \to \mathcal{D}$ be a functor of pointed categories for which $F(0_{\mathcal{C}})$ is a zero object of $\mathcal{D}$. Then for all objects X,Y we have

$$F(0_{X,Y}) = 0_{F(X),F(Y)}$$

2.6.7 Yoneda Lemma

The motivation for the following result becomes more clear in the next section. Roughly speaking properties of an object, X, are encoded in the functor $\mathrm{Mor}(X,-)$.

Proposition 2.6.58 (Yoneda Lemma)
Suppose $X,X' \in \mathcal{C}$ then there is a bijection between morphisms and natural transformation of functors

$$\begin{aligned} \mathrm{Mor}(X',X) &\xrightarrow{\sim} \mathrm{Nat}(\mathrm{Mor}(X,-),\mathrm{Mor}(X',-)) \\ f &\longrightarrow \mathrm{Mor}(f,-) : (\phi \to \phi \circ f) \\ \alpha_X(1_X) &\longleftarrow \alpha \end{aligned}$$

Observe that under this correspondence

- $f = \mathrm{Mor}(f,X)(1_X)$
- $\mathrm{Mor}(f \circ g,-) = \mathrm{Mor}(g,-) \circ \mathrm{Mor}(f,-)$
- *f is an isomorphism $\iff$ $\mathrm{Mor}(f,-)$ is a natural isomorphism.*

In the latter case $\mathrm{Mor}(f,-)^{-1} = \mathrm{Mor}(f^{-1},-)$

In many cases the set of morphisms $\mathrm{Mor}(X,X')$ has an additional structure (typically an abelian group or module). We encode this with the following definition

Definition 2.6.59 (Enriched Hom Functor)
Let $(\mathcal{S},U)$ be a concrete category (2.6.9) for which U reflects isomorphisms. We say a bifunctor

$$\mathrm{Hom} : \mathcal{C}^{op} \times \mathcal{C} \to \mathcal{S}$$

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is an ***enriched hom functor*** *if we have*

- $U(\mathrm{Hom}(X,Y)) = \mathrm{Mor}(X,Y)$
- $U(\mathrm{Hom}(f,Y)) = \mathrm{Mor}(f,Y)$
- $U(\mathrm{Hom}(X,g)) = \mathrm{Mor}(X,g)$

and bijections

$$\begin{array}{ccc} & \mathrm{Mor}(X',X) & \\ \mathrm{Hom}(f,-) \swarrow \sim & & \sim \searrow \mathrm{Mor}(f,-) \\ \mathrm{Nat}(\mathrm{Hom}(X,-),\mathrm{Hom}(X',-)) & \xrightarrow[\sim]{U} & \mathrm{Nat}(\mathrm{Mor}(X,-),\mathrm{Mor}(X',-)) \end{array}$$

Write the inverse of $f \to \mathrm{Hom}(f,-)$ *as* $\mathcal{Y}$*. Then for* $\alpha \in \mathrm{Nat}(\mathrm{Hom}(X,-),\mathrm{Hom}(X',-))$

a) α *is a natural isomorphism* $\iff U\alpha$ *is a natural isomorphism* $\iff \mathcal{Y}(\alpha)$ *is an isomorphism.*
b) $\mathcal{Y}(\alpha) = U(\alpha_X)(1_X)$

Note the right hand arrow is always a bijection by Yoneda's Lemma (2.6.58) and the diagram always commutes by assumption.

Proof. Suppose α is a natural isomorphism then $U\alpha^{-1} \circ U\alpha = U\left(1_{\mathrm{Hom}(X,-)}\right) = 1_{\mathrm{Mor}(X,-)}$ and similarly $U\alpha \circ U\alpha^{-1} = 1_{\mathrm{Mor}(X',-)}$, so $U\alpha$ is a natural isomorphism. ∎

Note this trivially includes the usual case $\mathcal{S} = \mathbf{Set}$.

2.6.8 Representable Functors

Many algebraic constructions (e.g. tensor product) may be formalised in terms of "representable" functors (2.6.60). The usual definition involves the set-valued functor $\mathrm{Mor}(X,-)$ as defined in (2.6.11).

As a generalisation follows we consider an enriched hom functor (2.6.59) $\mathrm{Hom}(-,-)$ taking values in the concrete category $(\mathcal{S},U)$ where U reflects isomorphisms. This clearly generalises the usual case.

Definition 2.6.60 (Representable Functor)
Let $F : \mathcal{C} \to \mathcal{S}$ *a covariant functor. We say* F *is* ***representable*** *if there is a pair* (X,Φ) *where* $X \in \mathcal{C}$ *and* Φ *is a natural isomorphism*

$$\mathrm{Hom}(X,-) \xrightarrow[\sim]{\Phi} F(-)$$

We say F *is* ***represented*** *by the pair* (X,Φ)*. Note this implies* $U \circ F$ *is represented by* $U\Phi$*.*

It is often useful to have another conception of representable functor in terms of universal properties as this is more intuitive for applications.

Definition 2.6.61 (Universal Element)
Let $F : \mathcal{C} \to \mathcal{S}$ *be a covariant functor, then we say that a pair* (X,i)*, where* $X \in \mathcal{C}, i \in (U \circ F)(X)$*, is a* ***universal element*** *for* F *if for all* $Y \in \mathcal{C}$ *there are bijections*

$$\begin{aligned} \mathrm{Mor}(X,Y) &\longrightarrow (U \circ F)(Y) \\ f &\to (U \circ F)(f)(i) \end{aligned}$$

In this case we may also say F *is* ***represented*** *by the universal element* (X,i)*.*

We show the equivalence of the two notions by some abstract nonsense

Proposition 2.6.62 (Representable Set-Valued Functor = Universal Element)
Let $F : \mathcal{C} \to \mathbf{Set}$ *be a covariant functor then there is a natural correspondence between representations and universal elements*

$$\begin{aligned} \{\text{representations of } F\} &\longleftrightarrow \{\text{universal elements of } F\} \\ (X,\Phi) &\longrightarrow (X,\Phi(1_X)) \\ (X, f \to F(f)(i)) &\longleftarrow (X,i) \end{aligned}$$

The following may in particular be used to show that representations are unique up to isomorphism.

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Proposition 2.6.63 (Morphisms Between Representations)
Suppose (X, Φ) and (X', Φ') represent the covariant functors $F, F' : \mathcal{C}^{op} \to \mathcal{S}$ respectively. Then there are bijections

$$\begin{aligned} \mathrm{Mor}(X, X') &\xrightarrow{\sim} \mathrm{Nat}(\mathrm{Hom}(X', -), \mathrm{Hom}(X, -)) \xrightarrow{\sim} \mathrm{Nat}(F'(-), F(-)) \\ f &\to \mathrm{Hom}(f, -) && \to \hat{f} \\ U(\eta_{X'})(1_{X'}) &\leftarrow \eta \qquad \alpha^\star && \leftarrow \alpha \end{aligned}$$

where we define the mutual inverses

$$\begin{aligned} \hat{f} &= \Phi \circ \mathrm{Hom}(f, -) \circ (\Phi')^{-1} \\ \alpha^\star &= U\big(\Phi_{X'}^{-1} \circ \alpha_{X'} \circ \Phi'_{X'}\big)(1_{X'}) \end{aligned}$$

Note this has the following properties

- $\widehat{1_X} = \mathrm{id}_F$
- $\widehat{g \circ f} = \hat{f} \circ \hat{g}$ *where* $g : X' \to X''$ *is morphism and* (X'', Φ'') *represents* F''
- f *is an isomorphism* $\iff \hat{f}$ *is a natural isomorphism and in this case* $\widehat{f^{-1}} = \hat{f}^{-1}$.

Further $\alpha^\star \in \mathrm{Mor}(X, X')$ is the unique morphism such that the following diagram of natural transformations commutes

$$\begin{array}{ccc} \mathrm{Hom}(X', -) & \xRightarrow{\Phi'} & F'(-) \\ {\scriptstyle \mathrm{Hom}(\alpha^\star, -)}\Vdots & & \Vdots{\scriptstyle \alpha} \\ \mathrm{Hom}(X, -) & \xRightarrow{\Phi} & F(-) \end{array}$$

and satisfies

- $\mathrm{id}_F^\star = 1_X$
- $(\beta \circ \alpha)^\star = \alpha^\star \circ \beta^\star$
- $\alpha^\star$ *is an isomorphism* $\iff \alpha$ *is a natural isomorphism*

Proof. The first bijection is simply the Yoneda Lemma and the second bijection is obvious. ∎

Corollary 2.6.64 (Representation is Unique)
Let $F : \mathcal{C} \to \mathcal{S}$ be a covariant functor which is represented by pairs (X, Φ) and (X', Φ'). Then they are isomorphic with two-sided inverses

$$X \underset{U(\Phi'^{-1})(i)}{\overset{U(\Phi^{-1})(i')}{\rightleftarrows}} X'$$

where $i \coloneqq (U\Phi)(1_X)$ and $i' \coloneqq (U\Phi')(1_{X'})$.

Proof. We may apply the previous result with $F = F'$ and the identity natural transformation id_F. ∎

In practice we typically have a family of representations, and we want to show the construction is functorial

Definition 2.6.65 (Representable Bifunctor)
Let $F : \mathcal{D}^{op} \times \mathcal{C} \to \mathcal{S}$ be a bifunctor. We say that it is ***representable*** *by (G, Φ) where $G : \mathcal{D} \to \mathcal{C}$ is a functor, if Φ is a natural isomorphism of bifunctors*

$$\Phi : \mathrm{Hom}(G(-), -) \xRightarrow{\sim} F(-, -)$$

The following shows we only need to construct the representation pointwise.

Proposition 2.6.66
Let $F : \mathcal{D}^{op} \times \mathcal{C} \to \mathcal{S}$ be a bifunctor such that $F(D, -)$ is representable for all $D \in \mathcal{D}$, by $(G(D), \Phi_D)$. Then G can be made into a covariant functor uniquely such that Φ constitutes a natural isomorphism of bifunctors

$$\Phi_{D,C} : \mathrm{Hom}(G(D), C) \xRightarrow{\sim} F(D, C)$$

Explicitly for any morphism $h : D \to D'$ we have the identity

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$$G(h) = \Phi^{-1}_{D,G(D')}\Big(\Phi_{D',G(D')}\big(1_{G(D')}\big) \circ h\Big)$$

Proof. It is straightforward to verify that for any covariant functor G, the mapping $\mathrm{Mor}(G(-),-)$ is a bifunctor (contravariant in the first "slot").

Let $h : D \to D'$ be a morphism then by (2.6.21) $F(h,-) : F(D',-) \to F(D,-)$ is a natural transformation. Therefore we may define

$$G(h) \coloneqq F(h,-)^\star$$

in the notation of (2.6.63); it is the unique morphism making the following diagram commute

$$\begin{array}{ccc} \mathrm{Hom}(G(D'),-) & \xRightarrow{\Phi_{D'}} & F(D',-) \\ \Big\Vert\!\Big\Vert\!\Big\Vert\!\

Proof. In order to demonstrate that $\mathrm{Hom}(f_D, 1_C)$ is a natural transformation of bifunctors it's enough by (2.6.21) to show the following diagram commutes for $h : D' \to D$

$$\begin{array}{ccc} \mathrm{Hom}(G'(D), C) & \xrightarrow{-\circ f_D} & \mathrm{Hom}(G(D), C) \\ \downarrow{\scriptstyle -\circ G'(h)} & & \downarrow{\scriptstyle -\circ G(h)} \\ \mathrm{Hom}(G'(D'), C) & \xrightarrow{-\circ f_{D'}} & \mathrm{Hom}(G(D'), C) \end{array}$$

However $f_D \circ G(h) = G'(h) \circ f_{D'}$ by definition of natural transformation, from which it follows immediately. The map is a pointwise bijection (for every $D \in \mathcal{D}$) with the inverse given by the usual Yoneda Lemma.

The first two properties are straightforward. From these it follows that if f is a natural isomorphism then so is $\mathrm{Hom}(f, -)$. Conversely if $\mathrm{Hom}(f, -)$ is a natural isomorphism then it has a two-sided inverse which is of the form $\mathrm{Hom}(g, -)$. It's then straightforward to argue that g is a two-sided inverse of f. ∎

2.6.9 Adjoint Functors

Some universal constructions may be expressed as an adjoint pair of functors. My view is the most illustrative are the module Tensor-Hom adjunction and the sheafification construction. Using this concept we can simplify the verification of universal properties by appealing to general criteria for adjoint functors as below.

Definition 2.6.69 (Adjoint Pair)
*Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors. We say that F is **left adjoint** to G if there is a bijection*

$$\psi_{C,D} : \mathrm{Mor}(F(C), D) \longrightarrow \mathrm{Mor}(C, G(D))$$

which is natural in C and D in the following sense. Let $\alpha : C' \to C$, $\beta : D \to D'$, then for all $f : F(C) \to D$ we have

$$\psi_{C',D'}(\beta \circ f \circ F(\alpha)) = G(\beta) \circ \psi_{C,D}(f) \circ \alpha(1)$$

or equivalently for all $g : C \to G(D)$

$$\beta \circ \psi^{-1}_{C,D}(g) \circ F(\alpha) = \psi^{-1}_{C',D'}(G(\beta) \circ g \circ \alpha)(2)$$

Equivalently $F(-)$ represents the bifunctor $\mathrm{Mor}(-, G(-))$ (2.6.65).

Proposition 2.6.70 (Adjoint⇒unit, counit)
Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be adjoint functors with relationship

$$\psi_{C,D} : \mathrm{Mor}(F(C), D) \xrightarrow{\sim} \mathrm{Mor}(C, G(D))$$

*Then we have two natural transformations (**unit** and **counit** respectively)*

$$\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow G \circ F$$
$$\epsilon : F \circ G \Rightarrow \mathrm{id}_{\mathcal{D}}$$

defined by

$$\eta_C = \psi_{C,F(C)}\left(1_{F(C)}\right)$$
$$\epsilon_D = \psi^{-1}_{G(D),D}\left(1_{G(D)}\right)$$

Furthermore we may recover the adjoint relationship via

$$\psi_{C,D}(f) = G(f) \circ \eta_C$$

$$\psi^{-1}_{C,D}(g) = \epsilon_D \circ F(g)$$

Proof. We show that the transformations given are natural. Suppose $\alpha : C \to C'$ then

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$$\begin{aligned} G(F(\alpha)) \circ \eta_C &= G(F(\alpha)) \circ \psi_{C,F(C)}\left(1_{F(C)}\right) \\ &= \psi_{C,F(C')}\left(F(\alpha) \circ 1_{F(C)}\right) \quad (1) \\ &= \psi_{C,F(C')}\left(1_{F(C')} \circ F(\alpha)\right) \\ &= \psi_{C',F(C')}\left(1_{F(C')}\right) \circ \alpha \quad (1) \\ &= \eta_{C'} \circ \alpha \end{aligned}$$

so η is a natural transformation. Furthermore

$$\psi_{C,D}(f) = \psi_{C,D}\left(f \circ 1_{F(C)}\right) = G(f) \circ \psi_{C,F(C)}(1) = G(f) \circ \eta_C$$

as required. Similarly for $\beta : D \to D'$

$$\begin{aligned} \beta \circ \epsilon_D &= \beta \circ \psi^{-1}_{G(D),D}\left(1_{G(D)}\right) \\ &= \psi^{-1}_{G(D),D'}\left(G(\beta) \circ 1_{G(D)}\right) \quad (2) \\ &= \psi^{-1}_{G(D),D'}\left(1_{G(D')} \circ G(\beta)\right) \\ &= \psi^{-1}_{G(D'),D'}\left(1_{G(D')}\right) \circ F(G(\beta)) \quad (2) \\ &= \epsilon_{D'} \circ F(G(\beta)) \end{aligned}$$

■

Given two natural transformations we may recover a corresponding adjoint

Proposition 2.6.71 (Adjoint from unit and counit)
Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors with two natural transformations

$$\begin{aligned} \eta &: \mathrm{id}_{\mathcal{C}} \Rightarrow G \circ F \\ \epsilon &: F \circ G \Rightarrow \mathrm{id}_{\mathcal{D}} \end{aligned}$$

Then the following are equivalent

a) F is left adjoint to G with unit and counit η, ϵ, and
*b) the so-called **triangular identities** are satisfied*

$$1_{G(D)} : G(D) \overset{\eta_{G(D)}}{\longrightarrow} GFG(D) \overset{G(\epsilon_D)}{\longrightarrow} G(D)$$

$$1_{F(C)} : F(C) \overset{F(\eta_C)}{\longrightarrow} FGF(C) \overset{\epsilon_{F(C)}}{\longrightarrow} F(C)$$

More precisely the adjunction is given by

$$\begin{aligned} \mathrm{Mor}(F(C), D) &\underset{\phi}{\overset{\psi}{\rightleftharpoons}} \mathrm{Mor}(C, G(D)) \\ f &\longrightarrow G(f) \circ \eta_C \\ \epsilon_D \circ F(g) &\longleftarrow g \end{aligned}$$

Proof. Let ψ, ϕ denote the proposed adjunction maps. We will use the triangular identities to show that these are mutually inverse. First observe by naturality of η and ϵ that

$$\begin{aligned} \psi\phi(g) &= G(\epsilon_D \circ F(g)) \circ \eta_C = G(\epsilon_D) \circ \eta_{G(D)} \circ g (3) \\ \phi\psi(f) &= \epsilon_D \circ F(G(f) \circ \eta_C) = f \circ \epsilon_{F(C)} \circ F(\eta_C) \end{aligned}$$

It's then immediate that these are mutually inverse maps if and only if the triangular identities are satisfied (one way is obvious, the other way consider $f = 1_{F(C)}$ and $g = 1_{G(D)}$).

Further one may easily verify that ψ, ϕ so-defined are natural in C and D

$$\begin{aligned} \psi(\beta \circ f \circ F(\alpha)) &= G(\beta) \circ G(f) \circ GF(\alpha) \circ \eta_C \\ &= G(\beta) \circ G(f) \circ \eta_{C'} \circ \alpha \\ &= G(\beta) \circ \psi(f) \circ \alpha \end{aligned}$$

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■

Proposition 2.6.72 (Criteria for right adjoint to be full and faithful)
Let $\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$ be adjoint functors with η, ϵ unit and counit transformations. Then

- *G is faithful if and only if ϵ is pointwise epic*
- *G is full if and only if ϵ is pointwise split-monic*
- *G is full and faithful if and only if ϵ is a pointwise isomorphism*

Proof. Consider the composite map

$$\mathrm{Mor}(D', D) \overset{\mathrm{Mor}(\epsilon_{D'}, D)}{\longrightarrow} \mathrm{Mor}(F(G(D')), D) \overset{\psi_{G(D'),D}}{\to} \mathrm{Mor}(G(D'), G(D))$$

which is natural in D and D'. Note that image of $\alpha \in \mathrm{Mor}(D', D)$ is

$$\psi_{G(D'),D}(\alpha \circ \epsilon_{D'}) = G(\alpha) \circ \psi_{G(D'),D'}(\epsilon_{D'}) = G(\alpha)$$

so the composite is just $G(-)$. The second map is bijective by the adjoint assumption. Therefore the first map is injective (resp. surjective) if and only if G is faithful (resp. full).

By (2.6.44) $\mathrm{Mor}(\epsilon_{D'}, D)$ is injective (resp. surjective) for all D, D' if and only if $\epsilon_{D'}$ is epic (resp. split-monic) for all D'.

Then the first two statements follow easily. The last statement follows from the previous two, combined with (2.6.36).■

Proposition 2.6.73 (Reflective Sub-Category)
Let $\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$ be adjoint functors such that G is fully faithful. Then for $C \in \mathcal{C}$ we have η_C is an isomorphism iff C is in the essential image of G (i.e. is isomorphic to $G(D)$ for some D).

*In particular let $\mathcal{C}'$ denote the **essential image** of $\mathcal{D}$ under G. Then this restricts to an equivalence of categories $\mathcal{C}' \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$*

Proof. If $\eta_C : C \to GF(C)$ is an isomorphism define $D \coloneqq F(C)$, and we are done.

Conversely we first show that $\eta_{G(D)}$ is an isomorphism. By the triangular identities (2.6.71) we have

$$G(\epsilon_D) \circ \eta_{G(D)} = 1_{G(D)}$$

so $\eta_{G(D)}$ has a left inverse. Recall by (2.6.72) ϵ_D is an isomorphism, whence so is $G(\epsilon_D)$ (2.6.42). Therefore we may show directly that it is also a right inverse, and therefore $\eta_{G(D)}$ is an isomorphism.

Now assume that we have an isomorphism $\psi : C \xrightarrow{\sim} G(D)$ for some $D \in \mathcal{D}$. Then by naturality we have a commutative diagram

$$\begin{array}{ccc} C & \overset{\psi}{\underset{\sim}{\longrightarrow}} & G(D) \\ {\scriptstyle \eta_C}\downarrow & & \sim\downarrow{\scriptstyle \eta_{G(D)}} \\ GF(C) & \overset{\sim}{\underset{GF(\psi)}{\longrightarrow}} & GFG(D) \end{array}$$

observing that $GF(\psi)$ is also an isomorphism. Then we conclude η_C is an isomorphism as required. ■

The following criteria will be useful

Proposition 2.6.74 (Adjoints can be constructed pointwise)
Let $G : \mathcal{D} \to \mathcal{C}$ be a functor. Suppose that for all $C \in \mathrm{ob}(\mathcal{C})$ there exists $F(C) \in \mathrm{ob}(\mathcal{D})$ such that there is a bijection natural in D

$$\psi_{C,D} : \mathrm{Mor}(F(C), D) \xrightarrow{\sim} \mathrm{Mor}(C, G(D))$$

Then there exists a unique functor F which ensures this is natural in C, more precisely for $g : C \to C'$ this is given by

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$$F(g) = \psi^{-1}_{C,F(C')}(\eta_{C'} \circ g)$$
$$\eta_C \coloneqq \psi_{C,F(C)}\left(1_{F(C)}\right)$$

This is also the unique functor which ensures η_C is a natural transformation.

Proof. The hypothesis is precisely that, for each $C \in \mathcal{C}$, $\left(F(C), \psi_{C,-}\right)$ represents the functor $\mathrm{Mor}(C, G(-)) : \mathcal{D} \to \mathcal{S}$; equivalently, this is the hypothesis of (2.6.66) applied to the (already well-defined, since G is a functor) bifunctor $\mathrm{Mor}(-, G(-)) : \mathcal{C}^{\mathrm{op}} \times \mathcal{D} \to \mathcal{S}$, with the roles of the two categories there swapped so that $F(C)$ plays the role of the representing object varying over $\mathcal{C}$. That proposition's conclusion is precisely that F can be made into a functor uniquely such that ψ is a natural isomorphism of bifunctors, giving the first uniqueness; its explicit formula for $g : C \to C'$ reads

$$F(g) = \psi^{-1}_{C,F(C')}\left(\psi_{C',F(C')}\left(1_{F(C')}\right) \circ g\right) = \psi^{-1}_{C,F(C')}(\eta_{C'} \circ g)$$

using the definition of $\eta_{C'}$ for the second equality.

We have already shown that if $\psi_{C,D}$ is natural in C then η_C is a natural transformation (2.6.70). Conversely suppose that η_C is a natural transformation. Since we have assumed that $\psi_{C,D}$ is natural in D, applying this to $\beta \coloneqq F(g)$ and $f \coloneqq 1_{F(C)}$ gives

$$\begin{aligned}
\psi_{C,F(C')}(F(g)) &= \psi_{C,F(C')}\left(F(g) \circ 1_{F(C)}\right) \\
&= G(F(g)) \circ \psi_{C,F(C)}\left(1_{F(C)}\right) \\
&= GF(g) \circ \eta_C \\
&= \eta_{C'} \circ g
\end{aligned}$$

using naturality of η for the last equality, whence $F(g) = \psi^{-1}_{C,F(C')}(\eta_{C'} \circ g)$. ∎

Proposition 2.6.75
Let $\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$ be a pair of adjoint functors. Then F preserves initial objects and G preserves terminal objects.

If $\mathcal{C}, \mathcal{D}$ are pointed categories then F and G also preserve zero objects and zero morphisms.

Proof. We only prove that F preserves initial objects, as the second statement follows dually. By hypothesis there is a bijection natural in D

$$\mathrm{Mor}(F(\emptyset), D) \xrightarrow{\sim} \mathrm{Mor}(\emptyset, G(D))$$

Therefore if $\emptyset$ is initial, then so is $F(\emptyset)$.

The last statement follows from (2.6.55) and (2.6.57). ∎

2.6.10 Limits

Definition 2.6.76
*A functor $F : I \to \mathcal{C}$ where I is a small category is referred to as a **diagram***

Definition 2.6.77 (Cones)
*Let $F : I \to \mathcal{C}$ be a diagram. We say that the pair $\left(L, \{\lambda_i : L \to F(i)\}_{i \in I}\right)$ is a **cone** over F if*

$$F(\alpha) \circ \lambda_i = \lambda_j \quad \text{for all morphisms} \quad \alpha : i \to j$$

*The cones for F constitute a **cone category** $(\Delta \downarrow F)$ where a morphism $\alpha : \left(L, \{\lambda_i\}_{i \in I}\right) \to \left(L', \{\lambda'_i\}_{i \in I}\right)$ is just a morphism $\alpha : L \to L'$ such that $\lambda'_i \circ \alpha = \lambda_i$ for all $i \in I$*

Definition 2.6.78 (Small Limit)
*Let $F : I \to \mathcal{C}$ be a diagram. A cone $\left(L, \{\lambda_i\}_{i \in I}\right)$ over F is a **limit** for F if it is a **terminal object** (2.6.50) in the cone category.*

Explicitly for every cone $\left(L', \{\lambda'_i\}_{i \in I}\right)$ there is a unique morphism $\alpha : L' \to L$ such that $\lambda_i \circ \alpha = \lambda'_i$.

We may denote this as $\lim F$.

Proposition 2.6.79
Let $F : I \to \mathcal{C}$ be a diagram, $\left(L, \{\lambda_i\}_{i \in I}\right)$ a limit of F and $\left(L', \{\lambda'_i\}_{i \in I}\right)$ a cone. Then the following are equivalent

a) L' is a limit for F

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b) there is an isomorphism of cones $L' \xrightarrow{\sim} L$

c) there is a morphism of cones $L' \to L$ *and precisely one (cone) endomorphism of* L'.

In this case the (cone) isomorphism is the unique cone morphism.

Proof. By definition $\left(L, \{\lambda_i\}_{i\in I}\right)$ is a terminal object of the cone category $(\Delta \downarrow F)$, so this is precisely (2.6.51) applied with $\star := \left(L, \{\lambda_i\}_{i\in I}\right)$ and $X := \left(L', \{\lambda'_i\}_{i\in I}\right)$. ∎

Proposition 2.6.80 (Uniqueness of Limits)
Let $F : I \to \mathcal{C}$ *be a diagram and* $\left(L, \{\lambda_i\}_{i\in I}\right)$ *and* $\left(L', \{\lambda'_i\}_{i\in I}\right)$ *limits of* F. *Then there is a unique morphism* $\psi : L \to L'$ *satisfying* $\lambda'_i \circ \psi = \lambda_i$, *which is an isomorphism.*

Proof. Since $\left(L', \{\lambda'_i\}_{i\in I}\right)$ is a limit for F, (2.6.79) gives a unique isomorphism of cones $\psi : L \to L'$, that is, a unique isomorphism ψ with $\lambda'_i \circ \psi = \lambda_i$ for all $i \in I$. ∎

Proposition 2.6.81 (Product)
Let I *be a discrete small category and* $\{X_i\}_{i\in I}$ *be a family of objects in* $\mathcal{C}$. *Then there is a unique functor* $F : I \to \mathcal{C}$ *such that* $F(i) = X_i$.

Consider an object $X \in \mathcal{C}$ *and family of morphisms* $\pi_i : X \to X_i$, *then the following are equivalent*

a) The cone $\left(X, \{\pi_i\}_{i\in I}\right)$ *is a limit for* F

b) For any family of morphisms $\lambda_i : X' \to X_i$ *there exists a unique* $\lambda : X' \to X$ *such that* $\pi_i \circ \lambda = \lambda_i$

c) For all objects X' *the map*

$$\Phi : \mathrm{Mor}(X', X) \longrightarrow \prod_{i\in I} \mathrm{Mor}(X', X_i)$$
$$f \to \pi_i \circ f$$

is a bijection, i.e. (X, Φ) *represents the contravariant functor* $\prod_{i\in I} \mathrm{Mor}(-, X_j)$.

In this case we say that $\left(X, \{\pi_i\}_{i\in I}\right)$ *is a* ***product*** *for* $\{X_i\}_{i\in I}$ *and denote this by* $\prod_{i\in I} X_i$.

For any family of morphisms $\lambda_i : X' \to X_i$ *we write the induced morphism* $\lambda : X' \to \prod_{i\in I} X_i$ *as* $(\lambda_i)_{i\in I}$.

Proposition 2.6.82 (Functoriality of Product)
Let I *be a discrete small category and suppose that* $\mathcal{C}$ *has* I*-shaped limits. Then we may define a map on objects*

$$\prod_{i\in I}(-) : \mathcal{C}^I \to \mathcal{C}$$

Recall that a morphism $\prod_{i\in I} X_i \to \prod_{i\in I} Y_i$ *in* $\mathcal{C}^I$ *is simply a family of morphisms* $(f_i : X_i \to Y_i)_{i\in I}$ *in* $\mathcal{C}$. *Then we define*

$$\prod_{i\in I} f_i := (f_i \circ \pi_i)_{i\in I}$$

which is the unique morphism satisfying

$$\pi_i \circ \prod_{i\in I} f_i = f_i \circ \pi_i$$

This makes $\prod_{i\in I}(-)$ *into a functor.*

Proposition 2.6.83 (Products of Monomorphisms)
Let I *be a discrete small category and* $\{f_i : X_i \to Y_i\}_{i\in I}$ *a family of monomorphisms such that* $\prod_{i\in I} X_i$ *and* $\prod_{i\in I} Y_i$ *exist. Then* $\prod_{i\in I} f_i : \prod_{i\in I} X_i \to \prod_{i\in I} Y_i$ *is a monomorphism.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Suppose $a, b : Z \to \prod_{i\in I} X_i$ satisfy $\left(\prod_{i\in I} f_i\right) \circ a = \left(\prod_{i\in I} f_i\right) \circ b$. For each $i \in I$, composing with the projection $\pi_i : \prod_{i\in I} Y_i \to Y_i$ and using $\pi_i \circ \prod_{i\in I} f_i = f_i \circ \pi_i$ gives

$$f_i \circ (\pi_i \circ a) = f_i \circ (\pi_i \circ b)$$

and since f_i is monic, $\pi_i \circ a = \pi_i \circ b$. As this holds for every $i \in I$, by the uniqueness clause of the universal property of the product $\prod_{i\in I} X_i$ (2.6.81) we conclude $a = b$. ∎

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Proposition 2.6.84 (Functoriality of Binary Product)
Let $X, X', Y, Y' \in \mathcal{C}$ be such that $X \times Y$ and $X' \times Y'$ exists. For $f : X \to X'$ and $g : Y \to Y'$ there exists a unique morphism

$$f \times g : X \times Y \to X' \times Y'$$

such that

$$\pi_{X'} \circ (f \times g) = f \circ \pi_X$$

and

$$\pi_{Y'} \circ (f \times g) = g \circ \pi_Y$$

If $\mathcal{C}$ has all binary products then this determines a bifunctor

$$\times : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$$

Proposition 2.6.85 (Limits exist in Set)
Limits exist in **Set** *more precisely for a diagram $F : I \to \mathbf{Set}$*

$$\lim F = \left\{ (x_i)_{i \in I} \in \prod_{i \in I} F(i) \;\middle|\; F(\alpha)(x_i) = x_j \text{ for all } \alpha : i \to j \right\}$$

Definition 2.6.86
Let $U : \mathcal{C} \to \mathcal{D}$ be a functor and I a small category.

*We say that U **creates limits** of shape I if for all diagrams $F : I \to \mathcal{C}$ such that $\lim U \circ F$ exists then $\lim F$ exists and is the unique cone such that $U(\lim F) = \lim(U \circ F)$.*

*We say that U **preserves limits** of shape I if whenever $\lim F$ exists in $\mathcal{C}$ we have $\lim U \circ F$ exists in $\mathcal{D}$ and is equal to $U(\lim F)$.*

If we restrict to the case I is discrete (resp. $I = \mathbf{2}$) then we say that U creates/preserves products (resp. equalisers).

Proposition 2.6.87 (Creates⇒Preserves)
Let $U : \mathcal{C} \to \mathcal{D}$ which creates limits of shape I and $\mathcal{D}$ has all limits of shape I. Then U preserves limits of shape I.

In particular this holds unconditionally for $\mathcal{D} = \mathbf{Set}$ since this has all limits.

2.6.11 Colimit

Definition 2.6.88 (Co-cones)
*Let $F : I \to \mathcal{C}$ be a diagram. We say that the pair $\left(L, \{\lambda_i : F(i) \to L\}_{i \in I}\right)$ is a **co-cone** over F if*

$$\lambda_j \circ F(\alpha) = \lambda_i \quad \text{for all morphisms} \quad \alpha : i \to j$$

*The cones for F constitute a **co-cone category** $(F \downarrow \Delta)$ where a morphism $\alpha : \left(L, \{\lambda_i\}_{i \in I}\right) \to \left(L', \{\lambda'_i\}_{i \in I}\right)$ is just a morphism $\alpha : L \to L'$ such that $\alpha \circ \lambda_i = \lambda'_i$ for all $i \in I$*

Definition 2.6.89 (Small Colimit)
*Let $F : I \to \mathcal{C}$ be a diagram. A co-cone $\left(L, \{\lambda_i\}_{i \in I}\right)$ over F is a **colimit** for F if it is an **initial object** (2.6.48) in the co-cone category.*

Explicitly for every co-cone $\left(L', \{\lambda'_i\}_{i \in I}\right)$ there is a unique morphism $\alpha : L \to L'$ such that $\lambda'_i \circ \alpha = \lambda_i$.

We may denote this as $\operatorname{colim} F$.

Proposition 2.6.90
Let $F : I \to \mathcal{C}$ be a diagram, $\left(L, \{\lambda_i\}_{i \in I}\right)$ a colimit of F and $\left(L', \{\lambda'_i\}_{i \in I}\right)$ a co-cone. Then the following are equivalent

a) L' is a colimit for F
b) there is an isomorphism of co-cones $L' \xrightarrow{\sim} L$
c) there is a morphism of co-cones $L' \to L$ and precisely one (co-cone) endomorphism of L'.

In this case the (co-cone) isomorphism is unique.

Proof. By definition $\left(L, \{\lambda_i\}_{i \in I}\right)$ is an initial object of the co-cone category $(F \downarrow \Delta)$, so this is precisely (2.6.49) applied with $\emptyset := \left(L, \{\lambda_i\}_{i \in I}\right)$ and $X := \left(L', \{\lambda'_i\}_{i \in I}\right)$. ∎

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Proposition 2.6.91 (Uniqueness of Colimits)
Let $F : I \to \mathcal{C}$ be a diagram and $\left(L, \{\lambda_i\}_{i\in I}\right)$ and $\left(L', \{\lambda'_i\}_{i\in I}\right)$ colimits of F. Then there is a unique morphism $\psi : L \to L'$ satisfying $\psi \circ \lambda_i = \lambda'_i$, which is an isomorphism.

Proof. Since $\left(L', \{\lambda'_i\}_{i\in I}\right)$ is a colimit for F, (2.6.90) gives a unique isomorphism of co-cones $\phi : L' \to L$, that is, a unique isomorphism ϕ with $\phi \circ \lambda'_i = \lambda_i$. Setting $\psi := \phi^{-1} : L \to L'$ gives the required unique isomorphism with $\psi \circ \lambda_i = \lambda'_i$. ∎

Proposition 2.6.92 (Limits and Colimits swap under $\mathcal{C}^{\mathrm{op}}$)
Let I be a small category and $F : I \to \mathcal{C}$ a diagram. Write $F^{op} : I^{op} \to \mathcal{C}^{op}$ for the diagram with $F^{op}(i) := F(i)$ on objects and $F^{op}(\alpha^{op}) := F(\alpha)^{op}$ for $\alpha^{op} : j \to i$ the morphism of I^{op} corresponding to $\alpha : i \to j$ of I. Then, regarding each $\lambda_i : L \to F(i)$ as a morphism $\lambda_i^{op} : F(i) \to L$ of $\mathcal{C}^{op}$, $\left(L, \{\lambda_i\}_{i\in I}\right)$ is a limit for F in $\mathcal{C}$ if and only if $\left(L, \{\lambda_i^{op}\}_{i\in I}\right)$ is a colimit for F^{op} in $\mathcal{C}^{op}$.

Dually, $\left(L, \{\lambda_i\}_{i\in I}\right)$ is a colimit for F in $\mathcal{C}$ if and only if $\left(L, \{\lambda_i^{op}\}_{i\in I}\right)$ is a limit for F^{op} in $\mathcal{C}^{op}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

We only prove the first statement, the second following by applying it to $\mathcal{C}^{\mathrm{op}}$ in place of $\mathcal{C}$, using $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$.

Define $\Sigma : (\Delta \downarrow F) \to (F^{\mathrm{op}} \downarrow \Delta)^{\mathrm{op}}$, $\left(L, \{\lambda_i\}_{i\in I}\right) \mapsto \left(L, \{\lambda_i^{\mathrm{op}}\}_{i\in I}\right)$, acting on a cone morphism $\mu : (L, \lambda) \to (L', \lambda')$ (i.e. $\mu : L \to L'$ with $\lambda'_i \circ \mu = \lambda_i$ for all i) by $\Sigma(\mu) := \mu^{\mathrm{op}}$. This is well-defined: upon reversing every arrow (using $\mathrm{Mor}_{\mathcal{C}^{\mathrm{op}}}(X, Y) = \mathrm{Mor}_{\mathcal{C}}(Y, X)$ and that composition in $\mathcal{C}^{\mathrm{op}}$ reverses that of $\mathcal{C}$ (2.6.4)), the cone condition $F(\alpha) \circ \lambda_i = \lambda_j$ for $\alpha : i \to j$ of I becomes exactly the co-cone condition $\lambda_i^{\mathrm{op}} \circ F^{\mathrm{op}}(\alpha^{\mathrm{op}}) = \lambda_j^{\mathrm{op}}$ for $\alpha^{\mathrm{op}} : j \to i$ of I^{op}, so $\{\lambda_i^{\mathrm{op}}\}$ is a co-cone over F^{op}; and the cone morphism condition $\lambda'_i \circ \mu = \lambda_i$ becomes $\mu^{\mathrm{op}} \circ (\lambda'_i)^{\mathrm{op}} = \lambda_i^{\mathrm{op}}$, so μ^{op} is a morphism $(L', \lambda'^{\mathrm{op}}) \to (L, \lambda^{\mathrm{op}})$ of $(F^{\mathrm{op}} \downarrow \Delta)$, that is, a morphism $(L, \lambda^{\mathrm{op}}) \to (L', \lambda'^{\mathrm{op}})$ of $(F^{\mathrm{op}} \downarrow \Delta)^{\mathrm{op}}$. Thus Σ is well-defined, and is manifestly functorial.

Dually define $\Sigma^{-1} : (F^{\mathrm{op}} \downarrow \Delta)^{\mathrm{op}} \to (\Delta \downarrow F)$, $\left(L, \{\mu_i\}_{i\in I}\right) \mapsto \left(L, \{\mu_i^{\mathrm{op}}\}_{i\in I}\right)$, acting on morphisms by $\nu \mapsto \nu^{\mathrm{op}}$. Since $(-)^{\mathrm{op}}$ is an involution on morphisms, $\Sigma^{-1}\Sigma = 1$ and $\Sigma\Sigma^{-1} = 1$, so Σ is an isomorphism of categories, in particular an equivalence of categories (2.6.25).

By (2.6.4), a category $\mathcal{D}$ has T as a terminal object if and only if $\mathcal{D}^{\mathrm{op}}$ has T as an initial object (since $\mathrm{Mor}_{\mathcal{D}^{\mathrm{op}}}(X, T) = \mathrm{Mor}_{\mathcal{D}}(T, X)$ for all X). So $(L, \{\lambda_i\})$ is a limit for F, i.e. terminal in $(\Delta \downarrow F)$ (2.6.78), iff, by (2.6.52), $\Sigma(L, \{\lambda_i\}) = (L, \{\lambda_i^{\mathrm{op}}\})$ is terminal in $(F^{\mathrm{op}} \downarrow \Delta)^{\mathrm{op}}$, iff (applying the noted equivalence to $\mathcal{D} := (F^{\mathrm{op}} \downarrow \Delta)$) $(L, \{\lambda_i^{\mathrm{op}}\})$ is initial in $(F^{\mathrm{op}} \downarrow \Delta)$, that is, a colimit for F^{op} (2.6.89). ∎

Proposition 2.6.93 (Co-product)
Let I be a discrete small category and $\{X_i\}_{i\in I}$ be a family of objects in $\mathcal{C}$. Then there is a unique functor $F : I \to \mathcal{C}$ such that $F(i) = X_i$.

Consider an object $X \in \mathcal{C}$ and family of morphisms $s_i : X_i \to X$, then the following are equivalent

a) The co-cone $\left(X, \{s_i\}_{i\in I}\right)$ is a colimit for F
b) For any family of morphisms $\lambda_i : X_i \to X'$ there exists a unique $\lambda : X \to X'$ such that $\lambda \circ s_i = \lambda_i$
c) For all objects X' the map

$$\Phi : \mathrm{Mor}(X, X') \longrightarrow \prod_{i\in I} \mathrm{Mor}(X_i, X')$$
$$\lambda \to (\lambda \circ s_i)_{i\in I}$$

is a bijection, i.e. (X, Φ) represents the covariant functor $\prod_{i\in I} \mathrm{Mor}(X_i, -)$.

*In this case we say that $\left(X, \{s_i\}_{i\in I}\right)$ is a **co-product** for $\{X_i\}_{i\in I}$ and denote this by $\coprod_{i\in I} X_i$.*

For any family of morphisms $\lambda_i : X_i \to X'$ we write the induced morphism $\lambda : \coprod_{i\in I} X_i \to X'$ as $[\lambda_i]_{i\in I}$.

Definition 2.6.94
Let $U : \mathcal{C} \to \mathcal{D}$ be a functor and I a small category.

*We say that U **creates co-limits** of shape I if for all diagrams $F : I \to \mathcal{C}$ such that $\lim U \circ F$ exists then $\operatorname{colim} F$ exists and is the unique cone such that $U(\operatorname{colim} F) = \operatorname{colim}(U \circ F)$.*

*We say that U **preserves co-limits** of shape I if whenever $\operatorname{colim} F$ exists in $\mathcal{C}$ we have $\operatorname{colim} U \circ F$ exists in $\mathcal{D}$ and is equal to $U(\operatorname{colim} F)$.*

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If we restrict to the case I is discrete (resp. $I = \mathbf{2}$) then we say that U creates/preserves coproducts (resp. coequalisers).

Proposition 2.6.95 (Creates⇒Preserves)
Let $U : \mathcal{C} \to \mathcal{D}$ which creates colimits of shape I and $\mathcal{D}$ has all colimits of shape I. Then U preserves colimits of shape I.

In particular this holds unconditionally for $\mathcal{D} = \mathbf{Set}$ since this has all colimits.

2.6.12 Functorial Limits

Having fixed a single diagram at a time so far, we now let the diagram vary. By (2.6.96), taking the limit in one variable of a bifunctor is functorial in the other – so a family of limits indexed by $\mathcal{C}$ assembles into a genuine functor $\mathcal{C} \to \mathcal{E}$. Specialising this to the evaluation bifunctor recovers the more familiar statement, (2.6.97): the limit $\lim_J(-)$ is itself a functor out of (a subcategory of) the functor category $\mathcal{C}^J$, so a natural transformation of diagrams induces a morphism of their limits, compatibly with composition and identities – and in particular carries isomorphisms of diagrams to isomorphisms of limits. We also record that limits are invariant under isomorphism of the indexing category, that a limit over a product of two index categories may be computed either simultaneously or iteratively one variable at a time (2.6.99), with the two iterated orders agreeing whenever both make sense (2.6.100), and dual statements for colimits, together with a comparison morphism relating a colimit of limits to a limit of colimits, which need not be an isomorphism in general.

Proposition 2.6.96 (Pointwise Limits are Functorial)
Let J be a small category and $F : \mathcal{C} \times J \to \mathcal{E}$ a bifunctor.

Suppose that for every $X \in \mathcal{C}$ there is a limit $\left(\lim_{j\in J} F(X,j), \{\lambda_{X,j}\}_{j\in J}\right)$ for $F(X,-)$.

Then $\lim_{j\in J} F(-,j)$ is functorial in the sense there is a unique morphism making the following diagram commute

$$\begin{array}{ccc} \lim_{j\in J} F(X,j) & \overset{\lim_{j\in J} F(f,j)}{\dashrightarrow} & \lim_{j\in J} F(X',j) \\ \downarrow{\scriptstyle \lambda_{X,j}} & & \downarrow{\scriptstyle \lambda_{X',j'}} \\ F(X,j) & \xrightarrow{F(f,\beta)} & F(X',j') \end{array}$$

for all $f : X \to X'$ and $\beta : j \to j'$.

Therefore this determines a functor $\lim_{j\in J} F(-,j) : \mathcal{C} \to \mathcal{E}$.

Proof. Recall that $\left(\lim_{j\in J} F(X,j), \{\lambda_{X,j}\}_{j\in J}\right)$ is a terminal object in the category of cones $(\Delta \downarrow F(X,-))$. Suppose $f : X \to X'$ is a morphism. Since F is a bifunctor (2.6.19), for every $\beta : j \to j'$ in J

$$F(X',\beta) \circ F\big(f, 1_j\big) \circ \lambda_{X,j} = F(f,\beta) \circ \lambda_{X,j} = F\big(f, 1_{j'}\big) \circ F(X,\beta) \circ \lambda_{X,j} = F\big(f, 1_{j'}\big) \circ \lambda_{X,j'}$$

so $\left(\lim_{j\in J} F(X,j), \{F\big(f,1_j\big) \circ \lambda_{X,j}\}_{j\in J}\right)$ is a cone over $F(X',-)$.

Therefore there is a unique morphism $h : \lim_{j\in J} F(X,j) \to \lim_{j\in J} F(X',j)$ into the terminal object of $(\Delta \downarrow F(X',-))$ with $\lambda_{X',j} \circ h = F\big(f,1_j\big) \circ \lambda_{X,j}$ for every $j \in J$. For general $\beta : j \to j'$, the target cone condition and the bifunctoriality identity above give

$$\lambda_{X',j'} \circ h = F(X',\beta) \circ \lambda_{X',j} \circ h = F(X',\beta) \circ F\big(f,1_j\big) \circ \lambda_{X,j} = F(f,\beta) \circ \lambda_{X,j}$$

as required.

Writing $\lim_{j\in J} F(f,j) \coloneqq h$, the identity and composition laws needed for functoriality of $\lim_{j\in J} F(-,j)$ follow directly from this uniqueness: for $f = 1_X$, $h = 1_{\lim_j F(X,j)}$ has the defining property of $\lim_j F(1_X,j)$, so equals it by uniqueness; and for $f : X \to X'$, $f' : X' \to X''$, the morphism $\lim_j F(f',j) \circ \lim_j F(f,j)$ satisfies the same defining property (compatibility with the $\lambda_{X'',j}$) as $\lim_j F(f'f,j)$, so the two agree by uniqueness (this is the same pattern used for image factorisations, see later). ∎

Corollary 2.6.97 (Limits are Functorial in the Diagram)
Let J be a small category and $\mathcal{C}$ a category, and let $\mathcal{C}^J_{\lim}$ be the full subcategory of the functor category $\mathcal{C}^J$ (2.6.14) spanned by those diagrams $D : J \to \mathcal{C}$ for which $\lim_J D$ exists. Then

$$\lim_J(-) : \mathcal{C}^J_{\lim} \to \mathcal{C}, \quad D \mapsto \lim_J D$$

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is a functor. Consequently, a natural transformation $\eta : D \Rightarrow D'$ of $\mathcal{C}_{\lim}^{J}$ induces $\lim_J \eta : \lim_J D \to \lim_J D'$, satisfying the composition and identity laws of a functor; and if η is a natural isomorphism, $\lim_J \eta$ is an isomorphism, with inverse $\lim_J(\eta^{-1})$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Define the **evaluation bifunctor** $\mathrm{ev} : \mathcal{C}_{\lim}^{J} \times J \to \mathcal{C}$ by $\mathrm{ev}(D, j) := D(j)$ on objects and, for a morphism $(\eta, \beta) : (D, j) \to (D', j')$ of $\mathcal{C}_{\lim}^{J} \times J$ (so $\eta : D \Rightarrow D'$ a natural transformation and $\beta : j \to j'$ in J), $\mathrm{ev}(\eta, \beta) := D'(\beta) \circ \eta_j$ $(= \eta_{j'} \circ D(\beta)$, by naturality of η at $\beta)$. This preserves identities $(\mathrm{ev}(1_D, 1_j) = D(1_j) \circ (1_D)_j = 1_{D(j)})$ and composition: for a further morphism $(\eta', \beta') : (D', j') \to (D'', j'')$,

$$\mathrm{ev}(\eta', \beta') \circ \mathrm{ev}(\eta, \beta) = D''(\beta') \circ \eta'_{j'} \circ D'(\beta) \circ \eta_j = D''(\beta') \circ D''(\beta) \circ \eta'_j \circ \eta_j = \mathrm{ev}((\eta' \eta, \beta' \beta)),$$

using naturality of η' at β (i.e. $\eta'_{j'} \circ D'(\beta) = D''(\beta) \circ \eta'_j$) and functoriality of D''; so ev is a bifunctor.

By definition of $\mathcal{C}_{\lim}^{J}$, $\lim_{j \in J} \mathrm{ev}(D, j) = \lim_J D$ exists for every $D \in \mathcal{C}_{\lim}^{J}$, so (2.6.96) applies to ev (with $\mathcal{C}_{\lim}^{J}$ in the role of the first factor), giving that $\lim_{j \in J} \mathrm{ev}(-, j) = \lim_J(-) : \mathcal{C}_{\lim}^{J} \to \mathcal{C}$ is a functor, with $\lim_J \eta$ its action on a morphism η, satisfying the composition and identity laws by functoriality. If η is a natural isomorphism, i.e. an isomorphism of $\mathcal{C}_{\lim}^{J}$, then $\lim_J \eta$ is an isomorphism with inverse $\lim_J(\eta^{-1})$ by (2.6.42). ∎

Proposition 2.6.98 (Limits are Invariant under Isomorphism of the Indexing Category)
Let $\sigma : J \to I$ be an isomorphism of categories with inverse σ^{-1}, and $F : I \to \mathcal{C}$ a diagram. Then $\left(L, \{\lambda_i\}_{i \in I}\right)$ is a limit for F if and only if $\left(L, \left\{\lambda_{\sigma(j)}\right\}_{j \in J}\right)$ is a limit for $F\sigma : J \to \mathcal{C}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Define $\Sigma : (\Delta \downarrow F) \to (\Delta \downarrow F\sigma)$, $\left(L, \{\lambda_i\}_{i \in I}\right) \mapsto \left(L, \left\{\lambda_{\sigma(j)}\right\}_{j \in J}\right)$, acting as the identity on the underlying morphism of apexes; this is well-defined since $\left\{\lambda_{\sigma(j)}\right\}_{j \in J}$ is a cone over $F\sigma$ whenever $\{\lambda_i\}_{i \in I}$ is a cone over F (the cone condition for $F\sigma$ at a morphism β of J is exactly the cone condition for F at $\sigma(\beta)$), and Σ is manifestly functorial.

Dually define $\Sigma^{-1} : (\Delta \downarrow F\sigma) \to (\Delta \downarrow F)$, $\left(L, \left\{\mu_j\right\}_{j \in J}\right) \mapsto \left(L, \left\{\mu_{\sigma^{-1}(i)}\right\}_{i \in I}\right)$. Since $\sigma^{-1}\sigma = 1_J$ and $\sigma\sigma^{-1} = 1_I$, we have $\Sigma^{-1}\Sigma = 1$ and $\Sigma\Sigma^{-1} = 1$ on the nose, so Σ is an isomorphism of categories, in particular an equivalence of categories (2.6.25) (with identity natural isomorphisms).

By (2.6.52), $(\Delta \downarrow F)$ has a terminal object iff $(\Delta \downarrow F\sigma)$ does, and Σ carries a terminal object of $(\Delta \downarrow F)$, i.e. a limit of F, to a terminal object of $(\Delta \downarrow F\sigma)$, i.e. a limit of $F\sigma$, as required. ∎

Proposition 2.6.99 (Iterated Limits)
Let I, J be small categories and $F : I \times J \to \mathcal{C}$ a bifunctor. Suppose that for every $i \in I$ there is a limit $\left(L(i), \left\{\lambda_{i,j}\right\}_{j \in J}\right)$ for $F(i, -)$, so that by (2.6.96) this determines a functor $L : I \to \mathcal{C}$.

Then $\lim_{(i,j) \in I \times J} F(i, j)$ exists if and only if $\lim_{i \in I} L(i)$ exists, and in this case

- *a) given a limit $\left(M, \{\nu_i\}_{i \in I}\right)$ for L, the cone $\left(M, \left\{\lambda_{i,j} \circ \nu_i\right\}_{(i,j) \in I \times J}\right)$ is a limit for F;*
- *b) conversely, given a limit $\left(N, \left\{\pi_{i,j}\right\}_{(i,j) \in I \times J}\right)$ for F, for each $i \in I$ there is a unique morphism $\sigma_i : N \to L(i)$ such that $\lambda_{i,j} \circ \sigma_i = \pi_{i,j}$ for all $j \in J$, and $\left(N, \{\sigma_i\}_{i \in I}\right)$ is a limit for L.*

Symmetrically, suppose that for every $j \in J$ there is a limit $\left(R(j), \left\{\mu_{i,j}\right\}_{i \in I}\right)$ for $F(-, j)$, so that by (2.6.96) this determines a functor $R : J \to \mathcal{C}$. Then $\lim_{(i,j) \in I \times J} F(i, j)$ exists if and only if $\lim_{j \in J} R(j)$ exists, and in this case

- *c) given a limit $\left(M', \left\{\xi_j\right\}_{j \in J}\right)$ for R, the cone $\left(M', \left\{\mu_{i,j} \circ \xi_j\right\}_{(i,j) \in I \times J}\right)$ is a limit for F;*
- *d) conversely, given a limit $\left(N, \left\{\pi_{i,j}\right\}_{(i,j) \in I \times J}\right)$ for F, for each $j \in J$ there is a unique morphism $\tau_j : N \to R(j)$ such that $\mu_{i,j} \circ \tau_j = \pi_{i,j}$ for all $i \in I$, and $\left(N, \left\{\tau_j\right\}_{j \in J}\right)$ is a limit for R.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

We construct mutually inverse functors $\Xi : (\Delta \downarrow L) \to (\Delta \downarrow F)$ and $\Theta : (\Delta \downarrow F) \to (\Delta \downarrow L)$; as $\Theta \circ \Xi = 1$, $\Xi \circ \Theta = 1$, Ξ is in particular an equivalence of categories (2.6.25) (with identity natural isomorphisms), so by (2.6.52), $(\Delta \downarrow F)$ has a terminal object iff $(\Delta \downarrow L)$ does, and Ξ carries a terminal object of $(\Delta \downarrow L)$, i.e. a limit of L, to a terminal object of $(\Delta \downarrow F)$, i.e. a limit of F, as required.

The functor Ξ. Given a cone $\left(X, \{\eta_i\}_{i \in I}\right)$ over L, define $\Xi(X, \{\eta_i\}) := \left(X, \left\{\lambda_{i,j} \circ \eta_i\right\}_{(i,j) \in I \times J}\right)$. This is a cone over F: for $(\alpha, \beta) : (i, j) \to (i', j')$, writing $(\alpha, \beta) = (\alpha, 1_{j'}) \circ (1_i, \beta)$,

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$$\begin{aligned} F(\alpha,\beta)\circ\lambda_{i,j}\circ\eta_i &= F\big(\alpha,1_{j'}\big)\circ F(1_i,\beta)\circ\lambda_{i,j}\circ\eta_i \\ &= F\big(\alpha,1_{j'}\big)\circ\lambda_{i,j'}\circ\eta_i \\ &= \lambda_{i',j'}\circ L(\alpha)\circ\eta_i \\ &= \lambda_{i',j'}\circ\eta_{i'} \end{aligned}$$

using in turn that $\lambda_{i,_}$ is a cone over $F(i,-)$, the defining property of $L(\alpha)$ (2.6.96), and that η is a cone over L. On morphisms Ξ is the identity on the underlying morphism of apexes, which is trivially functorial.

The functor Θ. Given a cone $\Big(X,\big\{\rho_{i,j}\big\}_{(i,j)\in I\times J}\Big)$ over F, for each $i\in I$ the family $\big\{\rho_{i,j}\big\}_{j\in J}$ is a cone over $F(i,-)$, so by the universal property of $\Big(L(i),\big\{\lambda_{i,j}\big\}_j\Big)$ there is a unique morphism $\theta_i : X\to L(i)$ such that $\lambda_{i,j}\circ\theta_i=\rho_{i,j}$ for all $j\in J$. Define $\Theta\big(X,\{\rho_{i,j}\}\big) := \big(X,\{\theta_i\}_{i\in I}\big)$.

We check $\{\theta_i\}$ is a cone over L. Fix $\alpha : i\to i'$ in I; then for every $j'\in J$

$$\lambda_{i',j'}\circ L(\alpha)\circ\theta_i = F\big(\alpha,1_{j'}\big)\circ\lambda_{i,j'}\circ\theta_i = F\big(\alpha,1_{j'}\big)\circ\rho_{i,j'} = \rho_{i',j'} = \lambda_{i',j'}\circ\theta_{i'}$$

using the defining property of $L(\alpha)$, the defining property of θ_i, that ρ is a cone over F applied to $\big(\alpha,1_{j'}\big):(i,j')\to(i',j')$, and the defining property of $\theta_{i'}$. As this holds for all $j'\in J$, the universal property of $\Big(L(i'),\big\{\lambda_{i',j'}\big\}_{j'}\Big)$ gives $L(\alpha)\circ\theta_i=\theta_{i'}$.

On morphisms Θ again acts as the identity on apexes: if $\phi:\big(X,\{\rho_{i,j}\}\big)\to\big(X',\{\rho'_{i,j}\}\big)$ is a morphism of cones, then $\lambda_{i,j}\circ\theta'_i\circ\phi=\rho'_{i,j}\circ\phi=\rho_{i,j}=\lambda_{i,j}\circ\theta_i$ for all j, so by uniqueness in the universal property of $L(i)$, $\theta'_i\circ\phi=\theta_i$, i.e. ϕ is also a morphism of cones $(X,\{\theta_i\})\to(X',\{\theta'_i\})$.

Mutually inverse. Given a cone $(X,\{\eta_i\})$ over L, applying Ξ then Θ recovers, for each i, the unique morphism $X\to L(i)$ factoring $\big(\lambda_{i,j}\circ\eta_i\big)_j$ through $\big\{\lambda_{i,j}\big\}_j$; since η_i already satisfies this by definition, uniqueness gives $\Theta(\Xi(X,\{\eta_i\}))=(X,\{\eta_i\})$. Conversely, given a cone $\big(X,\{\rho_{i,j}\}\big)$ over F, applying Θ then Ξ gives $\big(X,\{\lambda_{i,j}\circ\theta_i\}\big)=\big(X,\{\rho_{i,j}\}\big)$ directly from the defining property of θ_i. Both functors act as the identity on the underlying morphism of apexes, so Ξ and Θ are mutually inverse isomorphisms of categories.

Finally, the symmetric statement follows by applying the above to the bifunctor $F^T : J\times I\to\mathcal{C}$, $F^T(j,i):=F(i,j)$ (swapping the roles of I,J, with R playing the role of L), and reindexing via (2.6.98) along the swap isomorphism $I\times J\cong J\times I$. ∎

Corollary 2.6.100 (Comparison of Iterated Limits)
Let I,J be small categories and $F:I\times J\to\mathcal{C}$ a bifunctor, and suppose (as in (2.6.99)) that $L(i):=\lim_{j\in J}F(i,j)$ exists for every $i\in I$ and $R(j):=\lim_{i\in I}F(i,j)$ exists for every $j\in J$. If any one of $\lim_{(i,j)\in I\times J}F(i,j)$, $\lim_{i\in I}L(i)$, $\lim_{j\in J}R(j)$ exists, then all three exist, and there is a unique isomorphism Φ such that the following diagram commutes

$$\begin{array}{ccccc} \lim_{i\in I}L(i) & & \xrightarrow{\quad\Phi\quad} & & \lim_{j\in J}R(j) \\ {\scriptstyle\nu_{i'}}\downarrow & & & & \downarrow{\scriptstyle\xi_{j'}} \\ L(i') & \xrightarrow[\lambda_{i',j'}]{} & F(i',j') & \xleftarrow[\mu_{i',j'}]{} & R(j') \end{array}$$

for all $i'\in I$, $j'\in J$, where $\nu_{i'}:\lim_{i\in I}L(i)\to L(i')$ and $\xi_{j'}:\lim_{j\in J}R(j)\to R(j')$ are the respective limit cones.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (2.6.99), $\lim_{(i,j)\in I\times J}F(i,j)$ exists if and only if $\lim_{i\in I}L(i)$ exists, and if and only if $\lim_{j\in J}R(j)$ exists; so if any one of the three exists, all three do. In this case, taking $(M,\{\nu_i\}):=(\lim_{i\in I}L(i),\{\nu_i\})$ in part a), the cone $\Big(\lim_{i\in I}L(i),\big\{\lambda_{i,j}\circ\nu_i\big\}_{(i,j)\in I\times J}\Big)$ is a limit for F; likewise, taking $\big(M',\{\xi_j\}\big):=\big(\lim_{j\in J}R(j),\{\xi_j\}\big)$ in part c), the cone $\Big(\lim_{j\in J}R(j),\big\{\mu_{i,j}\circ\xi_j\big\}_{(i,j)\in I\times J}\Big)$ is also a limit for F. By (2.6.80) there is therefore a unique isomorphism Φ as required. ∎

Proposition 2.6.101 (Pointwise Colimits are Functorial)
Let I be a small category and $F:I\times\mathcal{D}\to\mathcal{E}$ a bifunctor.

Suppose that for every $X\in\mathcal{D}$ there is a colimit $\Big(\operatorname{colim}_{i\in I}F(i,X),\big\{\mu_{i,X}\big\}_{i\in I}\Big)$ for $F(-,X)$

Then $\operatorname{colim}_{i\in I}F(i,-)$ is functorial in the sense there is a unique morphism making the following diagram commute

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$$\begin{array}{ccc} F(i,X) & \xrightarrow{F(\alpha,g)} & F(i',X') \\ \downarrow{\scriptstyle \mu_{i,X}} & & \downarrow{\scriptstyle \mu_{i',X'}} \\ \operatorname{colim}_{i\in I} F(i,X) & \xrightarrow{\operatorname{colim}_{i\in I} F(i,g)} & \operatorname{colim}_{i\in I} F(i,X') \end{array}$$

for all $g : X \to X'$ *and* $\alpha : i \to i'$.

Proposition 2.6.102 (Comparison Map between Limits and Colimits)
Let I, J *be small categories and* $F : I \times J \to \mathcal{C}$ *be a bifunctor.*

Suppose that there exists

a) *limits* $i \to \lim_{j\in J} F(i,j)$ *(which is automatically functorial in* i *(2.6.96))*
b) *colimits* $j \to \operatorname{colim}_{i\in I} F(i,j)$ *(which us automatically functorial in* j *(2.6.101))*
c) *a colimit* $\operatorname{colim}_{i\in I}\big(\lim_{j\in J} F(i,j)\big)$
d) *a limit* $\lim_{j\in J} \operatorname{colim}_{i\in I} F(i,j)$

(which holds if $\mathcal{C}$ *has limits of shape* J *and colimits of shape* I*). Then there exists a unique "comparison morphism"*

$$\begin{array}{ccccc} \operatorname{colim}_{i\in I} \lim_{j\in J} F(i,j) & & \xrightarrow{\quad\Phi\quad} & & \lim_{j\in J} \operatorname{colim}_{i\in I} F(i,j) \\ {\scriptstyle \lambda_{i'}}\uparrow & & & & \downarrow{\scriptstyle \mu_{j'}} \\ \lim_{j\in J} F(i',j) & \xrightarrow[\lambda_{i',j'}]{} & F(i',j') & \xrightarrow[\mu_{i',j'}]{} & \operatorname{colim}_{i\in I} F(i,j') \end{array}$$

making the diagram commute for all $i' \in I$ *and* $j' \in J$.

Proof. The morphisms $\lambda_{i'}$, $\lambda_{i',j'}$, $\mu_{i',j'}$ and $\mu_{j'}$ are the usual cone morphisms arising from the limit and colimit constructions. Suppose $\beta : j' \to j''$ then we have a commutative diagram, using both functoriality of colimit (2.6.101) and definition of cone $\lambda_{i',-}$

$$\begin{array}{ccccc} \lim_{j\in J} F(i',j) & \xrightarrow{\lambda_{i',j'}} & F(i',j') & \xrightarrow{\mu_{i',j'}} & \operatorname{colim}_{i\in I} F(i,j') \\ & \searrow^{\lambda_{i',j''}} & \downarrow{\scriptstyle F(1,\beta)} & & \downarrow{\scriptstyle \operatorname{colim}_{i\in I} F(i,\beta)} \\ & & F(i',j'') & \xrightarrow{\mu_{i',j''}} & \operatorname{colim}_{i\in I} F(i,j'') \end{array}$$

By universal property of the (top-right) limit to $\mu_{i',j'} \circ \lambda_{i',j'}$ we may find for each $i' \in I$ a unique morphism

$$\Phi_{i'} : \lim_{j\in J} F(i',j) \longrightarrow \lim_{j\in J} \operatorname{colim}_{i\in I} F(i,j)$$

satisfying

$$\mu_{j'} \circ \Phi_{i'} = \mu_{i',j'} \circ \lambda_{i',j'} \quad \forall j' \in J$$

Then by functoriality of limit (2.6.96) and definition of the cocone $\mu_{-,j'}$ we have a commutative diagram for all $\alpha : i' \to i''$

$$\begin{array}{ccccc} \lim_{j\in J} F(i',j) & \xrightarrow{\lambda_{i',j'}} & F(i',j') & \xrightarrow{\mu_{i',j'}} & \operatorname{colim}_{i\in I} F(i,j') \\ {\scriptstyle \lim_{j\in J} F(\alpha,j)}\downarrow & & {\scriptstyle F(\alpha,1)}\downarrow & \nearrow_{\mu_{i',j''}} & \\ \lim_{j\in J} F(i'',j) & \xrightarrow{\lambda_{i'',j'}} & F(i',j'') & & \end{array}$$

so that for all $j' \in J$

$$\begin{aligned} \mu_{j'} \circ \Phi_{i''} \circ \lim_{j\in J} F(\alpha,j) &= \mu_{i'',j''} \circ \lambda_{i'',j''} \circ \lim_{j\in J} F(\alpha,j) \\ &= \mu_{i',j'} \circ \lambda_{i',j'} \end{aligned}$$

whence by uniqueness

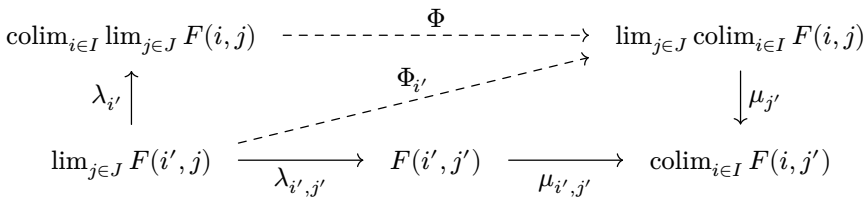
$$\Phi_{i''} \circ \lim_{j\in J} F(\alpha,j) = \Phi_{i'}$$

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Therefore by the definition of the (top-left) co-limit there is a unique morphism

$$\Phi : \operatorname{colim}_{i\in I} \lim_{j\in J} F(i,j) \longrightarrow \lim_{j\in J} \operatorname{colim}_{i\in I} F(i,j)$$

such that we have a commutative diagram



Then if there is another morphism Φ' which makes the square commute define $\Phi'_{i'} = \Phi' \circ \lambda_{i'}$, we may use uniqueness to deduce that $\Phi'_{i'} = \Phi_{i'}$ and $\Phi = \Phi'$. ∎

2.6.13 Biproduct

In **AbGrp** and A-**Mod** coproducts and products coincide, so we formalise this property.

Definition 2.6.103 (Finite Biproduct)
Let $\mathcal{C}$ be a pointed category, I a finite *set and $\{X_i\}_{i\in I}$ be a family of objects. We say a triple $\left(\bigoplus_{i\in I} X_i, \{s_i\}_{i\in I}, \{\pi_i\}_{i\in I}\right)$ is a **biproduct** for the family $\{X_i\}_{i\in I}$ if the following holds*

a) $\left(\bigoplus_{i\in I} X_i, \{\pi_i\}_{i\in I}\right)$ is a product
b) $\left(\bigoplus_{i\in I} X_i, \{s_i\}_{i\in I}\right)$ is a coproduct
c) The morphisms π_i, s_j satisfy the orthogonality conditions

$$\pi_i \circ s_j = \delta_{ij} := \begin{cases} 1_{X_i} & i = j \\ 0 & i \neq j \end{cases}$$

Proposition 2.6.104 (Criteria for Existence of Biproduct)
Let $\mathcal{C}$ be a pointed category, I a finite set and $\{X_i\}_{i\in I}$ be a family of objects. Suppose both the product $\prod_{i\in I} X_i$ and coproduct $\coprod_{i\in I} X_i$ exists. Then there is a unique comparison morphism

$$\psi : \coprod_{i\in I} X_i \longrightarrow \prod_{i\in I} X_i$$

such that

$$\pi_i \circ \psi \circ s_j = \delta_{ij}$$

Then the following are equivalent

a) ψ is an isomorphism
b) $\left(\coprod_{i\in I} X_i, \{s_i\}_{i\in I}, \{\pi_i \circ \psi\}_{i\in I}\right)$ is a biproduct
c) $\left(\prod_{i\in I} X_i, \{\psi \circ s_i\}_{i\in I}, \{\pi_i\}_{i\in I}\right)$ is a biproduct
d) a biproduct $\bigoplus_{i\in I} X_i$ exists

Proof. We may write $\psi = \left[\left(\delta_{ij}\right)_{i\in I}\right]_{j\in I}$ then ψ is the unique map such that $\psi \circ s_j = \left(\delta_{ij}\right)_{i\in I}$, which is the unique map such that $\pi_i \circ \psi \circ s_j = \delta_{ij}$. Suppose ψ' is another such map then $\psi' \circ s_j = \psi \circ s_j$ whence $\psi' = \psi$.

a) $\implies$ b) We show that ψ is a morphism of cones, and the result follows from (2.6.79).

a) $\implies$ c) We show that ψ^{-1} is a morphism of co-cones, and the result follows from (2.6.90).

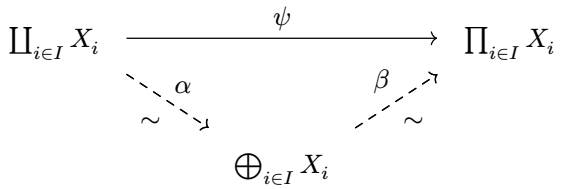
b) $\implies$ a) By (2.6.80) ψ is uniquely determined as the isomorphism of limits.

c) $\implies$ a) Similarly.

b) $\implies$ d) immediate

d) $\implies$ a) Let π'_i, s'_j be the biproduct maps for $\bigoplus_{i\in I} X_i$. Then by (2.6.80) (2.6.91) there is a diagram

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where α is the unique morphism satisfying $\alpha \circ s_i = s_i'$, β is the unique morphism satisfying $\pi_i \circ \beta = \pi_i'$, and both α and β are isomorphisms. Therefore $\pi_i \circ \beta \circ \alpha \circ s_j = \pi_i' \circ s_j' = \delta_{ij}$. Therefore by uniqueness $\psi = \beta \circ \alpha$ and is an isomorphism as required. ∎

Proposition 2.6.105 (Binary biproducts are enough)
Let $\mathcal{C}$ be a pointed category. Then it has all (finite) biproducts iff it has all binary biproducts (i.e. the case $\#I = 2$).

Proof. We prove that $\bigoplus_{i\in I} X_i$ exists for $I = \{1, ..., n\}$ by induction on n, where $n = 1$ is immediate and $n = 2$ is the base case. Consider a biproduct $\left(\bigoplus_{i=1}^n X_i, \{s_i\}, \{\pi_i\}\right)$ which we denote by Y, and the binary biproduct $(Y \oplus X_{n+1}, \{f, g\}, \{p, q\})$. Then we claim that

$$Z := (Y \oplus X_{n+1}, \{f \circ s_1, ..., f \circ s_n, g\}, \{\pi_1 \circ p, ..., \pi_n \circ p, q\})$$

is a biproduct. There is a natural isomorphism

$$\begin{aligned}\mathrm{Mor}(Z, -) &\xrightarrow{\sim} \mathrm{Mor}(Y, -) \times \mathrm{Mor}(X_{n+1}, -) \\ &\xrightarrow{\sim} \prod_{i=1}^{n+1} \mathrm{Mor}(X_i, -)\end{aligned}$$

which shows that Z is a product with the given projection maps. Similarly we may show that Z is a coproduct. Finally

$$\begin{aligned}\pi_i \circ p \circ f \circ s_j &= \pi_i \circ 1_Y \circ s_j = \delta_{ij} \\ \pi_i \circ p \circ g &= \pi_i \circ 0 = 0 \\ q \circ f \circ s_j &= 0 \circ s_j = 0\end{aligned}$$

so that the cone and co-cone morphisms satisfy the required orthogonality conditions. ∎

In **AbGrp** and A-**Mod**, a homomorphism between direct sums $\bigoplus_{j\in J} X_j \to \bigoplus_{i\in I} Y_i$ is determined by, and can be recovered from, a matrix of homomorphisms $X_j \to Y_i$; we record this categorically.

Definition 2.6.106 (Matrix of a Morphism)
*Let $\mathcal{C}$ be a pointed category with biproducts $\bigoplus_{j\in J} X_j$ (injections s_j, projections π_j^X) and $\bigoplus_{i\in I} Y_i$ (injections s_i^Y, projections π_i^Y), and $f : \bigoplus_{j\in J} X_j \to \bigoplus_{i\in I} Y_i$ a morphism. The **matrix** of f is the family $(f_{ij})_{i\in I, j\in J}$ of morphisms*

$$f_{ij} := \pi_i^Y \circ f \circ s_j : X_j \to Y_i$$

When $I = \{1, ..., m\}$ and $J = \{1, ..., n\}$ we display this as an $m \times n$ array (f_{ij}), as for ordinary matrices.

Remark 2.6.107
*Calling (f_{ij}) a **matrix** is not yet fully justified: for this notation to behave like ordinary matrix notation, composition should correspond to matrix multiplication, $(f \circ g)_{ik} = \sum_j f_{ij} \circ g_{jk}$, but this formula presupposes an addition on $\mathrm{Mor}(X_k, Y_i)$, which is no part of the data of a pointed category with biproducts. We can nonetheless justify composition whenever it requires no addition, namely when each row of f (or column of g) has at most one nonzero entry (2.6.114); addition on hom-sets is only introduced once biproducts are used to construct such an addition (2.6.118), after which the general composition formula above is recovered in full (2.6.122).*

Proposition 2.6.108 (Matrix Notation for Biproducts)
In the setting of (2.6.106), the assignment $f \mapsto (f_{ij})_{i\in I, j\in J}$ is a bijection

$$\mathrm{Mor}\left(\bigoplus_{j\in J} X_j, \bigoplus_{i\in I} Y_i\right) \xrightarrow{\sim} \prod_{i\in I, j\in J} \mathrm{Mor}(X_j, Y_i)$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Given a family $(g_{ij})_{i\in I, j\in J}$ of morphisms $g_{ij} : X_j \to Y_i$, for each $j \in J$ the family $(g_{ij})_{i\in I}$ determines, by the universal property of the product $\bigoplus_{i\in I} Y_i$ (2.6.81), a unique morphism $c_j : X_j \to \bigoplus_{i\in I} Y_i$ with $\pi_i^Y \circ c_j = g_{ij}$ for every $i \in I$. In

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turn, the family $(c_j)_{j\in J}$ determines, by the universal property of the coproduct $\bigoplus_{j\in J} X_j$ (2.6.93), a unique morphism $g : \bigoplus_{j\in J} X_j \to \bigoplus_{i\in I} Y_i$ with $g \circ s_j = c_j$ for every $j \in J$. We show $f \mapsto (f_{ij})$ and $(g_{ij}) \mapsto g$ are mutually inverse.

Starting from f, form its matrix (f_{ij}) and then c_j, g as above. Since $\pi_i^Y \circ (f \circ s_j) = f_{ij}$ for every i, $f \circ s_j$ satisfies the defining property of c_j, so $c_j = f \circ s_j$ by uniqueness; then f itself satisfies $f \circ s_j = c_j$ for every j, so $g = f$ by uniqueness.

Starting from a family (g_{ij}), form c_j, g as above and then the matrix of g: for every i, j, the (i, j) entry of the matrix of g is $\pi_i^Y \circ g \circ s_j = \pi_i^Y \circ c_j = g_{ij}$, using $g \circ s_j = c_j$ and then the defining property of c_j. So the matrix of g is again (g_{ij}). ∎

Remark 2.6.109
*The matrix of $1_{\bigoplus_{i\in I} X_i}$ is the **identity matrix** $\left(\delta_{ij} 1_{X_i}\right)_{i,j\in I}$, immediate from the orthogonality conditions $\pi_i \circ s_j = \delta_{ij}$ (2.6.103).*

Remark 2.6.110
Any object X is trivially a biproduct of the one-element family $\{X\}$, via $(X, \{1_X\}, \{1_X\})$ (the orthogonality condition reduces to $1_X \circ 1_X = 1_X$). This lets us apply (2.6.108) whenever one side of a morphism is a single object rather than a genuine biproduct, by regarding it as such.

Remark 2.6.111 (Injections and Projections as Matrices)
For the binary biproduct $X_1 \oplus X_2$, the injections and projections are themselves matrices in the sense of (2.6.106), namely the columns and rows of the identity matrix

$$s_1 = \begin{pmatrix} 1_{X_1} \\ 0 \end{pmatrix} \quad s_2 = \begin{pmatrix} 0 \\ 1_{X_2} \end{pmatrix} \quad \pi_1 = \begin{pmatrix} 1_{X_1} & 0 \end{pmatrix} \quad \pi_2 = \begin{pmatrix} 0 & 1_{X_2} \end{pmatrix}$$

immediate from $\pi_i \circ s_j = \delta_{ij}$ (2.6.103). More generally, for $\bigoplus_{i\in I} X_i$, regarding X_{i_0} as the trivial biproduct of $\{X_{i_0}\}$, s_{i_0} is the i_0-th column and π_{i_0} the i_0-th row of the identity matrix $\left(\delta_{ij} 1_{X_i}\right)_{i,j\in I}$.

Lemma 2.6.112 (A Matrix is a Row of Columns and a Column of Rows)
Let $M : X_1 \oplus X_2 \to Y_1 \oplus Y_2$ have matrix $\begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}$. Then

$$M = \begin{pmatrix} \begin{pmatrix} M_{11} \\ M_{21} \end{pmatrix} & \begin{pmatrix} M_{12} \\ M_{22} \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} M_{11} & M_{12} \end{pmatrix} \\ \begin{pmatrix} M_{21} & M_{22} \end{pmatrix} \end{pmatrix}$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $c_1 \coloneqq M \circ s_1$, $c_2 \coloneqq M \circ s_2$ for the columns of M. The morphism $\begin{pmatrix} c_1 & c_2 \end{pmatrix} : X_1 \oplus X_2 \to Y_1 \oplus Y_2$ is, by definition of the coproduct-induced notation (2.6.93), the unique morphism satisfying $\begin{pmatrix} c_1 & c_2 \end{pmatrix} \circ s_1 = c_1$ and $\begin{pmatrix} c_1 & c_2 \end{pmatrix} \circ s_2 = c_2$. Since M itself satisfies $M \circ s_1 = c_1$ and $M \circ s_2 = c_2$ by definition of c_1, c_2, uniqueness gives $M = \begin{pmatrix} c_1 & c_2 \end{pmatrix}$.

Dually, writing $r_1 \coloneqq \pi_1 \circ M$, $r_2 \coloneqq \pi_2 \circ M$ for the rows of M, $\begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$ is the unique morphism with $\pi_1 \circ \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = r_1$ and $\pi_2 \circ \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = r_2$ (2.6.81), and M satisfies these by definition of r_1, r_2, so $M = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$. ∎

Lemma 2.6.113 (Composing a Column or Row on the Outside)
This is a general fact about products and coproducts, not requiring biproducts: let $p : Z \to Y_1$, $q : Z \to Y_2$ and $N : W \to Z$. Then

$$\begin{pmatrix} p \\ q \end{pmatrix} \circ N = \begin{pmatrix} p \circ N \\ q \circ N \end{pmatrix}$$

Dually, for $a : X_1 \to Z$, $b : X_2 \to Z$ and $N : Z \to W$

$$N \circ \begin{pmatrix} a & b \end{pmatrix} = \begin{pmatrix} N \circ a & N \circ b \end{pmatrix}$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write π_1, π_2 for the projections of $Y_1 \oplus Y_2$. Since $\pi_1 \circ \begin{pmatrix} p \\ q \end{pmatrix} = p$ and $\pi_2 \circ \begin{pmatrix} p \\ q \end{pmatrix} = q$ (2.6.106), for any $N : W \to Z$, by associativity of composition

$$\pi_1 \circ \left(\begin{pmatrix} p \\ q \end{pmatrix} \circ N \right) = p \circ N \qquad \pi_2 \circ \left(\begin{pmatrix} p \\ q \end{pmatrix} \circ N \right) = q \circ N$$

so $\begin{pmatrix} p \\ q \end{pmatrix} \circ N$ satisfies the defining property of $\begin{pmatrix} p \circ N \\ q \circ N \end{pmatrix}$, and the two are equal by the uniqueness clause of (2.6.81).

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Dually, writing s_1, s_2 for the injections of $X_1 \oplus X_2$, since $(a\ \ b) \circ s_1 = a$ and $(a\ \ b) \circ s_2 = b$, for any $N : Z \to W$

$$(N \circ (a\ \ b)) \circ s_1 = N \circ a \qquad (N \circ (a\ \ b)) \circ s_2 = N \circ b$$

so $N \circ (a\ \ b)$ satisfies the defining property of $(N \circ a\ \ N \circ b)$, and the two are equal by the uniqueness clause of (2.6.93).■

2.6.14 Biproduct Additivity

Following [Fre64] [Lan98], our eventual goal is to show that in a pointed category with (binary) biproducts, each hom-set $\mathrm{Mor}(X, Y)$ carries a canonical addition, making composition bilinear – so that additivity, usually imposed as an axiom, is instead a theorem, derived here purely from the existence of biproducts, via matrix notation.

Proposition 2.6.114 (Composition Without Cross Terms)
Let $\mathcal{C}$ be a pointed category with binary biproducts, hence all finite biproducts (2.6.105), I, J finite sets, $\sigma : I \to J$ a function, and $M : \bigoplus_{j\in J} X_j \to \bigoplus_{i\in I} Y_i$ a morphism whose matrix satisfies $M_{ij} = 0$ for every $i \in I$, $j \neq \sigma(i)$. Then for any $N : \bigoplus_{k\in K} W_k \to \bigoplus_{j\in J} X_j$ with matrix (N_{jk})

$$(M \circ N)_{ik} = M_{i,\sigma(i)} \circ N_{\sigma(i),k} \quad \text{for all } i \in I, k \in K$$

Dually, if $\tau : I \to J$ and $M' : \bigoplus_{i\in I} X_i \to \bigoplus_{j\in J} Y_j$ has matrix satisfying $M'_{ji} = 0$ for every $i \in I$, $j \neq \tau(i)$, then for any $N : \bigoplus_{j\in J} Y_j \to \bigoplus_{k\in K} Z_k$ with matrix (N_{kj})

$$(N \circ M')_{ki} = N_{k,\tau(i)} \circ M'_{\tau(i),i} \quad \text{for all } k \in K, i \in I$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

For $i \in I$ and any $j \in J$, $a_i \circ \pi_{\sigma(i)} \circ s_j = a_i \circ \delta_{\sigma(i),j}$ (using $\pi_{\sigma(i)} \circ s_j = \delta_{\sigma(i),j}$ (2.6.103)), which is $a_i = M_{i,\sigma(i)}$ if $j = \sigma(i)$ and $0 = M_{ij}$ otherwise (by hypothesis on M); this matches $\pi_i \circ M \circ s_j = M_{ij}$ for every $j \in J$, so by the uniqueness clause of (2.6.93) (applied to maps out of $\bigoplus_{j\in J} X_j$), $\pi_i \circ M = a_i \circ \pi_{\sigma(i)}$.

Hence, for every $i \in I$, $k \in K$

$$\pi_i \circ M \circ N \circ s_k = a_i \circ \pi_{\sigma(i)} \circ N \circ s_k = a_i \circ N_{\sigma(i),k}$$

giving the matrix of $M \circ N$.

Dually, for $i \in I$ and any $j \in J$, $\pi_j \circ s_{\tau(i)} \circ c_i = \delta_{j,\tau(i)} c_i$, which matches $\pi_j \circ M' \circ s_i = M'_{ji}$ for every $j \in J$ (equal to c_i if $j = \tau(i)$ and 0 otherwise, by hypothesis), so by the uniqueness clause of (2.6.81) (applied to maps into $\bigoplus_{j\in J} Y_j$), $M' \circ s_i = s_{\tau(i)} \circ c_i$; hence, for every $k \in K$, $i \in I$

$$\pi_k \circ N \circ M' \circ s_i = \pi_k \circ N \circ s_{\tau(i)} \circ c_i = N_{k,\tau(i)} \circ c_i$$

giving the matrix of $N \circ M'$. ■

Example 2.6.115
(2.6.114) recovers familiar matrix identities, at various arities, without needing addition, since none of the matrices below have cross terms. For $a : X \to Y$, regarding X (resp. Y) as the trivial biproduct of $\{X\}$ (resp. $\{Y\}$) (2.6.108)

$$(a\ \ 0) = a \circ (1_X\ \ 0) \qquad \begin{pmatrix} a \\ 0 \end{pmatrix} = \begin{pmatrix} 1_Y \\ 0 \end{pmatrix} \circ a$$

for $a : X_1 \to Y_1$, $b : X_2 \to Y_2$, $p : W \to X_1$ and $q : W \to X_2$

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \circ \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} a \circ p \\ b \circ q \end{pmatrix}$$

and for $u : X_1 \to Z$

$$(u\ \ 0) \circ \begin{pmatrix} p \\ q \end{pmatrix} = u \circ p$$

Definition 2.6.116 (Diagonal and Codiagonal)
*Let $\mathcal{C}$ be a pointed category with binary biproduct $X \oplus X$, with injections s_1, s_2 and projections π_1, π_2. Regarding X as the trivial biproduct of $\{X\}$, (2.6.108) identifies $\mathrm{Mor}(X, X \oplus X)$ with 2×1 matrices and $\mathrm{Mor}(X \oplus X, X)$ with 1×2 matrices over $\mathrm{Mor}(X, X)$. The **diagonal** $\Delta_X : X \to X \oplus X$ and **codiagonal** (or **summation map**) $\nabla_X : X \oplus X \to X$ are the morphisms with matrices*

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$$\Delta_X = \begin{pmatrix} 1_X \\ 1_X \end{pmatrix} \qquad \nabla_X = \begin{pmatrix} 1_X & 1_X \end{pmatrix}$$

that is, the unique morphisms with $\pi_1 \circ \Delta_X = \pi_2 \circ \Delta_X = 1_X$ *and* $\nabla_X \circ s_1 = \nabla_X \circ s_2 = 1_X$ *respectively.*

Remark 2.6.117
In **AbGrp**, *where* $X \oplus X$ *is the direct sum* $X \times X$ *with* $s_1(x) = (x, 0)$, $s_2(x) = (0, x)$, $\pi_1(x, y) = x$, $\pi_2(x, y) = y$, *the codiagonal is exactly the addition map:* $\nabla_X(x, 0) = x$ *and* $\nabla_X(0, y) = y$ *force, by additivity of the homomorphism* ∇_X, $\nabla_X(x, y) = \nabla_X(x, 0) + \nabla_X(0, y) = x + y$. *Dually the diagonal is* $\Delta_X(x) = (x, x)$.

Definition 2.6.118 (Sum of Parallel Morphisms)
Let $\mathcal{C}$ *be a pointed category with binary biproducts and* $f, g : X \to Y$. *We define*

$$f + g := \begin{pmatrix} f & g \end{pmatrix} \circ \Delta_X = \nabla_Y \circ \begin{pmatrix} f & 0 \\ 0 & g \end{pmatrix} \circ \Delta_X = \nabla_Y \circ \begin{pmatrix} f \\ g \end{pmatrix} : X \to Y$$

the three expressions agreeing by (2.6.114), since $\begin{pmatrix} f & 0 \\ 0 & g \end{pmatrix}$ *has no cross terms.*

Proposition 2.6.119 (Interchange Law)
Let $\mathcal{C}$ *be a pointed category with binary biproducts and* $w, x, y, z : X \to Y$. *Then* $(w + x) + (y + z) = (w + y) + (x + z)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino. Following [Fre64] Theorem 2.38.

Let $M := \begin{pmatrix} w & x \\ y & z \end{pmatrix} : X \oplus X \to Y \oplus Y$ (2.6.108). Then

$$M \circ \Delta_X \overset{(2.6.112)}{=} \begin{pmatrix} w & x \\ y & z \end{pmatrix} \circ \Delta_X \overset{(2.6.113)}{=} \begin{pmatrix} w & x \\ y & z \end{pmatrix} \circ \Delta_X \overset{(2.6.118)}{=} \begin{pmatrix} w + x \\ y + z \end{pmatrix}$$

$$\nabla_Y \circ M \overset{(2.6.112)}{=} \nabla_Y \circ \left(\begin{pmatrix} w \\ y \end{pmatrix} \begin{pmatrix} x \\ z \end{pmatrix} \right) \overset{(2.6.113)}{=} \left(\nabla_Y \circ \begin{pmatrix} w \\ y \end{pmatrix} \quad \nabla_Y \circ \begin{pmatrix} x \\ z \end{pmatrix} \right) \overset{(2.6.118)}{=} \begin{pmatrix} w + y & x + z \end{pmatrix}$$

so, applying $\nabla_Y \circ -$ to the first display and $- \circ \Delta_X$ to the second (each a further instance of (2.6.118))

$$(w + x) + (y + z) = \nabla_Y \circ (M \circ \Delta_X) = (\nabla_Y \circ M) \circ \Delta_X = (w + y) + (x + z)$$

■

Remark 2.6.120 (Eckmann--Hilton Phenomenon)
The Interchange Law is an instance of the classical ***Eckmann–Hilton argument****: whenever a set carries two unital binary operations* $*, \cdot$ *satisfying the interchange law* $(a * b) \cdot (c * d) = (a \cdot c) * (b \cdot d)$ *and sharing a common unit, the two operations coincide and are commutative and associative. Here the two operations are the row-first and column-first ways of combining the block* $\begin{pmatrix} w & x \\ y & z \end{pmatrix}$ *into a single morphism via* Δ_X, ∇_Y; *(2.6.118) already records that they coincide (as a single well-defined* $+$*), and (2.6.119) is exactly the interchange law for it. Commutativity and associativity, proved below, are the usual Eckmann–Hilton consequences, recovered by the substitutions* $w = z := 0$ *and* $y := 0$ *respectively.*

Proposition 2.6.121 ($\mathrm{Mor}(X, Y)$ **is a Commutative Monoid, and Composition is Bilinear)**
Let $\mathcal{C}$ *be a pointed category with binary biproducts. For objects* X, Y, $(\mathrm{Mor}(X, Y), +)$ *is a commutative monoid with identity* 0. *Moreover, for* $f, g : X \to Y$, $h : Y \to Z$ *and* $k : W \to X$

$$h \circ (f + g) = h \circ f + h \circ g \qquad (f + g) \circ k = f \circ k + g \circ k$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Identity. Since $\begin{pmatrix} f & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & f \end{pmatrix}$ each have a single nonzero entry, (2.6.114) gives

$$f + 0 = \begin{pmatrix} f & 0 \end{pmatrix} \circ \begin{pmatrix} 1_X \\ 1_X \end{pmatrix} = f \circ 1_X = f \qquad 0 + f = \begin{pmatrix} 0 & f \end{pmatrix} \circ \begin{pmatrix} 1_X \\ 1_X \end{pmatrix} = f \circ 1_X = f$$

Commutativity. By (2.6.119), setting $w := 0$, $x := f$, $y := g$, $z := 0$ gives $(0 + f) + (g + 0) = (0 + g) + (f + 0)$, that is $f + g = g + f$ using the identity just proved.

Associativity. For a third morphism $h : X \to Y$, setting $w := f$, $x := g$, $y := 0$, $z := h$ in (2.6.119) gives $(f + g) + (0 + h) = (f + 0) + (g + h)$, that is $(f + g) + h = f + (g + h)$.

Bilinearity.

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$$h \circ (f+g) \overset{(2.6.118)}{=} h \circ (f \;\; g) \circ \Delta_X \overset{(2.6.113)}{=} (h \circ f \;\; h \circ g) \circ \Delta_X \overset{(2.6.118)}{=} h \circ f + h \circ g$$

and dually

$$(f+g) \circ k \overset{(2.6.118)}{=} \nabla_Y \circ \begin{pmatrix} f \\ g \end{pmatrix} \circ k \overset{(2.6.113)}{=} \nabla_Y \circ \begin{pmatrix} f \circ k \\ g \circ k \end{pmatrix} \overset{(2.6.118)}{=} f \circ k + g \circ k$$

■

Proposition 2.6.122 (Composition is Matrix Multiplication)
Let $\mathcal{C}$ be a pointed category with binary biproducts, I, J, K finite sets, $M : \bigoplus_{j\in J} X_j \to \bigoplus_{i\in I} Y_i$ with matrix (M_{ij}) and $N : \bigoplus_{k\in K} W_k \to \bigoplus_{j\in J} X_j$ with matrix (N_{jk}). Then

$$(M \circ N)_{ik} = \sum_{j\in J} M_{ij} \circ N_{jk} \quad \text{for all } i \in I, k \in K$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

We first show that any column $C : Z \to \bigoplus_{j\in J} X_j$, with entries $C_j \coloneqq \pi_j \circ C$, satisfies $C = \sum_{j\in J} s_j \circ C_j$. Indeed, for each $i \in J$, by (2.6.121) (bilinearity, iterated over the finite sum)

$$\pi_i \circ \sum_{j\in J} s_j \circ C_j = \sum_{j\in J} \pi_i \circ s_j \circ C_j = \sum_{j\in J} \delta_{ij} C_j = C_i$$

using $\pi_i \circ s_j = \delta_{ij}$ (2.6.103) and that every term with $j \neq i$ vanishes (composing with the zero morphism), so this matches the matrix of C entrywise, whence $\sum_{j\in J} s_j \circ C_j = C$ by (2.6.108). Dually, any row $R : \bigoplus_{i\in I} Y_i \to Z$ satisfies $R = \sum_{i\in I} R_i \circ \pi_i$, where $R_i \coloneqq R \circ s_i$.

Now fix $i \in I$, $k \in K$. The k-th column of N is $N \circ s_k$, with entries $N_{jk} = \pi_j \circ N \circ s_k$, so by the column decomposition, $N \circ s_k = \sum_{j\in J} s_j \circ N_{jk}$. Hence, using bilinearity to distribute $M \circ -$ and then $\pi_i \circ -$ over the finite sum

$$(M \circ N)_{ik} = \pi_i \circ M \circ N \circ s_k = \pi_i \circ M \circ \sum_{j\in J} s_j \circ N_{jk} = \sum_{j\in J} \pi_i \circ M \circ s_j \circ N_{jk} = \sum_{j\in J} M_{ij} \circ N_{jk}$$

■

2.6.15 Direct Limit / Directed Colimit

Here we consider only a certain type of colimit, which is referred to variously as **direct limit** [AM69], **limite inductive** [Gro64] or **filtered colimit** [Lan98]. We also restrict to the case where the indexing category is "thin", which is more properly refered to as a directed colimit.

Definition 2.6.123 (Directed Preorder)
*We say a category I is a **directed preorder** if it is small, thin and filtered, that is for any $i, j \in I$*

- *there exists at most one morphism $i \to j$, and*
- *there exists $k \in I$ and morphisms $i \to k$ and $j \to k$*

If there exists a morphism $i \to j$ then we write $i \prec j$.

Example 2.6.124
A typical example is any family of subsets of a set X ordered by reversion inclusion which are closed under finite intersection.

Definition 2.6.125 (Direct limit / Directed colimit)
Let I be a directed preorder and $F : I \to \mathcal{C}$ a diagram.

*A **cocone** over F is a pair $(L, \{\mu : F(i) \to L\}_{i\in I})$ such that*

$$i \prec j \implies \mu_j \circ F(i \to j) = \mu_i$$

The cocones over F constitute a category $(F \downarrow \Delta)$ where morphisms $\alpha : (L, \{\mu_i\}) \to (L', \{\mu'_i\})$ are simply morphisms $\alpha : L \to L'$ such that $\alpha \circ \mu_i = \mu'_i$ for all $i \in I$.

*A cone $\left(L, \{\mu_i\}_{i\in I}\right)$ over F is a **direct limit** for F if it is an **initial object** (2.6.48) in the cocone category $(F \downarrow \Delta)$. Explicitly for every cocone $\left(L', \{\mu'_i\}_{i\in I}\right)$ there is a unique morphism $\alpha : L \to L'$ such that $\alpha \circ \mu_i = \mu'_i$ for all $i \in I$.*

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We denote this by $\varinjlim_{i\in I} F(i)$.

Definition 2.6.126 (Cofinal Subpreorder)
Let I be a directed preorder. Then a full subcategory $I' \subset I$ is said to be ***cofinal*** *if*

$$\forall i \in I \quad \exists i' \in I' \text{ s.t. } i \leq i'$$

In this case I' is also directed.

Proposition 2.6.127 (Cofinal Direct Limit)
Let I be a directed preorder and I' a full subcategory which is also directed. Suppose $F : I \to \mathcal{C}$ is a diagram then there is a unique morphism of cocones

$$\varinjlim_{i\in I'} F(i) \longrightarrow \varinjlim_{i\in I} F(i)$$

provided both these exist.

If I' is cofinal in I then this is an isomorphism.

Proposition 2.6.128 (Direct Limits of Sets)
Let I be a directed preorder and $F : I \to \mathbf{Set}$ a diagram. Consider

$$L := \{(i, x) \mid i \in I, x \in F(i)\}/\sim$$

where $(i, x) \sim (j, y)$ if for some $k \in I$ we have $i \prec k, j \prec k$ and $F(i \to k)(x) = F(j \to k)(y)$, and let $\mu_i : F(i) \to L$, $\mu_i(x) := [(i, x)]$.

Then $\sim$ is an equivalence relation, and $\left(L, \{\mu_i\}_{i\in I}\right)$ is a direct limit for F (2.6.125), that is $L = \varinjlim_{i\in I} F(i)$. Moreover the μ_i are jointly surjective, that is every element of L is of the form $\mu_i(x)$ for some $i \in I$, $x \in F(i)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\sim$ is an equivalence relation. Reflexivity and symmetry are immediate, using $i \prec i$ (via the identity morphism). For transitivity, suppose $(i, x) \sim (j, y)$ via k (so $i, j \prec k$, $F(i \to k)(x) = F(j \to k)(y)$) and $(j, y) \sim (l, z)$ via m (so $j, l \prec m$, $F(j \to m)(y) = F(l \to m)(z)$). As I is directed there is n with $k, m \prec n$, and applying $F(k \to n)$, resp. $F(m \to n)$, to the two witnessing equations and using functoriality of F,

$$F(i \to n)(x) = F(k \to n)(F(i \to k)(x)) = F(k \to n)(F(j \to k)(y)) = F(j \to n)(y)$$

$$F(j \to n)(y) = F(m \to n)(F(j \to m)(y)) = F(m \to n)(F(l \to m)(z)) = F(l \to n)(z)$$

so $F(i \to n)(x) = F(l \to n)(z)$, that is $(i, x) \sim (l, z)$ via n.

Joint surjectivity. Immediate, as every element of L is by construction the class $[(i, x)] = \mu_i(x)$ of some pair.

$(L, \{\mu_i\})$ is a cocone. For $i \prec j$, taking $k := j$ in the equivalence relation (using $F(j \to j) = 1_{F(j)}$) gives $(i, x) \sim (j, F(i \to j)(x))$, that is $\mu_j(F(i \to j)(x)) = \mu_i(x)$, as required by (2.6.125).

$(L, \{\mu_i\})$ is initial. Let $(L', \{\mu'_i\})$ be any cocone over F. Define $\alpha : L \to L'$ by $\alpha([(i, x)]) := \mu'_i(x)$; this is well-defined, for if $(i, x) \sim (j, y)$ via k then, using the cocone condition for μ',

$$\mu'_i(x) = \mu'_k(F(i \to k)(x)) = \mu'_k(F(j \to k)(y)) = \mu'_j(y)$$

and by construction $\alpha \circ \mu_i = \mu'_i$ for all i. Any morphism of cocones $\beta : L \to L'$ satisfies $\beta(\mu_i(x)) = \mu'_i(x)$ for all i, x, so $\beta = \alpha$ by joint surjectivity of the μ_i; hence α is the unique such morphism, and $(L, \{\mu_i\})$ is initial in $(F \downarrow \Delta)$, that is a direct limit for F. ∎

2.6.16 Inverse Limit

Definition 2.6.129 (Inverse Limit / Directed Limit)
Let I be a directed preorder and $F : I^{op} \to \mathcal{C}$ a diagram – equivalently, an ***inverse system*** *indexed by I: for $i \prec j$ in I, a* ***restriction*** *morphism $\rho_{ij} : F(j) \to F(i)$ in $\mathcal{C}$, with $\rho_{ii} = 1_{F(i)}$ and $\rho_{ik} = \rho_{ij} \circ \rho_{jk}$ whenever $i \prec j \prec k$.*

A limit for F in the sense of (2.6.78) (applied to the small category I^{op}) is called an ***inverse limit*** *for the system, denoted $\varprojlim_{i\in I} F(i)$: explicitly, a cone $\left(L, \{\lambda_i : L \to F(i)\}_{i\in I}\right)$ with $\rho_{ij} \circ \lambda_j = \lambda_i$ for every $i \prec j$, terminal among such cones.*

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Proposition 2.6.130 (Inverse Limits of Sets)
Let I be a directed preorder and $F : I^{op} \to \mathbf{Set}$ an inverse system, with restriction maps $\rho_{ij} : F(j) \to F(i)$ $(i \prec j)$. Then

$$\varprojlim_{i\in I} F(i) = \left\{ (x_i)_{i\in I} \in \prod_{i\in I} F(i) \;\middle|\; \rho_{ij}(x_j) = x_i \text{ for all } i \prec j \right\},$$

with projections $\lambda_i\Big((x_j)_j\Big) := x_i$.

Proof. This is (2.6.85) applied to the diagram $F : I^{\mathrm{op}} \to \mathbf{Set}$ (I^{op} a small category, as I is). ∎

Proposition 2.6.131 (Cofinal Inverse Limit)
Let I be a directed preorder, $I' \subset I$ a full subcategory which is also directed, and $F : I^{\mathrm{op}} \to \mathcal{C}$ an inverse system, with restriction maps ρ_{ij} ($i \prec j$ in I). Then, provided both exist, there is a unique morphism of cones

$$\varprojlim_{i\in I} F(i) \longrightarrow \varprojlim_{i\in I'} F(i)$$

(restricting the limit cone over I to the subcategory I').

If I' is cofinal in I (2.6.126) then this is an isomorphism.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $L' := \varprojlim_{i\in I'} F(i)$, with projections $\lambda'_i : L' \to F(i)$ $(i \in I')$. The cone $\left(L', \{\lambda'_i\}_{i\in I'}\right)$ over $F|_{I'^{\mathrm{op}}}$ is in particular a cone over all of F restricted along the inclusion $I'^{\mathrm{op}} \subset I^{\mathrm{op}}$, so composing with the (terminal, by definition of $\varprojlim_{i\in I} F(i)$) limit cone's universal property in the other direction gives nothing yet – rather, the stated morphism is simply the unique one afforded by terminality of L' among cones over $F|_{I'^{\mathrm{op}}}$, applied to the cone $\left(\varprojlim_{i\in I} F(i), \{\lambda_i\}_{i\in I'}\right)$ (the limit cone over I, restricted in its indexing set to I' – still a cone over $F|_{I'^{\mathrm{op}}}$, by the cone condition on I restricted to $i \prec j$ both in I'): this gives the unique morphism $\varprojlim_{i\in I} F(i) \to L'$ of the statement.

Now suppose I' is cofinal. For $i \in I$, choose (by cofinality) $i' \in I'$ with $i \prec i'$, and define $\mu_i := \rho_{i,i'} \circ \lambda'_{i'} : L' \to F(i)$.

$\boldsymbol{\mu_i}$ is independent of the choice of $\boldsymbol{i'}$. If $i \prec i'_1, i'_2$ with $i'_1, i'_2 \in I'$, directedness of I' gives $i'_3 \in I'$ with $i'_1, i'_2 \prec i'_3$; then, using the cone condition for λ' (within I', at $i'_1 \prec i'_3$) and the restriction composition law of (2.6.129) (at $i \prec i'_1 \prec i'_3$),

$$\rho_{i,i'_1} \circ \lambda'_{i'_1} = \rho_{i,i'_1} \circ \rho_{i'_1,i'_3} \circ \lambda'_{i'_3} = \rho_{i,i'_3} \circ \lambda'_{i'_3},$$

and symmetrically $\rho_{i,i'_2} \circ \lambda'_{i'_2} = \rho_{i,i'_3} \circ \lambda'_{i'_3}$; so both choices agree.

$\boldsymbol{\left(L', \{\mu_i\}_{i\in I}\right)}$ is a cone over $\boldsymbol{F}$. Let $i \prec j$ in I. Choose $j' \in I'$ with $j \prec j'$; then also $i \prec j'$, so j' may be used to compute both μ_i and μ_j:

$$\rho_{ij} \circ \mu_j = \rho_{ij} \circ \rho_{j,j'} \circ \lambda'_{j'} = \rho_{i,j'} \circ \lambda'_{j'} = \mu_i,$$

using the restriction composition law at $i \prec j \prec j'$. Moreover for $i \in I'$ itself, taking $i' := i$ (with $\rho_{ii} = 1_{F(i)}$) gives $\mu_i = \lambda'_i$: the cone $\left(L', \{\mu_i\}_{i\in I}\right)$ restricts, on I', to the original cone $\left(L', \{\lambda'_i\}_{i\in I'}\right)$.

$\boldsymbol{\left(L', \{\mu_i\}_{i\in I}\right)}$ is terminal. Let $\left(N, \{\nu_i\}_{i\in I}\right)$ be any cone over F. Restricting to I' gives a cone over $F|_{I'^{\mathrm{op}}}$, so by terminality of L' there is a unique morphism $\gamma : N \to L'$ with $\lambda'_i \circ \gamma = \nu_i$ for all $i \in I'$. For general $i \in I$, choosing $i' \in I'$ with $i \prec i'$,

$$\mu_i \circ \gamma = \rho_{i,i'} \circ \lambda'_{i'} \circ \gamma = \rho_{i,i'} \circ \nu_{i'} = \nu_i$$

using the cone condition for ν (over all of I) at $i \prec i'$; so γ is a morphism of cones $\left(N, \{\nu_i\}_{i\in I}\right) \to \left(L', \{\mu_i\}_{i\in I}\right)$. Any such morphism δ satisfies, in particular for $i \in I'$ (where $\mu_i = \lambda'_i$), $\lambda'_i \circ \delta = \nu_i$, so $\delta = \gamma$ by the uniqueness already established for γ. Hence $\left(L', \{\mu_i\}_{i\in I}\right)$ is terminal among cones over F on all of I, that is, $\left(L', \{\mu_i\}_{i\in I}\right)$ is $\varprojlim_{i\in I} F(i)$.

By uniqueness of limits up to unique isomorphism of cones (2.6.78), there is a (unique) isomorphism $\varprojlim_{i\in I} F(i) \xrightarrow{\sim} L'$ compatible with the projections $\{\mu_i\}_{i\in I}$ (on the L' side) and $\{\lambda_i\}_{i\in I}$ (on the $\varprojlim_{i\in I} F(i)$ side); restricting this compatibility to $i \in I'$ (where $\mu_i = \lambda'_i$) shows it is compatible with $\{\lambda'_i\}_{i\in I'}$ too, so by the uniqueness clause established in the first paragraph, this isomorphism is exactly the morphism of cones $\varprojlim_{i\in I} F(i) \to \varprojlim_{i\in I'} F(i) = L'$ of the statement. ∎

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Corollary 2.6.132 (Functoriality of the Inverse Limit)
Let I be a directed preorder, and let $\mathcal{C}^{I^{op}}_{\lim}$ denote the full subcategory of the functor category $\mathcal{C}^{I^{op}}$ (2.6.14) spanned by those inverse systems $F : I^{op} \to \mathcal{C}$ for which $\varprojlim_i F(i)$ exists. Then

$$\varprojlim_i (-) : \mathcal{C}^{I^{op}}_{\lim} \longrightarrow \mathcal{C}, \quad F \mapsto \varprojlim_i F(i)$$

is a functor.

More precisely: for a natural transformation $\sigma = (\sigma_i : F(i) \to G(i))_{i \in I}$ there is a unique morphism

$$\varprojlim_i \sigma_i : \varprojlim_i F(i) \longrightarrow \varprojlim_i G(i)$$

compatible with the projections; and

a) for a further natural transformation $\tau = (\tau_i : G(i) \to K(i))_i$ (so $\tau\sigma := (\tau_i \sigma_i)_i$, itself a natural transformation $F \Rightarrow K$),

$$\varprojlim_i (\tau_i \sigma_i) = \varprojlim_i \tau_i \circ \varprojlim_i \sigma_i;$$

b) $\varprojlim_i 1_{F(i)} = 1_{\varprojlim_i F(i)}$.

Consequently, if every σ_i is an isomorphism, so is $\varprojlim_i \sigma_i$ (2.6.42).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

A natural transformation $\sigma = (\sigma_i)_i$ satisfying $\sigma_i \circ \rho^F_{ij} = \rho^G_{ij} \circ \sigma_j$ for all $i \prec j$ is exactly a morphism $F \Rightarrow G$ of the functor category $\mathcal{C}^{I^{op}}$ (2.6.14) (the naturality square for σ at the morphism $j \to i$ of I^{op} corresponding to $i \prec j$), invertible in $\mathcal{C}^{I^{op}}$ exactly when every σ_i is an isomorphism (with inverse $(\sigma_i^{-1})_i$). So this is precisely the case $J := I^{op}$ of (2.6.97). ∎

Corollary 2.6.133 (Inverse Limits Commute with Products)
Let I be a directed preorder, K a set, and $(F_k)_{k \in K}$ inverse systems $F_k : I^{op} \to \mathcal{C}$, with limit-cone projections $\lambda_{k,i} : \varprojlim_i F_k(i) \to F_k(i)$, such that $\varprojlim_i F_k(i)$ exists for every $k \in K$; and suppose $\prod_{k \in K} F_k(i)$ exists for every $i \in I$, with product projections $\pi_{k,i} : \prod_{k' \in K} F_{k'}(i) \to F_k(i)$. If $\prod_{k \in K} \varprojlim_i F_k(i)$ exists, with product projections p_k, then $\varprojlim_i \left(\prod_{k \in K} F_k(i)\right)$ exists too, with limit-cone projections Λ_i, and there is a unique morphism

$$\Phi : \prod_{k \in K} \varprojlim_i F_k(i) \longrightarrow \varprojlim_i \left(\prod_{k \in K} F_k(i) \right)$$

such that the following commutes for every $k \in K$, $i \in I$

$$\begin{array}{ccccc}
\prod_{k \in K} \varprojlim_i F_k(i) & & \xrightarrow{\quad\Phi\quad} & & \varprojlim_i \left(\prod_{k \in K} F_k(i)\right) \\
\Big\downarrow{\scriptstyle p_k} & & & & \Big\downarrow{\scriptstyle \Lambda_i} \\
\varprojlim_i F_k(i) & \xrightarrow[\lambda_{k,i}]{} & F_k(i) & \xleftarrow[\pi_{k,i}]{} & \prod_{k \in K} F_k(i)
\end{array}$$

and Φ is an isomorphism.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Regard K as a discrete small category and define a bifunctor $H : K \times I^{op} \to \mathcal{C}$ by $H(k,i) := F_k(i)$ (a morphism of $K \times I^{op}$ is a pair $(1_k, \alpha)$ for α a morphism of I^{op}, since K is discrete, and $H(1_k, \alpha) := F_k(\alpha)$; this is a bifunctor as each F_k is a functor). By (2.6.81) (limits over a discrete category are products), $\left(\prod_{k \in K} F_k(i), \{\pi_{k,i}\}_k\right)$ is $\lim_{k \in K} H(k,i)$ for every $i \in I^{op}$, and by (2.6.129), $\left(\varprojlim_i F_k(i), \{\lambda_{k,i}\}_i\right)$ is $\lim_{i \in I^{op}} H(k,i)$ for every $k \in K$; both exist by hypothesis, so (writing $L(k) := \varprojlim_i F_k(i)$, $R(i) := \prod_{k \in K} F_k(i)$) (2.6.100) applies to H.

As $\left(\prod_{k \in K} \varprojlim_i F_k(i), \{p_k\}_k\right) = \left(\lim_{k \in K} L(k), \{p_k\}_k\right)$ exists by hypothesis, (2.6.100) gives that $\lim_{i \in I^{op}} R(i) = \varprojlim_i \left(\prod_{k \in K} F_k(i)\right)$ also exists, with limit-cone projections Λ_i say, together with a unique isomorphism Φ making the stated diagram commute for every k, i (that proposition's diagram, with $\nu_k := p_k$, $\xi_i := \Lambda_i$, $\lambda_{k,i}$ as here, and $\mu_{k,i} := \pi_{k,i}$). ∎

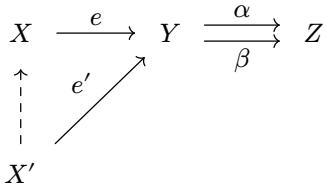
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2.6.17 Equaliser and Coequaliser

Definition 2.6.134 (Equaliser)
Consider a commutative diagram in a category $\mathcal{C}$

$$X \xrightarrow{e} Y \underset{\beta}{\overset{\alpha}{\rightrightarrows}} Z$$

*such that $\alpha \circ e = \beta \circ e$. We say that (X, e) is an **equaliser** for (α, β) if for all $e' : X' \to Y$ such that $\alpha \circ e' = \beta \circ e'$ there is a unique factorisation*



This is equivalent to (X, e) being a limit for the obvious functor $F : \mathbf{2} \to \mathcal{C}$ where $\mathbf{2}$ denotes the category $\cdot \rightrightarrows \cdot$.

Proposition 2.6.135
Every equaliser is a monomorphism, and every coequaliser is an epimorphism.

Proof. Suppose that (X, e) is an equaliser for $\alpha, \beta : Y \rightrightarrows Z$, and $e \circ f = e \circ g = h$. Then evidentally $\alpha \circ h = \beta \circ h$ so there is a unique morphism k such that $e \circ k = h$, whence $k = f = g$.

The second statement follows by duality. ∎

Proposition 2.6.136
Suppose $(X, e) = \mathrm{eq}(\alpha, \beta)$. Then $\alpha = \beta$ if and only if e is an isomorphism.

Proof. By hypothesis $\alpha \circ e = \beta \circ e$ so if e is an isomorphism $\alpha = \alpha \circ e \circ e^{-1} = \beta \circ e \circ e^{-1} = \beta$ as required.

Conversely if $\alpha = \beta$, then $\alpha \circ 1_Y = \beta \circ 1_Y$ so there is a morphism $i : Y \to X$ such that $e \circ i = 1_Y$. Therefore e is split-epic and monic (2.6.135) and therefore an isomorphism (2.6.36). ∎

Proposition 2.6.137
Let (X, e) be an equaliser for $\alpha, \beta : Y \rightrightarrows Z$. Suppose $\phi : X' \xrightarrow{\sim} X$ be an isomorphism. Then $(X', e \circ \phi)$ is also an equaliser for (α, β).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By definition of Equaliser, (X, e) is a limit for the diagram $F : \mathbf{2} \to \mathcal{C}$ determined by (α, β), and since $\alpha \circ e = \beta \circ e$, $(X', e \circ \phi)$ is a cone for F. As $e \circ \phi = e \circ \phi$, the isomorphism $\phi : X' \to X$ is (tautologically) an isomorphism of cones $(X', e \circ \phi) \to (X, e)$, so by (2.6.79) $(X', e \circ \phi)$ is also a limit for F, that is, an equaliser for (α, β). ∎

Proposition 2.6.138
Let (X, e) be a coequaliser for $\alpha, \beta : Y \rightrightarrows Z$. Suppose $\phi : X \xrightarrow{\sim} X'$ is an isomorphism. Then $(X', \phi \circ e)$ is also a coequaliser for (α, β).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By definition of Coequaliser, (X, e) is a colimit for the diagram $F : \mathbf{2} \to \mathcal{C}$ determined by (α, β), and since $e \circ \alpha = e \circ \beta$, $(X', \phi \circ e)$ is a co-cone for F. As $\phi \circ e = \phi \circ e$, the isomorphism $\phi : X \to X'$ is (tautologically) an isomorphism of co-cones $(X, e) \to (X', \phi \circ e)$, so by (2.6.90) $(X', \phi \circ e)$ is also a colimit for F, that is, a coequaliser for (α, β). ∎

Proposition 2.6.139
A diagram in **Set**

$$X \xrightarrow{i} Y \underset{\beta}{\overset{\alpha}{\rightrightarrows}} Z$$

is an equaliser if and only if i is injective and

$$\mathrm{Im}(i) = \{y \in Y \mid \alpha(y) = \beta(y)\}$$

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Definition 2.6.140 (Coequaliser)
Consider a commutative diagram in a category $\mathcal{C}$

$$X \underset{\beta}{\overset{\alpha}{\rightrightarrows}} Y \xrightarrow{\pi} Z$$

*such that $\pi \circ \alpha = \pi \circ \beta$. We say that π is a **co-equaliser** for (α, β) if for all $\pi' : Y \to Z'$ such that $\pi' \circ \alpha = \pi' \circ \beta$ there is a unique factorisation*

$$\begin{array}{ccccc} X & \underset{\beta}{\overset{\alpha}{\rightrightarrows}} & Y & \xrightarrow{\pi} & Z \\ & & & \searrow^{\pi'} & \downarrow \\ & & & & Z' \end{array}$$

This is equivalent to (Z, π) being a colimit for the obvious functor $F : \mathbf{2} \to \mathcal{C}$ where $\mathbf{2}$ denotes the category $\cdot \rightrightarrows \cdot$.

Proposition 2.6.141
The following diagram is a coequaliser

$$X \underset{\beta}{\overset{\alpha}{\rightrightarrows}} Y \xrightarrow{\pi} Z$$

if and only if the following is an equaliser in **Set** *for all $W \in \mathcal{C}$*

$$\mathrm{Mor}(Z, W) \xrightarrow{\mathrm{Mor}(\pi, W)} \mathrm{Mor}(Y, W) \underset{\mathrm{Mor}(\beta, W)}{\overset{\mathrm{Mor}(\alpha, W)}{\rightrightarrows}} \mathrm{Mor}(X, W)$$

Proof. We see from the definition that π is a coequaliser for (α, β) if and only if $\mathrm{Mor}(\pi, W)$ is injective and

$$\mathrm{Im}(\mathrm{Mor}(\pi, W)) = \{\pi' : Y \to W \mid \pi' \circ \alpha = \pi' \circ \beta\}$$

which is exactly the criteria (2.6.139) for an equaliser in **Set**. ∎

Proposition 2.6.142 (Left Adjoints Preserve Coequalisers)
Let $\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$ a pair of adjoint functors. Then F preserves coequalisers and G preserves equalisers.

Proof. Consider a coequaliser

$$X \underset{\beta}{\overset{\alpha}{\rightrightarrows}} Y \xrightarrow{\pi} Z$$

By the adjoint relationship we have a commutative diagram for all $D \in \mathcal{D}$

$$\begin{array}{ccccc} \mathrm{Mor}(F(Y), D) & \xrightarrow{\mathrm{Mor}(F(\pi), D)} & \mathrm{Mor}(F(X), D) & \underset{\mathrm{Mor}(F(\beta), D)}{\overset{\mathrm{Mor}(F(\alpha), D)}{\rightrightarrows}} & \mathrm{Mor}(F(Z), D) \\ \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ \mathrm{Mor}(Y, G(D)) & \xrightarrow{\mathrm{Mor}(\pi, G(D))} & \mathrm{Mor}(X, G(D)) & \underset{\mathrm{Mor}(\beta, G(D))}{\overset{\mathrm{Mor}(\alpha, G(D))}{\rightrightarrows}} & \mathrm{Mor}(Y, G(D)) \end{array}$$

By (2.6.141) the bottom row is an equaliser, and so by the isomorphisms so is the top row. Therefore we conclude by the same result that the following is an equaliser as required

$$F(X) \underset{F(\beta)}{\overset{F(\alpha)}{\rightrightarrows}} F(Y) \xrightarrow{F(\pi)} F(Z)$$

∎

Proposition 2.6.143 (Coequalisers in a Reflective Subcategory)
Let $\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$ be adjoint functors with G fully faithful and unit η; by (2.6.73) we may regard $\mathcal{D}$ as a full subcategory

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of $\mathcal{C}$ via G. Let $\alpha, \beta : X \rightrightarrows Y$ be morphisms of $\mathcal{D}$ and suppose $\pi : Y \to Z$ is a coequaliser of (α, β) in $\mathcal{C}$. Then $\eta_Z \circ \pi : Y \to F(Z)$ is a coequaliser of (α, β) in $\mathcal{D}$.

In particular if every pair of morphisms of $\mathcal{D}$ has a coequaliser in $\mathcal{C}$ then $\mathcal{D}$ has all coequalisers.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $q \coloneqq \eta_Z \circ \pi$. Since $\pi \circ \alpha = \pi \circ \beta$ we have $q \circ \alpha = \eta_Z \circ \pi \circ \alpha = \eta_Z \circ \pi \circ \beta = q \circ \beta$.

Suppose $q' : Y \to H$ is a morphism of $\mathcal{D}$ with $q' \circ \alpha = q' \circ \beta$. Regarding q' as a morphism of $\mathcal{C}$ (as $\mathcal{D}$ is a subcategory), the universal property of the coequaliser π gives a unique morphism $\tilde{q} : Z \to H$ of $\mathcal{C}$ such that $\tilde{q} \circ \pi = q'$.

Since F is left adjoint to G with unit η (2.6.71), there is a bijection $\mathrm{Mor}(F(Z), H) \xrightarrow{\sim} \mathrm{Mor}(Z, H)$, $\phi \mapsto \phi \circ \eta_Z$, so there is a unique morphism $\overline{q} : F(Z) \to H$ of $\mathcal{D}$ such that $\overline{q} \circ \eta_Z = \tilde{q}$. Then

$$\overline{q} \circ q = \overline{q} \circ \eta_Z \circ \pi = \tilde{q} \circ \pi = q'$$

so $\overline{q}$ is a factorisation of q' through q.

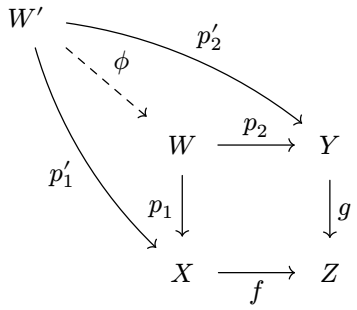
Suppose $\overline{q}'$ is another morphism of $\mathcal{D}$ with $\overline{q}' \circ q = q'$, that is $(\overline{q}' \circ \eta_Z) \circ \pi = q' = \tilde{q} \circ \pi$. As π is a coequaliser it is an epimorphism (2.6.135), whence $\overline{q}' \circ \eta_Z = \tilde{q}$, and by the uniqueness of the factorisation above $\overline{q}' = \overline{q}$.

Therefore $q = \eta_Z \circ \pi$ satisfies the universal property of the coequaliser of (α, β) in $\mathcal{D}$.

For the final statement, if every pair of morphisms of $\mathcal{D}$ has a coequaliser in $\mathcal{C}$ then the above applies to every such pair, so $\mathcal{D}$ has all coequalisers. ∎

2.6.18 Pullback and Kernel Pairs

Definition 2.6.144 (Pullback)
*Let $f : X \to Z$, $g : Y \to Z$ be morphisms in $\mathcal{C}$. We say (W, p_1, p_2), with $p_1 : W \to X$ and $p_2 : W \to Y$, is a **pullback** of (f, g) if $f \circ p_1 = g \circ p_2$, and for every (W', p_1', p_2') with $f \circ p_1' = g \circ p_2'$ there is a unique morphism $\phi : W' \to W$ such that $p_1 \circ \phi = p_1'$ and $p_2 \circ \phi = p_2'$*



This is equivalent to $(W, \{p_1, p_2\})$ being a limit for the evident diagram $F : \mathbf{\Lambda} \to \mathcal{C}$ determined by (f, g), where $\mathbf{\Lambda} \coloneqq \{a \to c \leftarrow b\}$ denotes the category with three objects and, besides identities, two further morphisms $a \to c$, $b \to c$. We write $W =: X \times_Z Y$.

Proposition 2.6.145 (Pullback from Products and Equalisers)
Let $f : X \to Z$, $g : Y \to Z$ be morphisms such that $X \times Y$ exists with projections π_X, π_Y, and such that $(W, e) \coloneqq \mathrm{eq}(f \circ \pi_X, g \circ \pi_Y : X \times Y \to Z)$ exists. Then $(W, \pi_X \circ e, \pi_Y \circ e)$ is a pullback of (f, g).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $p_1 \coloneqq \pi_X \circ e$, $p_2 \coloneqq \pi_Y \circ e$. By definition of the equaliser, $f \circ \pi_X \circ e = g \circ \pi_Y \circ e$, that is $f \circ p_1 = g \circ p_2$.

Suppose (W', p_1', p_2') satisfies $f \circ p_1' = g \circ p_2'$. By the universal property of the product there is a unique $h : W' \to X \times Y$ with $\pi_X \circ h = p_1'$, $\pi_Y \circ h = p_2'$. Then $f \circ \pi_X \circ h = f \circ p_1' = g \circ p_2' = g \circ \pi_Y \circ h$, so h equalises $(f \circ \pi_X, g \circ \pi_Y)$, whence by the universal property of the equaliser there is a unique $\phi : W' \to W$ with $e \circ \phi = h$. Then $p_1 \circ \phi = \pi_X \circ e \circ \phi = \pi_X \circ h = p_1'$, and similarly $p_2 \circ \phi = p_2'$.

If ϕ' also satisfies $p_1 \circ \phi' = p_1'$, $p_2 \circ \phi' = p_2'$, then $\pi_X \circ (e \circ \phi') = p_1' = \pi_X \circ h$ and $\pi_Y \circ (e \circ \phi') = p_2' = \pi_Y \circ h$, so by uniqueness in the universal property of the product, $e \circ \phi' = h$; as e is monic (2.6.135), $\phi' = \phi$. ∎

Definition 2.6.146 (Kernel Pair)
*Let $f : X \to Y$ be a morphism. A **kernel pair** of f is a pullback (Z, p_1, p_2) of (f, f), that is $p_1, p_2 : Z \to X$ with $f \circ p_1 = f \circ p_2$, universal among such pairs*

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$$\begin{array}{ccc} Z & \xrightarrow{p_2} & X \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle f} \\ X & \xrightarrow[f]{} & Y \end{array}$$

In **Set**, *the kernel pair of f is precisely the relation $\{(x,x') \in X \times X \mid f(x) = f(x')\} \subseteq X \times X$, together with its two projections: it records exactly which pairs of elements f identifies. So f is injective if and only if this relation is trivial, i.e. reduces to the diagonal $\{(x,x) \mid x \in X\}$ – this is the picture behind (2.6.147) below, which characterises monomorphisms in general as those morphisms whose kernel pair is trivial.*

Proposition 2.6.147 (Monomorphisms via Kernel Pairs)
Let $\mu : X \to Y$ be a morphism with a kernel pair (Z, p_1, p_2). Then μ is monic if and only if $p_1 = p_2$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

($\Leftarrow$) Suppose $p_1 = p_2$. If $a, b : W \to X$ satisfy $\mu \circ a = \mu \circ b$, then by the universal property of the kernel pair there is a unique $\psi : W \to Z$ with $p_1 \circ \psi = a$ and $p_2 \circ \psi = b$; since $p_1 = p_2$ this gives $a = p_1 \circ \psi = p_2 \circ \psi = b$. So μ is monic.

($\Rightarrow$) Suppose μ is monic. Since $\mu \circ p_1 = \mu \circ p_2$ by definition of the kernel pair (2.6.146), monicity of μ gives $p_1 = p_2$. ■

Corollary 2.6.148 (Monomorphisms have Trivial Kernel Pair)
Let $\mu : X \to Y$ be a morphism. Then μ is monic if and only if $(X, 1_X, 1_X)$ is a kernel pair of μ.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

($\Leftarrow$) This is the case $Z \coloneqq X$, $p_1 = p_2 \coloneqq 1_X$ of (2.6.147) ($\Leftarrow$).

($\Rightarrow$) Suppose μ is monic. Given $a, b : W \to X$ with $\mu \circ a = \mu \circ b$, monicity gives $a = b$; then $\phi \coloneqq a$ satisfies $1_X \circ \phi = a$ and $1_X \circ \phi = b$ (as $a = b$), and is the unique such morphism, since any ϕ' with $1_X \circ \phi' = a$ must equal a. So $(X, 1_X, 1_X)$ satisfies the universal property of a kernel pair of μ (2.6.146). ■

Proposition 2.6.149 (Regular Epimorphism is the Coequaliser of its Kernel Pair)
Let $e : X \to Y$ be a regular epimorphism with kernel pair (Z, p_1, p_2). Then e is the coequaliser of (p_1, p_2).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Since e is regular, by (2.6.158) there are morphisms $\gamma, \delta : W \rightrightarrows X$ such that e is the coequaliser of (γ, δ). Since $e \circ \gamma = e \circ \delta$, the universal property of the kernel pair (Z, p_1, p_2) gives a unique morphism $h : W \to Z$ with $p_1 \circ h = \gamma$ and $p_2 \circ h = \delta$.

By definition of the kernel pair, $e \circ p_1 = e \circ p_2$. Suppose $k : X \to V$ satisfies $k \circ p_1 = k \circ p_2$. Then

$$k \circ \gamma = k \circ p_1 \circ h = k \circ p_2 \circ h = k \circ \delta$$

so, as e is the coequaliser of (γ, δ), there is a unique morphism $l : Y \to V$ such that $l \circ e = k$. Since e is (in particular) an epimorphism (2.6.158), this l is the unique morphism satisfying $l \circ e = k$ among all morphisms $Y \to V$, in particular the unique one factoring k through e. So e satisfies the universal property of the coequaliser of (p_1, p_2). ■

2.6.19 Kernels and Cokernels

Proposition 2.6.150 (Kernel)
Let $\mathcal{C}$ be a pointed category (2.6.54). Suppose we have a sequence

$$\ker(\alpha) \xrightarrow{i} X \xrightarrow{\alpha} Y$$

then the following are equivalent

a) i is the equaliser of $(\alpha, 0)$
b) $\alpha \circ i = 0$ and $\alpha \circ i' = 0 \Longrightarrow \exists!\, e$ s.t. $i \circ e = i'$

In this case we say that $(i, \ker(\alpha))$ is the ***kernel*** *of α*

Lemma 2.6.151 (f monic $\Longrightarrow \ker(f) = 0$)
Let $\mathcal{C}$ be a pointed category and $f : X \to Y$ monic. Then $\ker(f) = 0$.

Dually f epic $\Longrightarrow \operatorname{coker}(f) = 0$.

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Proof. Evidentally $f \circ 0 = 0_{0,X}$ and for any object X' we have a unique morphism $0_{X',0} : X' \to 0$, so satisfies the uniqueness property. ∎

Remark 2.6.152
The converse does not hold, for we may define a monoid homomorphism $f : \mathbb{N} \to \mathbb{N}$ given by $f(n) := \min(n, 1)$. Then f is a homomorphism, $\ker(f) = \{0\}$ but f is not injective.

Proposition 2.6.153 (Cokernel)
Let $\mathcal{C}$ be a pointed category (2.6.54). Suppose we have a sequence

$$X \xrightarrow{\alpha} Y \xrightarrow{\pi} \operatorname{coker}(\alpha)$$

then the following are equivalent

a) π is the coequaliser of $(\alpha, 0)$,
b) $\pi \circ \alpha = 0$ and $\pi' \circ \alpha = 0 \implies \exists!\, h$ s.t. $h \circ \pi = \pi'$,

*In this case we say that $(\pi, \operatorname{coker}(\alpha))$ is the **cokernel** of f*

Lemma 2.6.154 (Kernels and Cokernels swap under $\mathcal{C}^{\mathrm{op}}$)
Let $\mathcal{C}$ be a pointed category and $\alpha : X \to Y$ a morphism, with kernel $\ker(\alpha) \xrightarrow{i} X$ and cokernel $Y \xrightarrow{\pi} \operatorname{coker}(\alpha)$. Regarding i and π as morphisms of $\mathcal{C}^{op}$,

$$(\ker(\alpha), i) \text{ is a cokernel of } \alpha^{op} \text{ in } \mathcal{C}^{op}, \quad (\operatorname{coker}(\alpha), \pi) \text{ is a kernel of } \alpha^{op} \text{ in } \mathcal{C}^{op}.$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The defining property of $(\ker(\alpha), i)$ (2.6.150) – namely $\alpha \circ i = 0$, and for every $i' : X' \to X$ with $\alpha \circ i' = 0$ there is a unique $e : X' \to \ker(\alpha)$ with $i \circ e = i'$ – is, upon reversing every arrow, exactly the defining property (2.6.153) of $(\ker(\alpha), i)$ as a cokernel of α^{op} in $\mathcal{C}^{\mathrm{op}}$. The dual statement follows similarly. ∎

Proposition 2.6.155 (Left Adjoints Preserve Cokernels)
Let $\mathcal{C}, \mathcal{D}$ be pointed categories and $\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$ a pair of adjoint functors.

Then F preserves cokernels and G preserves kernels.

Proof. By (2.6.75) we see that F and G preserve zero morphisms. Therefore we are done by (2.6.142) since a cokernel of f is just the coequaliser of $(f, 0)$. ∎

2.6.20 Normal, Regular and Extremal Morphisms

In a category which is not balanced, monic and epic are not always the "right" notions to characterise isomorphisms; the following refinements repair this.

Definition 2.6.156 (Extremal Epimorphism / Monomorphism)
*An epimorphism $e : X \to Y$ is **extremal** if whenever $e = m \circ f$ for some monomorphism m, then m is an isomorphism.*

*Dually, a monomorphism $m : X \to Y$ is **extremal** if whenever $m = f \circ e$ for some epimorphism e, then e is an isomorphism.*

Proposition 2.6.157
Let $f : X \to Y$ be a morphism. Then the following are equivalent

a) f is an isomorphism
b) f is a monomorphism and an extremal epimorphism
c) f is an extremal monomorphism and an epimorphism

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Being an **isomorphism** is a self-dual notion, and monic/epic swap under passing to $\mathcal{C}^{\mathrm{op}}$ (2.6.33), so condition c) is precisely condition b) applied in $\mathcal{C}^{\mathrm{op}}$. Hence it suffices to prove a) $\iff$ b); applying this to $\mathcal{C}^{\mathrm{op}}$ then gives a) $\iff$ c).

a) $\implies$ b) An isomorphism is trivially monic and epic; if f is an isomorphism and $f = m \circ g$ then by (2.6.37), m is an isomorphism, so f is an extremal epimorphism.

b) $\implies$ a) Suppose f is monic and an extremal epimorphism. The trivial factorisation $f = f \circ 1_X$ exhibits f as a monomorphism through which f factors, so by extremality f is an isomorphism. ∎

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Definition 2.6.158 (Regular Epimorphism / Monomorphism)
*An epimorphism $e : X \to Y$ is **regular** if there exist morphisms $\alpha, \beta : W \rightrightarrows X$ such that e is the coequaliser of (α, β).*

*Dually, a monomorphism $m : X \to Y$ is **regular** if there exist morphisms $\alpha, \beta : Y \rightrightarrows Z$ such that m is the equaliser of (α, β).*

Definition 2.6.159 (Normal Epimorphism / Monomorphism)
*Let $\mathcal{C}$ be a pointed category (2.6.54). An epimorphism $e : X \to Y$ is **conormal** if (e, Y) is the cokernel of some morphism $\alpha : W \to X$.*

*Dually, a monomorphism $m : X \to Y$ is **normal** if (m, X) is the kernel of some morphism $\alpha : Y \to Z$.*

Proposition 2.6.160 (Normal, Regular and Extremal Morphisms)
Let $\mathcal{C}$ be a pointed category. Then every normal monomorphism (resp. conormal epimorphism) is regular.

In a general category $\mathcal{C}$ every regular monomorphism (resp. epimorphism) is an extremal monomorphism (resp. epimorphism).

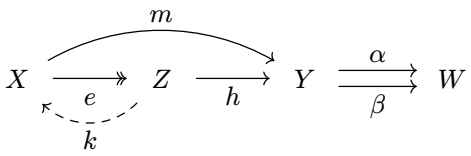
In summary

$$\textbf{normal}\ \ (\textbf{conormal}) \implies \textbf{regular} \implies \textbf{extremal} \implies \textbf{monic}\ (\textbf{epic})$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

(Normal $\implies$ regular) If e is a normal monomorphism, then by (2.6.150) e is the equaliser of $(\alpha, 0)$ for some α, so e is regular.

(Regular $\implies$ extremal) Suppose $m : X \to Y$ is the equaliser of $\alpha, \beta : Y \rightrightarrows W$, and $m = h \circ e$ for some epimorphism $e : X \to Z$. Since $\alpha \circ m = \beta \circ m$ we get $(\alpha \circ h) \circ e = (\beta \circ h) \circ e$, and as e is epic, $\alpha \circ h = \beta \circ h$. Hence h is equalised by (α, β), so by the universal property of the equaliser there is a unique $k : Z \to X$ with $m \circ k = h$.



Then $m \circ (k \circ e) = h \circ e = m = m \circ 1_X$, and since equalisers are monic (2.6.135) we may cancel m to get $k \circ e = 1_X$, so e is split-monic. As e is also epic, (2.6.36) gives that e is an isomorphism, and m is extremal.

Dually, every conormal epimorphism is regular and every regular epimorphism is extremal. ∎

For balanced categories we can reverse the implication

Proposition 2.6.161 (Balanced Categories via Extremal Morphisms)
Let $\mathcal{C}$ be a category. Then the following are equivalent

a) $\mathcal{C}$ is balanced
b) every monomorphism in $\mathcal{C}$ is extremal
c) every epimorphism in $\mathcal{C}$ is extremal

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Being **balanced** is a self-dual notion: monic and epic swap under passing to $\mathcal{C}^{\mathrm{op}}$ (2.6.33) while isomorphisms remain isomorphisms, so $\mathcal{C}$ is balanced iff $\mathcal{C}^{\mathrm{op}}$ is. Likewise, by the "Dually" clause of (2.6.156), a morphism is an extremal epimorphism in $\mathcal{C}$ iff it is an extremal monomorphism in $\mathcal{C}^{\mathrm{op}}$. So it suffices to prove a) $\iff$ b); applying this to $\mathcal{C}^{\mathrm{op}}$ then gives a) $\iff$ c).

a) $\implies$ b) Suppose $\mathcal{C}$ is balanced and $m : X \to Y$ is monic. If $m = f \circ e$ for some epimorphism $e : X \to W$, then e is also monic: if $eg_1 = eg_2$ for $g_1, g_2 : Z \to X$ then $f \circ e \circ g_1 = f \circ e \circ g_2$, that is $mg_1 = mg_2$, whence $g_1 = g_2$ as m is monic. As e is both monic and epic, balancedness gives that e is an isomorphism, so m is extremal.

b) $\implies$ a) Suppose every monomorphism is extremal and f is both monic and epic. Then f is an extremal monomorphism, so by (2.6.157) (with f extremal monic and epic) f is an isomorphism. So $\mathcal{C}$ is balanced. ∎

Lemma 2.6.162
Let $\mathcal{C}$ be a pointed category and $f : X \to Y$ a morphism for which $\ker(\mathrm{coker}(f))$ exists. Then the following are equivalent

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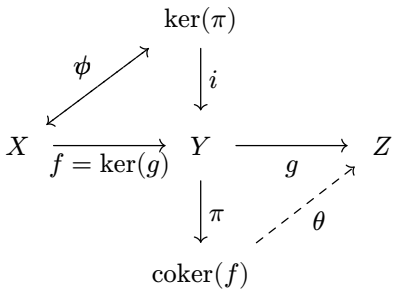
a) f *is normal*
b) $f = \ker(\operatorname{coker}(f))$

Dually, suppose $\operatorname{coker}(\ker(f))$ *exists then the following are equivalent*

a) f *is conormal*
b) $f = \operatorname{coker}(\ker(f))$

Proof. b) $\implies$ a) immediate.

a) $\implies$ b) By assumption $\pi : Y \to \operatorname{coker}(f)$ and $i : \ker(\pi) \to Y$ exists so we have a commutative diagram.



By definition of the cokernel $\pi \circ f = 0$ whence there is a unique $\phi : X \to \ker(\pi)$ such that $i \circ \phi = f$.

As $g \circ f = 0$ then by definition there is a unique $\theta : \operatorname{coker}(f) \to Z$ such that $\theta \circ \pi = g$.

Therefore $g \circ i = \theta \circ \pi \circ i = 0$. So by definition there is a unique $\psi : \ker(\pi) \to X$ such that $f \circ \psi = i$.

Then $i \circ \phi \circ \psi = f \circ \psi = i$ and since i is a monomorphism (2.6.135) $\phi \circ \psi = 1_{\ker(\pi)}$. Similarly $f \circ \psi \circ \phi = i \circ \phi = f$ and since $f = \ker(g)$ is a monomorphism (2.6.135) $\psi \circ \phi = 1_X$. Therefore X and $\ker(\pi)$ are isomorphic via ϕ and ψ, and since $i \circ \phi = f$, by (2.6.137) X is also a kernel for π as required.

Dually, $\mathcal{C}^{\mathrm{op}}$ is again pointed with the same zero object (2.6.54), monic/epic swap under passing to $\mathcal{C}^{\mathrm{op}}$ (2.6.33), and $\ker(f)$, $\operatorname{coker}(f)$ computed in $\mathcal{C}$ are precisely $\operatorname{coker}(f)$, $\ker(f)$ computed in $\mathcal{C}^{\mathrm{op}}$ (2.6.154). So applying the equivalence just proved to f regarded as a morphism of $\mathcal{C}^{\mathrm{op}}$ shows f is conormal iff $f = \operatorname{coker}(\ker(f))$. ∎

Definition 2.6.163
We say a pointed category $\mathcal{C}$ *is*

- ***normal*** *if every monomorphism is normal (i.e. a kernel of some morphism).*
- ***conormal*** *if every epimorphism is conormal (i.e. the cokernel of some morphism).*
- ***binormal*** *if it is both normal and conormal.*

Corollary 2.6.164
A normal (resp. conormal, binormal) category is balanced.

Example 2.6.165
We will see that the categories **AbGrp** *and* A**-Mod** *are binormal (and therefore every epi and mono is regular), whereas* **Grp** *is only conormal. All three are balanced.*

We also see may verify directly that in **Set** *every monomorphism and epimorphism is regular.*

Proposition 2.6.166
Let $\mathcal{C}$ *be a pointed category and* f *a morphism. Then the following are equivalent*

a) $\ker(f) = 0$
b) $f \circ g = 0 \implies g = 0$

Dually, the following are equivalent

a) $\operatorname{coker}(f) = 0$
b) $g \circ f = 0 \implies g = 0$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $f : X \to Y$ and $i : \ker(f) \to X$ for the kernel of f, so by (2.6.150), $f \circ i = 0$ and for every $i' : W \to X$ with $f \circ i' = 0$ there is a unique $e : W \to \ker(f)$ with $i \circ e = i'$.

a) $\implies$ b) Suppose $\ker(f) = 0$ and $f \circ g = 0$ for some $g : W \to X$. By the universal property there is a unique $e : W \to 0$ with $i \circ e = g$; as $i = 0_{0,X}$ (since $\ker(f) = 0$) and $e = 0_{W,0}$ (since 0 is terminal), $g = i \circ e = 0_{W,X} = 0$.

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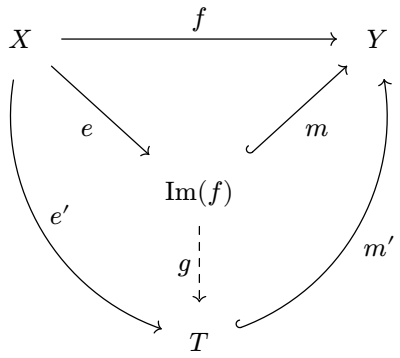
b) $\implies$ a) Let $\iota : 0 \to X$ be the canonical morphism. Since 0 is initial, $f \circ \iota = 0$ automatically. Given any $i' : W \to X$ with $f \circ i' = 0$, the hypothesis gives $i' = 0$, that is $i' = \iota \circ 0_{W,0}$, and $0_{W,0} : W \to 0$ is the unique such morphism since 0 is terminal. So $(0, \iota)$ satisfies the universal property of $\ker(f)$ (2.6.150), whence $\ker(f) = 0$.

Dually, $\mathcal{C}^{\mathrm{op}}$ is again pointed with the same zero object (2.6.54), and $\ker(f)$, $\operatorname{coker}(f)$ computed in $\mathcal{C}$ are precisely $\operatorname{coker}(f)$, $\ker(f)$ computed in $\mathcal{C}^{\mathrm{op}}$ (2.6.154), so applying the above to f regarded as a morphism of $\mathcal{C}^{\mathrm{op}}$ gives the dual equivalence. ■

2.6.21 Images and Coimages

In order to define exact sequence we need a categorical notion of image. We also define its dual.

Definition 2.6.167 (Image)
*Let $f : X \to Y$ be a morphism in $\mathcal{C}$. An **image** for f is a triple $(\mathrm{Im}(f), e, m)$ such that m is monic, we have a commutative triangle*



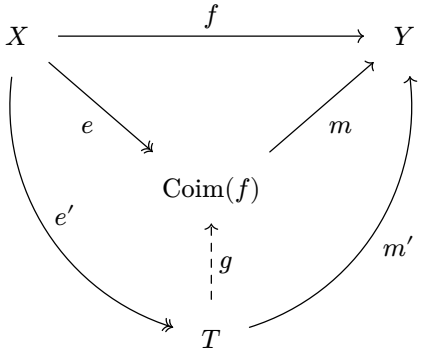
and for any other factorisation $f = m' \circ e'$ with m' monic there exists a unique morphism $g : \mathrm{Im}(f) \to T$ such that $m' \circ g = m$.

Note that any such g satisfies $g \circ e = e'$, since $m' \circ (g \circ e) = m \circ e = f = m' \circ e'$ and m' is monic.

The factorisations of f through a monomorphism constitute a category $\mathrm{SubObjFactors}(f)$, with objects the triples (T, e', m') as above and a morphism $(T, e', m') \to (T'', e'', m'')$ given by a morphism $g : T \to T''$ such that $m'' \circ g = m'$. So an image for f is precisely an initial object (2.6.48) of $\mathrm{SubObjFactors}(f)$, and is therefore unique up to unique isomorphism.

There is a dual notion

Definition 2.6.168 (Coimage)
*Let $f : X \to Y$ be a morphism in $\mathcal{C}$. A **coimage** for f is a triple $(\mathrm{Coim}(f), e, m)$ such that e is epic and we have a commutative triangle*



and for any other factorisation $f = m' \circ e'$ with e' epic there exists a unique morphism $g : T \to \mathrm{Coim}(f)$ such that $g \circ e' = e$.

Note that any such g satisfies $m \circ g = m'$, since $(m \circ g) \circ e' = m \circ e = f = m' \circ e'$ and e' is epic.

Dually, the factorisations of f through an epimorphism constitute a category $\mathrm{QuotObjFactors}(f)$, with objects the triples (T, e', m') as above and a morphism $(T, e', m') \to (T'', e'', m'')$ given by a morphism $g : T \to T''$ such that $g \circ e' = e''$. So a coimage for f is precisely a terminal object (2.6.50) of $\mathrm{QuotObjFactors}(f)$, and is therefore unique up to unique isomorphism.

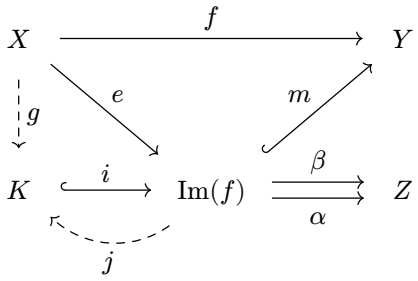
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Remark 2.6.169
We note that (I, e, m) is an image in $\mathcal{C}$ if and only if (I, m^{op}, e^{op}) is a coimage in $\mathcal{C}^{op}$.

Lemma 2.6.170
Let $f : X \to Y$ be a morphism and (I, e, m) an image. Suppose $\mathcal{C}$ has equalisers, then e is epic.

Dually if (I, e, m) is a coimage for $f : X \to Y$, and suppose $\mathcal{C}$ has coequalisers. Then m is monic.

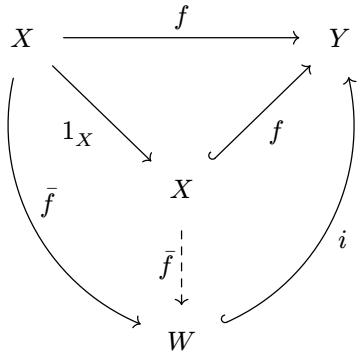
Proof. Consider the diagram



and suppose $\alpha \circ e = \beta \circ e$. Let $K = \text{coeq}(\alpha, \beta)$, then by definition there is a morphism $g : X \to K$ such that $i \circ g = e$. Then $(m \circ i) \circ g = f$ is a factorisation with $m \circ i$ a monomorphism (2.6.135). Then by the universal property of images there is a morphism $j : \text{Im}(f) \to K$ such that $\left(m_f \circ i\right) \circ j = m$. As m is monic $i \circ j = 1_{\text{Im}(f)}$ and so i is split-epic and therefore an isomorphism (2.6.36). ■

Lemma 2.6.171
Let $\mathcal{C}$ be a pointed category and $f : X \to Y$ a monomorphism. Then the triple $(X, 1_X, f)$ is an image for f.

Proof. Denote the morphism by f and suppose we have another factorisation $X \xrightarrow{\bar{f}} W \xrightarrow{i} Y$ with i monic, then we have a commutative diagram



and we may show by the argument in (2.6.167) that the factorisation is unique. ■

Lemma 2.6.172 (Images and Coimages are Preserved under Isomorphism)
Let (T, e, m) be an image of $f : X \to Y$ and $\theta : T \xrightarrow{\sim} T'$ an isomorphism. Then $(T', \theta \circ e, m \circ \theta^{-1})$ is also an image of f.

Dually, if (T, e, m) is a coimage of f and $\theta : T \xrightarrow{\sim} T'$ an isomorphism, then $(T', \theta \circ e, m \circ \theta^{-1})$ is also a coimage of f.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

For the image case θ determines a well-defined isomorphism in $\text{SubObjFactors}(f)$

$$(T, e, m) \xrightarrow{\sim} (T', \theta \circ e, m \circ \theta^{-1})$$

As isomorphisms preserve initial objects (2.6.49) we see that the right hand side is an image.

Dually, the coimage case follows by applying the image case to f^{op} in $\mathcal{C}^{\text{op}}$ (2.6.169). ■

2.6.22 Epi-Mono Factorisations

We saw that under mild conditions image and coimage yield an epi-mono factorisation. Strengthening this factorisation can ensure functorial properties; concretely, in an algebraic category, upgrading an epimorphism to a regular epimorphism is exactly upgrading from the (possibly too weak) categorical notion of epimorphism to the familiar notion of a surjective homomorphism (2.6.195).

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Definition 2.6.173
Let $\mathcal{C}$ be a category and $f : X \to Y$ a morphism. We say that $f = m \circ e$ is

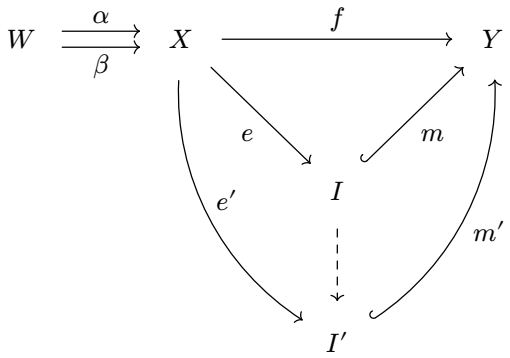
- *a **regular epi-mono factorisation** if m is a monomorphism and e is a regular epimorphism.*
- *an **epi-regular mono factorisation** if m is a regular monomorphism and e is an epimorphism*

We note that $f = m \circ e$ is a regular epi-mono (resp. epi-regular mono) factorisation if and only if $f^{op} = e^{op} \circ m^{op}$ is an epi-regular mono (resp. regular epi-mono) factorisation.

Proposition 2.6.174 (Regular Epi-Mono Factorisation)
Let $f : X \to Y$ be a morphism with regular epi-mono factorisation $f = m \circ e$. Then (I, e, m) is an image and every image is a regular epi-mono factorisation.

Dually, let $f : X \to Y$ be a morphism with epi-regular mono factorisation $f = m \circ e$. Then (I, e, m) is a coimage and every coimage is an epi-regular mono factorisation.

Proof. Consider the diagram



where e is a coequaliser for (α, β), m' is a monomorphism and e' is a general morphism. Then we need to prove the existence of a unique morphism $(I, e, m) \to (I', e', m')$. Since e coequalises (α, β), $e \circ \alpha = e \circ \beta$, so

$$f \circ \alpha = m \circ e \circ \alpha = m \circ e \circ \beta = f \circ \beta$$

and as $f = m' \circ e'$ with m' monic, $m' \circ e' \circ \alpha = f \circ \alpha = f \circ \beta = m' \circ e' \circ \beta$ gives $e' \circ \alpha = e' \circ \beta$. Therefore by definition of the coequaliser there is a unique ψ such that $\psi \circ e = e'$. Further

$$m' \circ \psi \circ e = m' \circ e' = f = m \circ e$$

and as e is an epimorphism (2.6.135) we conclude that $m' \circ \psi = m$, as required for a morphism $(I, e, m) \to (I', e', m')$ in $\mathrm{SubObjFactors}(f)$ (2.6.167). If ψ' also satisfies $m' \circ \psi' = m$, then $m' \circ \psi = m = m' \circ \psi'$, and since m' is monic $\psi = \psi'$; so ψ is unique.

Let (I', e', m') be an image, so m' is in particular monic and the above applies. As images are unique up to isomorphism (2.6.167) there is an isomorphism $\theta : I \xrightarrow{\sim} I'$ such that $m' \circ \theta = m$; then

$$m' \circ (\theta \circ e) = (m' \circ \theta) \circ e = m \circ e = f = m' \circ e'$$

and as m' is monic, $\theta \circ e = e'$. Since e is a coequaliser of (α, β) and θ is an isomorphism with $\theta \circ e = e'$, post-composing the coequalising cocone for e with θ shows e' is again a coequaliser of (α, β), as colimits are preserved under isomorphisms (2.6.90); hence e' is regular (2.6.158), as required. ∎

Proposition 2.6.175 (Regular Epi-Mono)
Set *has regular epi-mono factorisations.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $f : X \to Y$ be a morphism in **Set**. Set $\mathrm{Im}(f) \coloneqq \{f(x) \mid x \in X\}$ and let $m : \mathrm{Im}(f) \hookrightarrow Y$ denote the inclusion, which is injective and therefore monic by (2.6.38). Let $e : X \to \mathrm{Im}(f)$ be the corestriction of f, so that $f = m \circ e$.

Define $R = \{(x, x') \in X \times X \mid f(x) = f(x')\}$ with projections $\alpha, \beta : R \rightrightarrows X$, $\alpha(x, x') = x$, $\beta(x, x') = x'$. We claim e is the coequaliser of (α, β).

First $e \circ \alpha = e \circ \beta$: for $(x, x') \in R$ we have $f(x) = f(x')$ by definition of R, that is $e(x) = e(x')$.

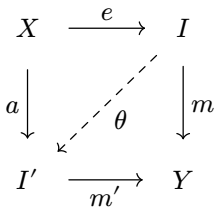
Now let $\pi' : X \to Z$ satisfy $\pi' \circ \alpha = \pi' \circ \beta$, that is $f(x) = f(x') \implies \pi'(x) = \pi'(x')$ for all $x, x' \in X$. Since every $y \in \mathrm{Im}(f)$ is of the form $f(x)$ for some $x \in X$, we may define $\psi : \mathrm{Im}(f) \to Z$ by $\psi(y) = \pi'(x)$ for any such x; this is well-

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defined precisely because $f(x) = f(x') = y$ implies $\pi'(x) = \pi'(x')$. By construction $\psi \circ e = \pi'$, and since e is surjective onto $\mathrm{Im}(f)$ by definition, any ψ' with $\psi' \circ e = \pi'$ agrees with ψ at every point of $\mathrm{Im}(f)$, so ψ is the unique such factorisation.

Hence e is a coequaliser of (α, β), so is a regular epimorphism (2.6.158). ∎

Lemma 2.6.176 (Diagonal Fill-In Lemma)
Let $e : X \to I$ be a regular epimorphism and $m' : I' \to Y$ a monomorphism. Then for every commutative square



there is a unique θ such that $\theta \circ e = a$. Furthermore it satisfies $m' \circ \theta = m$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Since e is a regular epimorphism, by (2.6.158) there are morphisms $\alpha, \beta : W \rightrightarrows X$ such that e is the coequaliser of (α, β).

We have $e \circ \alpha = e \circ \beta$, so $m \circ e \circ \alpha = m \circ e \circ \beta$, whence by commutativity of the square $m' \circ a \circ \alpha = m' \circ a \circ \beta$. As m' is monic we may cancel it to get $a \circ \alpha = a \circ \beta$.

By the universal property of the coequaliser e there is therefore a unique morphism $\theta : I \to I'$ such that

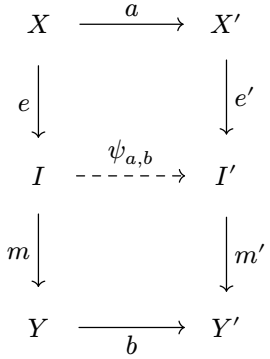
$$\theta \circ e = a$$

We check that θ also satisfies the remaining identity: $m' \circ \theta \circ e = m' \circ a = m \circ e$, and since e is an epimorphism (2.6.135) we may cancel it to get

$$m' \circ \theta = m$$

∎

Proposition 2.6.177 (Regular Epi-Mono Factorisation is Functorial)
Suppose $\mathcal{C}$ has regular epi-mono factorisations $f = m \circ e : X \to Y$, $f' = m' \circ e' : X' \to Y'$ and morphisms $a : X \to X'$, $b : Y \to Y'$ such that $b \circ f = f' \circ a$. Then there is a unique morphism $\psi_{a,b} : I \to I'$ which makes the top square commute



Furthermore the whole diagram commutes and the construction is functorial: $\psi_{1_X, 1_Y} = 1_I$ (taking $f' = f$, $a = 1_X$, $b = 1_Y$), and given a further regular epi-mono factorisation $f'' = m'' \circ e'' : X'' \to Y''$ and morphisms $c : X' \to X''$, $d : Y' \to Y''$ such that $d \circ f' = f'' \circ c$ (with induced comparison morphism $\psi_{c,d} : I' \to I''$), we have $\psi_{c,d} \circ \psi_{a,b} = \psi_{c \circ a, d \circ b}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Since e is a regular epimorphism, by (2.6.158) there are morphisms $\alpha, \beta : W \rightrightarrows X$ such that e is the coequaliser of (α, β).

We first show $e' \circ a$ coequalises (α, β). Since $e \circ \alpha = e \circ \beta$ we have $f \circ \alpha = m \circ e \circ \alpha = m \circ e \circ \beta = f \circ \beta$, and so by hypothesis $f' \circ a \circ \alpha = b \circ f \circ \alpha = b \circ f \circ \beta = f' \circ a \circ \beta$, that is $m' \circ e' \circ a \circ \alpha = m' \circ e' \circ a \circ \beta$. As m' is monic we may cancel it to get $(e' \circ a) \circ \alpha = (e' \circ a) \circ \beta$.

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By the universal property of the coequaliser e there is therefore a unique morphism $\psi_{a,b} : I \to I'$ such that

$$\psi_{a,b} \circ e = e' \circ a$$

We check that $\psi_{a,b}$ also satisfies the second required identity: $m' \circ \psi_{a,b} \circ e = m' \circ e' \circ a = f' \circ a = b \circ f = b \circ m \circ e$, and since e is an epimorphism (2.6.135) we may cancel it to get

$$m' \circ \psi_{a,b} = b \circ m$$

so the diagram commutes.

Taking $f' = f$, $a = 1_X$, $b = 1_Y$ (a valid compatible pair since $1_Y \circ f = f = f \circ 1_X$), 1_I satisfies $1_I \circ e = e = e \circ 1_X$ and $m \circ 1_I = m = 1_Y \circ m$, so by the uniqueness established above $\psi_{1_X,1_Y} = 1_I$.

For the composition law, let $c : X' \to X''$, $d : Y' \to Y''$ satisfy $d \circ f' = f'' \circ c$, with induced comparison morphism $\psi_{c,d} : I' \to I''$. Then

$$\left(\psi_{c,d} \circ \psi_{a,b}\right) \circ e = \psi_{c,d} \circ (e' \circ a) = \left(\psi_{c,d} \circ e'\right) \circ a = e'' \circ c \circ a = e'' \circ (c \circ a)$$

Note $d \circ b \circ f = d \circ f' \circ a = f'' \circ c \circ a = f'' \circ (c \circ a)$, so $(c \circ a, d \circ b)$ is again a compatible pair, with induced comparison morphism $\psi_{c\circ a, d\circ b} : I \to I''$ satisfying $\psi_{c\circ a, d\circ b} \circ e = e'' \circ (c \circ a)$ by definition. Since both $\psi_{c,d} \circ \psi_{a,b}$ and $\psi_{c\circ a, d\circ b}$ satisfy this same equation, and $\psi_{c\circ a, d\circ b}$ is, by the uniqueness established above, the unique morphism $I \to I''$ doing so, we conclude

$$\psi_{c,d} \circ \psi_{a,b} = \psi_{c\circ a, d\circ b}$$

■

Proposition 2.6.178 (Regular Epi-Mono Factorisation is Unique)
Let $f = m \circ e = m' \circ e'$ be two regular epi-mono factorisations, with $e : X \to I$, $e' : X \to I'$. Then there is a unique morphism $\theta : I \to I'$ such that $e' = \theta \circ e$. Furthermore it is an isomorphism and satisfies $m' \circ \theta = m$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Applying (2.6.177) to the factorisations (I, e, m) and (I', e', m') of f, with $f' = f$, $a = 1_X$, $b = 1_Y$, gives a unique morphism $\theta : I \to I'$ such that $\theta \circ e = e'$ and $m' \circ \theta = m$.

Symmetrically, applying (2.6.177) to the factorisations (I', e', m') and (I, e, m) of f (i.e. with the roles of the two factorisations swapped) gives a unique morphism $\theta' : I' \to I$ such that $\theta' \circ e' = e$ and $m \circ \theta' = m'$.

By the composition law of (2.6.177), applied to the chain of factorisations $(I, e, m) \to (I', e', m') \to (I, e, m)$ (i.e. $f' = f$, $f'' = f$, with $c = 1_X$, $d = 1_Y$, and θ, θ' as the two successive comparison morphisms), $\theta' \circ \theta$ is the comparison morphism from (I, e, m) to itself, and so equals 1_I by the identity law of (2.6.177).

Symmetrically $\theta \circ \theta' = 1_{I'}$, so θ is an isomorphism with inverse θ'. ■

2.6.23 Algebraic Categories

Drafted by Claude Sonnet, reviewed by David Rufino.

The categories of groups, rings and modules all share a common shape: their objects are sets equipped with some finitary operations, subject to some equational axioms. The following makes this precise, following the classical treatment of universal algebra [Bir35] [BS81] and its later categorical formulation [ARV10].

Definition 2.6.179 (Signature)
*A **signature** Ω is a set of **operation symbols**, each $\omega \in \Omega$ assigned a natural number n_ω, its **arity**.*

Definition 2.6.180 (Ω-algebra)
*Let Ω be a signature. An **Ω-algebra** is a set S together with, for each $\omega \in \Omega$, a function $\omega_S : S^{n_\omega} \to S$ (when $n_\omega = 0$ this is simply a distinguished element of S, since S^0 is a singleton).*

*A **homomorphism** of Ω-algebras $f : S \to T$ is a function such that*

$$f\left(\omega_S\left(s_1, ..., s_{n_\omega}\right)\right) = \omega_T\left(f(s_1), ..., f\left(s_{n_\omega}\right)\right) \quad \forall \omega \in \Omega, s_1, ..., s_{n_\omega} \in S$$

*This determines a concrete category Ω-**Alg** (2.6.9).*

Example 2.6.181
Taking $\Omega = \{\cdot\,(2), {}^{-1}(1), e(0)\}$ (writing the arity of each symbol in brackets), Ω-algebras are exactly sets equipped

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with a binary, a unary and a nullary operation; the subcategory of those satisfying the group axioms, (2.6.184) below, is the category of groups.

Definition 2.6.182 (Terms)
*Let Ω be a signature and X a set, whose elements we call **variables**. The set $T_\Omega(X)$ of **Ω-terms** over X is defined recursively by*

- *every $x \in X$ is a term;*
- *if $\omega \in \Omega$ and $t_1, ..., t_{n_\omega} \in T_\Omega(X)$ then $\omega\big(t_1, ..., t_{n_\omega}\big) \in T_\Omega(X)$.*

*Given an Ω-algebra S and a function $\theta : X \to S$, there is a unique **evaluation** $\overline{\theta} : T_\Omega(X) \to S$ such that $\overline{\theta}(x) = \theta(x)$ for $x \in X$ and*

$$\overline{\theta}\big(\omega\big(t_1, ..., t_{n_\omega}\big)\big) = \omega_S\big(\overline{\theta}(t_1), ..., \overline{\theta}\big(t_{n_\omega}\big)\big)$$

Remark 2.6.183 (The term algebra)
$T_\Omega(X)$ is itself an Ω-algebra, via $\omega_{T_\Omega(X)}\big(t_1, ..., t_{n_\omega}\big) := \omega\big(t_1, ..., t_{n_\omega}\big)$; with this structure, the defining recursion for $\overline{\theta}$ is exactly the condition that $\overline{\theta} : T_\Omega(X) \to S$ is a homomorphism, and it is the unique homomorphism $T_\Omega(X) \to S$ restricting to θ on X.

Consequently, for any homomorphism $h : S \to S'$ and $\theta : X \to S$, both $h \circ \overline{\theta}$ and $\overline{h \circ \theta}$ are homomorphisms $T_\Omega(X) \to S'$ restricting to $h \circ \theta$ on X, so by uniqueness

$$h \circ \overline{\theta} = \overline{h \circ \theta}$$

Definition 2.6.184 (Identity, Algebraic category)
*Let Ω be a signature. An **Ω-identity** is a pair $p, q \in T_\Omega(X)$ for some set of variables X, written $p = q$. An Ω-algebra S **satisfies** $p = q$ if $\overline{\theta}(p) = \overline{\theta}(q)$ for every $\theta : X \to S$ (2.6.182).*

*Given a set E of Ω-identities, the **algebraic category** determined by (Ω, E) – also called an **equational category** or **variety** in the literature – is the full subcategory of Ω-**Alg** of those Ω-algebras satisfying every identity in E, again concrete over* **Set**.

Example 2.6.185
Grp, **AbGrp**, **Rng** *are algebraic categories, taking E to consist of the usual group, abelian group and ring axioms respectively, each expressed as an identity. By contrast, the category of fields is* not *an algebraic category: the axiom that every nonzero element has an inverse cannot be expressed as an identity holding at every element, since it fails at 0.*

Example 2.6.186 (A-Mod as an algebraic category)
Fix a ring A. Take $\Omega = \{+, -, 0\} \cup \{\lambda_a\}_{a \in A}$, with $+$ binary, $-$ unary, 0 nullary and each λ_a unary; note Ω is infinite (one symbol per element of A) but every operation remains finitary. Take E to consist of the abelian group axioms for $(+, -, 0)$ together with, for all $a, a' \in A$,

$$\lambda_a(x + y) = \lambda_a(x) + \lambda_a(y), \quad \lambda_1(x) = x, \quad \lambda_{a+a'}(x) = \lambda_a(x) + \lambda_{a'}(x), \quad \lambda_{aa'}(x) = \lambda_a(\lambda_{a'}(x))$$

*Setting $\lambda_a(m) := a \cdot m$, these are exactly the A-module axioms, so A-**Mod** is the algebraic category determined by (Ω, E).*

Example 2.6.187 ($\mathbf{Alg}_k$ as an algebraic category)
*Fix a commutative ring A (so $\mathbf{Alg}_k$ is the case $A = k$ a field). Take $\Omega = \{+, \times, -, 0, 1\} \cup \{\lambda_a\}_{a \in A}$, with $+, \times$ binary, $-$ unary, $0, 1$ nullary and each λ_a unary, and take E to consist of the ring axioms for $(+, \times, -, 0, 1)$ (*not *imposing commutativity of $\times$) together with the four identities above and, in addition,*

$$\lambda_a(x) = \lambda_a(1) \times x \quad \forall a \in A$$

Given such an Ω-algebra B, set $i_B(a) := \lambda_a(1)$. Additivity and unitality of i_B are immediate from the identities above at $x = 1$, and multiplicativity follows from the last two identities,

$$i_B(aa') = \lambda_{aa'}(1) = \lambda_a(\lambda_{a'}(1)) = \lambda_a(1) \times \lambda_{a'}(1) = i_B(a) i_B(a')$$

so i_B is a ring homomorphism, and the final identity reads $\lambda_a(b) = i_B(a) \cdot b$ for all $b \in B$: this is exactly an A-algebra structure on B. Conversely every A-algebra satisfies these identities, and a ring homomorphism commutes with every λ_a if and only if it respects the structural morphisms (apply the final identity at $b = 1$), so $\mathbf{Alg}_A$, and in particular $\mathbf{Alg}_k$, is the algebraic category determined by (Ω, E).

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Remark 2.6.188
Birkhoff's theorem characterises the algebraic (equational) subcategories of Ω**-Alg** *as exactly those closed under homomorphic images, subalgebras and products [Bir35]; see [BS81] for a modern account, and [ARV10] for the categorical perspective via algebraic theories.*

Proposition 2.6.189 (Properties of Algebraic Categories)
Let $\mathcal{C}$ *be an algebraic category determined by* (Ω, E) *(2.6.184), with forgetful functor* $U : \mathcal{C} \to \mathbf{Set}$*. Then*

a) U *reflects isomorphisms*
b) U *creates limits and directed colimits*
c) U *has a left adjoint (i.e. free algebras exist)*

In particular U *also preserves limits and directed colimits by (2.6.87).*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Throughout, for $\omega \in \Omega$ we write $r = 1, ..., n_\omega$ for the arguments of ω, reserving i, j, k for diagram indices.

***U* creates limits.** Let $F : I \to \mathcal{C}$ be a diagram and suppose $L := \lim(U \circ F)$ exists in **Set**, with structure maps $\pi_i : L \to U(F(i))$; explicitly, by (2.6.85), $L = \left\{(x_i)_{i\in I} \in \prod_i U(F(i)) \;\middle|\; U(F(\alpha))(x_i) = x_j \quad \forall \alpha : i \to j\right\}$.

For $\omega \in \Omega$ and $x^{(1)}, ..., x^{(n_\omega)} \in L$, define $\omega_L\big(x^{(1)}, ..., x^{(n_\omega)}\big)_i := \omega_{F(i)}\Big(x_i^{(1)}, ..., x_i^{(n_\omega)}\Big)$ for each $i \in I$. This again lies in L: for $\alpha : i \to j$, since $F(\alpha)$ is a homomorphism, $F(\alpha)\Big(\omega_{F(i)}\Big(x_i^{(1)}, ...\Big)\Big) = \omega_{F(j)}\Big(F(\alpha)\Big(x_i^{(1)}\Big), ...\Big) = \omega_{F(j)}\Big(x_j^{(1)}, ...\Big)$, using that each $x^{(r)} \in L$. So L is an Ω-algebra and each π_i is, by construction, a homomorphism; this is the unique such structure on L, since $\omega_L(...)$ is pinned down coordinatewise by the requirement that every π_i be a homomorphism, and $L \subseteq \prod_i U(F(i))$.

Given $p = q \in E$ with variables in a set X, and $\theta : X \to L$: for each i, $\pi_i \circ \theta : X \to U(F(i))$, and by (2.6.183), $\pi_i\big(\overline{\theta}(t)\big) = \overline{\pi_i \circ \theta}(t)$ for every term t. As $F(i) \in \mathcal{C}$ satisfies $p = q$, $\overline{\pi_i\theta}(p) = \overline{\pi_i\theta}(q)$, so $\pi_i\big(\overline{\theta}(p)\big) = \pi_i\big(\overline{\theta}(q)\big)$ for every $i \in I$, hence $\overline{\theta}(p) = \overline{\theta}(q)$ as elements of $L \subseteq \prod_i U(F(i))$. So $L \in \mathcal{C}$.

Finally $(L, \{\pi_i\})$ is a limit for F in $\mathcal{C}$: given a cone $(M, \{\mu_i\})$ in $\mathcal{C}$, the unique **Set**-map $\varphi : U(M) \to L$ with $\pi_i \circ \varphi = \mu_i$ (afforded by $L = \lim(U \circ F)$) is a homomorphism, since for $\omega \in \Omega$, $m^{(1)}, ..., m^{(n_\omega)} \in U(M)$ and every i,

$$\pi_i\big(\varphi\big(\omega_M\big(m^{(1)}, ...\big)\big)\big) = \mu_i\big(\omega_M\big(m^{(1)}, ...\big)\big) = \omega_{F(i)}\big(\mu_i\big(m^{(1)}\big), ...\big) = \omega_{F(i)}\big(\pi_i\varphi\big(m^{(1)}\big), ...\big) = \pi_i\big(\omega_L\big(\varphi\big(m^{(1)}\big), ...\big)\big)$$

using that μ_i is a homomorphism, so $\varphi\big(\omega_M\big(m^{(1)}, ...\big)\big) = \omega_L\big(\varphi\big(m^{(1)}\big), ...\big)$. As $\mathcal{C}$ is a full subcategory of Ω**-Alg**, φ is the unique morphism of $\mathcal{C}$ with $\pi_i \circ \varphi = \mu_i$, so $(L, \{\pi_i\})$ is a limit for F in $\mathcal{C}$ with $U(L) = L$, as required.

***U* creates directed colimits.** Let I be a directed preorder and $F : I \to \mathcal{C}$ a diagram, and suppose $L := \varinjlim(U \circ F)$ exists in **Set** (2.6.128), with structure maps $\mu_i : U(F(i)) \to L$.

For $\omega \in \Omega$ and $\ell^{(1)}, ..., \ell^{(n_\omega)} \in L$: as I is directed and n_ω is finite, there is a common $i \in I$ and $y^{(1)}, ..., y^{(n_\omega)} \in U(F(i))$ with $\mu_i\big(y^{(r)}\big) = \ell^{(r)}$ for all r (each $\ell^{(r)}$ is represented at some stage, and finitely many stages admit a common upper bound); define $\omega_L\big(\ell^{(1)}, ..., \ell^{(n_\omega)}\big) := \mu_i\big(\omega_{F(i)}\big(y^{(1)}, ..., y^{(n_\omega)}\big)\big)$. This is independent of the choice of i and the $y^{(r)}$: given another such i', $\big(y'^{(r)}\big)$, for each r there is, by (2.6.128), a stage $k_r \succeq i, i'$ with $F(i \to k_r)\big(y^{(r)}\big) = F(i' \to k_r)\big(y'^{(r)}\big)$; as I is directed and there are finitely many r, there is a common $j \succeq k_1, ..., k_{n_\omega}$, and applying $F(k_r \to j)$ to each equality gives $F(i \to j)\big(y^{(r)}\big) = F(i' \to j)\big(y'^{(r)}\big)$ for every r simultaneously. As $F(i \to j), F(i' \to j)$ are homomorphisms,

$$\mu_i\big(\omega_{F(i)}\big(y^{(1)}, ...\big)\big) = \mu_j\big(\omega_{F(j)}\big(F(i \to j)\big(y^{(1)}\big), ...\big)\big) = \mu_j\big(\omega_{F(j)}\big(F(i' \to j)\big(y'^{(1)}\big), ...\big)\big) = \mu_{i'}\big(\omega_{F(i')}\big(y'^{(1)}, ...\big)\big)$$

So L is an Ω-algebra and each μ_i is, by construction, a homomorphism; as the μ_i are jointly surjective (2.6.128), this is again the unique such structure.

Given $p = q \in E$ with (finitely many) variables X, and $\theta : X \to L$: as X is finite and I directed, there is $i \in I$ and $\theta_i : X \to U(F(i))$ with $\theta = \mu_i \circ \theta_i$ (as above); by (2.6.183), $\overline{\theta} = \mu_i \circ \overline{\theta_i}$, so $\overline{\theta}(p) = \mu_i\big(\overline{\theta_i}(p)\big) = \mu_i\big(\overline{\theta_i}(q)\big) = \overline{\theta}(q)$, using $F(i) \in \mathcal{C}$. So $L \in \mathcal{C}$.

Finally $(L, \{\mu_i\})$ is a colimit for F in $\mathcal{C}$: given a cocone $(M, \{\nu_i\})$ in $\mathcal{C}$, the unique **Set**-map $\psi : L \to U(M)$ with $\psi \circ \mu_i = \nu_i$ is a homomorphism (for $\ell^{(r)} = \mu_i\big(y^{(r)}\big)$ at a common i, $\psi\big(\omega_L\big(\ell^{(1)}, ...\big)\big) = \psi\big(\mu_i\big(\omega_{F(i)}\big(y^{(1)}, ...\big)\big)\big) = \nu_i\big(\omega_{F(i)}\big(y^{(1)}, ...\big)\big) = \omega_M\big(\nu_i\big(y^{(1)}\big), ...\big) = \omega_M\big(\psi\big(\ell^{(1)}\big), ...\big)$, using that ν_i is a homomorphism), and is the unique morphism of $\mathcal{C}$ with this property as $\mathcal{C}$ is full in Ω**-Alg**.

***U* reflects isomorphisms.** Let $f : X \to Y$ in $\mathcal{C}$ with $U(f)$ bijective. For $\omega \in \Omega$ and $y^{(1)}, ..., y^{(n_\omega)} \in U(Y)$, writing $x^{(r)} := U(f)^{-1}\big(y^{(r)}\big)$, since f is a homomorphism $f\big(\omega_X\big(x^{(1)}, ...\big)\big) = \omega_Y\big(f\big(x^{(1)}\big), ...\big) = \omega_Y\big(y^{(1)}, ...\big)$, so applying $U(f)^{-1}$,

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$$U(f)^{-1}\big(\omega_Y(y^{(1)},\dots,y^{(n_\omega)})\big)=\omega_X\big(U(f)^{-1}(y^{(1)}),\dots,U(f)^{-1}(y^{(n_\omega)})\big)$$

so $U(f)^{-1}$ is a homomorphism $Y\to X$, hence a morphism of $\mathcal{C}$ (full subcategory), inverse to f on underlying sets; as U is faithful, it is a two-sided inverse to f in $\mathcal{C}$, so f is an isomorphism.

***U* has a left adjoint.** Fix a set X. Define a relation $\sim$ on $T_\Omega(X)$ by $t\sim t'$ if $\overline{\theta}(t)=\overline{\theta}(t')$ for every $A\in\mathcal{C}$ and $\theta: X\to U(A)$; this is manifestly an equivalence relation, and a congruence: if $t^{(r)}\sim t'^{(r)}$ for $r=1,\dots,n_\omega$, then for every A,θ, $\overline{\theta}\big(\omega(t^{(1)},\dots)\big)=\omega_A\big(\overline{\theta}(t^{(1)}),\dots\big)=\omega_A\big(\overline{\theta}(t'^{(1)}),\dots\big)=\overline{\theta}\big(\omega(t'^{(1)},\dots)\big)$, so $\omega\big(t^{(1)},\dots,t^{(n_\omega)}\big)\sim\omega\big(t'^{(1)},\dots,t'^{(n_\omega)}\big)$.

Let $\mathrm{Free}(X):=T_\Omega(X)/\sim$, with $\omega_{\mathrm{Free}(X)}\big([t^{(1)}],\dots,[t^{(n_\omega)}]\big):=\big[\omega(t^{(1)},\dots,t^{(n_\omega)})\big]$ (well-defined as $\sim$ is a congruence), and let $\pi: T_\Omega(X)\to\mathrm{Free}(X)$, $t\mapsto[t]$, an (evidently surjective) homomorphism. Write $\iota: X\to U(\mathrm{Free}(X))$, $\iota(x):=[x]$.

$\mathrm{Free}(X)\in\mathcal{C}$: let $p=q\in E$ with variable set Y, and $\psi: Y\to U(\mathrm{Free}(X))$; choose $\tilde{\psi}: Y\to T_\Omega(X)$ with $\pi\circ\tilde{\psi}=\psi$ (possible as π is surjective). Viewing $T_\Omega(X)$ as an Ω-algebra (2.6.183), $\overline{\tilde{\psi}}: T_\Omega(Y)\to T_\Omega(X)$ is a homomorphism, and for every $A\in\mathcal{C}$, $\theta: X\to U(A)$, (2.6.183) gives $\overline{\theta}\circ\overline{\tilde{\psi}}=\overline{\overline{\theta}\circ\tilde{\psi}}$; as $\overline{\theta}\circ\tilde{\psi}: Y\to U(A)$ and $A\vDash p=q$, $\overline{\overline{\theta}\tilde{\psi}}(p)=\overline{\overline{\theta}\tilde{\psi}}(q)$, so $\overline{\theta}\Big(\overline{\tilde{\psi}}(p)\Big)=\overline{\theta}\Big(\overline{\tilde{\psi}}(q)\Big)$. As A,θ were arbitrary, $\overline{\tilde{\psi}}(p)\sim\overline{\tilde{\psi}}(q)$, that is $\pi\Big(\overline{\tilde{\psi}}(p)\Big)=\pi\Big(\overline{\tilde{\psi}}(q)\Big)$. Again by (2.6.183), $\pi\circ\overline{\tilde{\psi}}=\overline{\psi}$ (both extend $\pi\circ\tilde{\psi}=\psi$ on Y), so $\overline{\psi}(p)=\overline{\psi}(q)$. As ψ was arbitrary, $\mathrm{Free}(X)\vDash p=q$; as $p=q\in E$ was arbitrary, $\mathrm{Free}(X)\in\mathcal{C}$.

Universal property: given $A\in\mathcal{C}$ and $\theta: X\to U(A)$, define $\hat{\theta}([t]):=\overline{\theta}(t)$; this is well-defined ($t\sim t'\Longrightarrow\overline{\theta}(t)=\overline{\theta}(t')$, by definition of $\sim$) and a homomorphism, since $\hat{\theta}\big(\omega_{\mathrm{Free}(X)}\big([t^{(1)}],\dots\big)\big)=\hat{\theta}\big([\omega(t^{(1)},\dots)]\big)=\overline{\theta}\big(\omega(t^{(1)},\dots)\big)=\omega_A\big(\overline{\theta}(t^{(1)}),\dots\big)=\omega_A\big(\hat{\theta}([t^{(1)}]),\dots\big)$, and satisfies $U\big(\hat{\theta}\big)\circ\iota=\theta$. If $\varphi:\mathrm{Free}(X)\to A$ is any morphism of $\mathcal{C}$ with $U(\varphi)\circ\iota=\theta$, then $\varphi\circ\pi$ is a homomorphism $T_\Omega(X)\to A$ restricting to θ on X, so $\varphi\circ\pi=\overline{\theta}$ by (2.6.183); as π is surjective, $\varphi=\hat{\theta}$. So $\mathrm{Free}(X)$ is free on X, and $X\mapsto\mathrm{Free}(X)$ is left adjoint to U by (2.6.74). ∎

Remark 2.6.190 (Limits are computed pointwise)
Unwinding the construction in the proof of (2.6.189): for a diagram $F: I\to\mathcal{C}$ admitting a limit, $U(\lim F)=\lim(U\circ F)$ (2.6.85), and each $\omega\in\Omega$ acts on $\lim F$ coordinatewise, $\omega_{\lim F}\big(x^{(1)},\dots\big)_i=\omega_{F(i)}\Big(x_i^{(1)},\dots\Big)$ for $i\in I$. In particular this applies to products.

Corollary 2.6.191 (Inverse Limits in Algebraic Categories)
Let $\mathcal{C}$ be an algebraic category determined by (Ω,E) (2.6.184), with forgetful functor $U:\mathcal{C}\to\mathbf{Set}$, I a directed preorder, and $F: I^{op}\to\mathcal{C}$ an inverse system. Then $\varprojlim_i F(i)$ exists in $\mathcal{C}$, with

$$U\Big(\varprojlim_i F(i)\Big)=\varprojlim_i U(F(i))=\left\{(x_i)_{i\in I}\in\prod_i U(F(i))\;\middle|\;U\big(\rho_{ij}\big)(x_j)=x_i\quad\forall i\prec j\right\}$$

(2.6.130), and each $\omega\in\Omega$ acts coordinatewise: $\omega_{\varprojlim_i F(i)}\big(x^{(1)},\dots\big)_i=\omega_{F(i)}\Big(x_i^{(1)},\dots\Big)$ for every $i\in I$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (2.6.189), U creates limits, and $\lim_{I^{op}}(U\circ F)=\varprojlim_i U(F(i))$ exists in **Set** for every diagram, being computed as in (2.6.130) (a special case of (2.6.85)); so $\varprojlim_i F(i):=\lim_{I^{op}}F$ exists in $\mathcal{C}$ too, with U carrying it to this set-level description. The coordinatewise formula for $\omega\in\Omega$ is then the case $\lim F:=\varprojlim_i F(i)$ of (2.6.190). ∎

Definition 2.6.192 (Subalgebra)
*Let A be an Ω-algebra. A subset $S\subseteq U(A)$ is a **subalgebra** if $\omega_A\Big(s_1,\dots,s_{n_\omega}\Big)\in S$ for all $\omega\in\Omega$ and $s_1,\dots,s_{n_\omega}\in S$.*

Proposition 2.6.193 (Subalgebra criterion)
Let A be an Ω-algebra and $S\subseteq U(A)$ a subalgebra (2.6.192). Then S carries a unique Ω-algebra structure making the inclusion $\iota: S\hookrightarrow A$ a homomorphism, namely $\omega_S:=\omega_A\,|_{S^{n_\omega}}$.

If moreover A satisfies an identity $p=q$, then so does S; in particular, if A lies in the algebraic category $\mathcal{C}$ determined by (Ω,E), then so does S.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Existence and uniqueness of ω_S are immediate: S being closed under ω_A means $\omega_A\,|_{S^{n_\omega}}$ takes values in S, and this is forced by the requirement that ι be a homomorphism.

Let $p=q$ be an identity with variables Y, satisfied by A, and $\theta: Y\to S$. Writing $\iota\circ\theta: Y\to U(A)$, (2.6.183) gives $\iota\Big(\overline{\theta}(p)\Big)=\overline{\iota\theta}(p)=\overline{\iota\theta}(q)=\iota\Big(\overline{\theta}(q)\Big)$, using that A satisfies $p=q$; as ι is injective, $\overline{\theta}(p)=\overline{\theta}(q)$, so S satisfies $p=q$. ∎

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Proposition 2.6.194 (Injective $\iff$ Monic)
Let $\mathcal{C}$ be an algebraic category. Then a morphism $f : X \to Y$ is a monomorphism if and only if $U(f)$ is injective.

Proof. If $U(f)$ is injective and $f \circ g = f \circ h$ then $U(f) \circ U(g) = U(f) \circ U(h)$ whence $U(g) = U(h)$ and by faithful-ness $g = h$ as required.

Conversely if $f : X \to Y$ is monic, consider the left adjoint F to U (2.6.189), then by definition we have a natural bijection

$$\Psi_X : \mathrm{Mor}(F(\{\star\}), X) \xrightarrow{\sim} U(X)$$

Suppose $U(f)(x) = U(f)(y)$ for $x, y \in U(X)$. Let g, h be morphisms corresponding to x and y under this bijection, that is $x = \Psi_X(g)$ and $y = \Psi_X(h)$.

Then by naturality of Ψ

$$U(f)(x) = \Psi_Y(f \circ g)$$

and

$$U(f)(y) = \Psi_Y(f \circ h)$$

so by assumption $f \circ g = f \circ h$. As f is monic we have $g = h$ whence $x = y$. Therefore $U(f)$ is injective. ∎

Proposition 2.6.195 (Regular Epimorphism $\iff$ Surjective)
Let $\mathcal{C}$ be an algebraic category (2.6.184) and $e : X \to Y$ a morphism of $\mathcal{C}$. Then e is a regular epimorphism if and only if $U(e)$ is surjective.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

($\Longrightarrow$) Since e is regular, there are $\alpha, \beta : W \rightrightarrows X$ such that e is the coequaliser of (α, β) (2.6.158).

The image is a subalgebra. Let $I \coloneqq U(e)(U(X)) \subseteq U(Y)$. For $\omega \in \Omega$ and $y_1, ..., y_{n_\omega} \in I$, writing $y_r = U(e)(x_r)$ for $x_r \in U(X)$, since e is a homomorphism $\omega_Y(y_1, ...) = \omega_Y(U(e)(x_1), ...) = U(e)(\omega_X(x_1, ...)) \in I$; so I is a subalgebra (2.6.192). By (2.6.193), I carries an Ω-algebra structure making the inclusion $\iota : I \hookrightarrow Y$ a homomorphism, and $I \in \mathcal{C}$ (as $Y \in \mathcal{C}$); as $\mathcal{C}$ is full in Ω-**Alg**, ι is a morphism of $\mathcal{C}$.

Write $e' : X \to I$ for the corestriction of e, that is $U(e') \coloneqq U(e)$ regarded as a map into I; since ω_I is just ω_Y restricted to I, e' is again a homomorphism, hence a morphism of $\mathcal{C}$, with $\iota \circ e' = e$ and $U(e')$ surjective onto I by construction.

(I, e') is a coequaliser of (α, β). Since $\iota \circ (e'\alpha) = e\alpha = e\beta = \iota \circ (e'\beta)$ and ι is monic (2.6.194), $e'\alpha = e'\beta$. Given $h : X \to Z$ with $h\alpha = h\beta$, the universal property of e gives a unique $k : Y \to Z$ with $k \circ e = h$; then $k' \coloneqq k \circ \iota$ satisfies $k' \circ e' = k \circ \iota \circ e' = k \circ e = h$. If k'' also satisfies $k'' \circ e' = h$, then $U(k') \circ U(e') = U(h) = U(k'') \circ U(e')$; as $U(e')$ is surjective, $U(k') = U(k'')$, so $k' = k''$ (U faithful). So (I, e') is a coequaliser of (α, β) (2.6.140), in particular e' is epic (2.6.135).

ι is an isomorphism. Applying the universal property of e to e' (which satisfies $e'\alpha = e'\beta$) gives a unique $\psi : Y \to I$ with $\psi \circ e = e'$. Then $(\iota \circ \psi) \circ e = \iota \circ e' = e = 1_Y \circ e$, and as e is epic (2.6.135), $\iota \circ \psi = 1_Y$. Also $(\psi \circ \iota) \circ e' = \psi \circ e = e' = 1_I \circ e'$, and as e' is epic, $\psi \circ \iota = 1_I$. So ι is an isomorphism, with inverse ψ.

So $U(\iota)$ is bijective; but $U(\iota)$ is, by construction, the inclusion $I \hookrightarrow U(Y)$, so $I = U(Y)$. As $I = U(e)(U(X))$ by definition, $U(e)$ is surjective.

($\Longleftarrow$) Suppose $U(e)$ is surjective.

Construction of the quotient. Write $\sim$ for the relation $a \sim b :\iff U(e)(a) = U(e)(b)$ on $U(X)$, an equivalence relation, and $Q \coloneqq U(X)/\sim$ with quotient map $\pi : U(X) \to Q$, $a \mapsto [a]$; π is surjective by construction. For $\omega \in \Omega$ and $a_1, ..., a_{n_\omega}, b_1, ..., b_{n_\omega} \in U(X)$ with $a_r \sim b_r$ for every r: since e is a homomorphism, $U(e)(\omega_X(a_1, ...)) = \omega_Y(U(e)(a_1), ...) = \omega_Y(U(e)(b_1), ...) = U(e)(\omega_X(b_1, ...))$, that is $\omega_X(a_1, ...) \sim \omega_X(b_1, ...)$; so $\sim$ is a congruence, and $\omega_Q\big([a_1], ..., [a_{n_\omega}]\big) \coloneqq \big[\omega_X\big(a_1, ..., a_{n_\omega}\big)\big]$ is well-defined, making Q into an Ω-algebra with π a homomorphism.

$Q \in \mathcal{C}$. Let $p = q \in E$ with variables W, and $\theta : W \to Q$; as π is surjective, choose $\tilde{\theta} : W \to U(X)$ with $\pi \circ \tilde{\theta} = \theta$. Since $X \in \mathcal{C}$, X satisfies $p = q$, that is $\overline{\tilde{\theta}}(p) = \overline{\tilde{\theta}}(q)$ in $U(X)$; applying the homomorphism π and (2.6.183) ($\pi \circ \overline{\tilde{\theta}} = \overline{\pi \circ \tilde{\theta}} = \overline{\theta}$),

$$\overline{\theta}(p) = \pi\Big(\overline{\tilde{\theta}}(p)\Big) = \pi\Big(\overline{\tilde{\theta}}(q)\Big) = \overline{\theta}(q)$$

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As θ was arbitrary, Q satisfies $p = q$; as $p = q \in E$ was arbitrary, $Q \in \mathcal{C}$, and $\pi : X \to Q$ is a morphism of $\mathcal{C}$ (full subcategory of Ω-**Alg**).

The kernel relation is a subalgebra. Let $R \coloneqq \{(a,b) \in U(X)^2 \mid a \sim b\}$. For $\omega \in \Omega$ and $(a_1,b_1),...,\left(a_{n_\omega},b_{n_\omega}\right) \in R$, since e is a homomorphism, $U(e)(\omega_X(a_1,...)) = \omega_Y(U(e)(a_1),...) = \omega_Y(U(e)(b_1),...) = U(e)(\omega_X(b_1,...))$, so $(\omega_X(a_1,...),\omega_X(b_1,...)) \in R$; as $U(X \times X) = U(X)^2$ (2.6.190), R is a subalgebra of $X \times X$ (2.6.192). By (2.6.193), $R \in \mathcal{C}$, and $\alpha \coloneqq \pi_1 \mid_R$, $\beta \coloneqq \pi_2 \mid_R$ (restrictions of the product projections) are morphisms of $\mathcal{C}$.

(Q,π) is a coequaliser of (α,β). For $(a,b) \in R$, i.e. $a \sim b$, $\pi(\alpha(a,b)) = \pi(a) = [a] = [b] = \pi(\beta(a,b))$; as U is faithful, $\pi \circ \alpha = \pi \circ \beta$. Suppose $h : X \to Z$ satisfies $h \circ \alpha = h \circ \beta$; then for every $(a,b) \in R$ (i.e. every $a \sim b$), $U(h)(a) = U(h)(b)$, so $\overline{h} : Q \to U(Z)$, $[a] \mapsto U(h)(a)$, is a well-defined map, and a homomorphism, since $\overline{h}(\omega_Q([a_1],...)) = \overline{h}([\omega_X(a_1,...)]) = U(h)(\omega_X(a_1,...)) = \omega_Z(U(h)(a_1),...) = \omega_Z\left(\overline{h}([a_1]),...\right)$, using that h is a homomorphism. So $\overline{h}$ is a morphism of $\mathcal{C}$ with $\overline{h} \circ \pi = h$ (U faithful). If $k : Q \to Z$ also satisfies $k \circ \pi = h$, then $U(k) \circ U(\pi) = U(h) = U\left(\overline{h}\right) \circ U(\pi)$; as $U(\pi)$ is surjective, $U(k) = U\left(\overline{h}\right)$, so $k = \overline{h}$ (U faithful). So (Q,π) is a coequaliser of (α,β) (2.6.140), in particular a regular epimorphism.

Conclusion. Define $\overline{e} : Q \to Y$, $[a] \mapsto U(e)(a)$; well-defined as $a \sim b \implies U(e)(a) = U(e)(b)$, and a homomorphism by the same computation as for π, hence a morphism of $\mathcal{C}$, with $\overline{e} \circ \pi = e$. As $\overline{e}([a]) = \overline{e}([b]) \implies U(e)(a) = U(e)(b) \implies a \sim b \implies [a] = [b]$, $U(\overline{e})$ is injective; as $e = \overline{e} \circ \pi$ and $U(e)$ is surjective (hypothesis), $U(\overline{e})$ is surjective. So $U(\overline{e})$ is bijective, hence $\overline{e}$ is an isomorphism, as $\mathcal{C}$ is algebraic and U reflects isomorphisms (2.6.189).

Since π is the coequaliser of (α,β) and $\overline{e}$ is an isomorphism, $e = \overline{e} \circ \pi$ is again a coequaliser of (α,β), as colimits are preserved under isomorphism (2.6.90); hence e is a regular epimorphism (2.6.158). ■

Proposition 2.6.196 (Algebraic Categories have Regular Epi-Mono Factorisations)
Let $\mathcal{C}$ be an algebraic category (2.6.184). Then $\mathcal{C}$ has regular epi-mono factorisations for every morphism (2.6.173).

More precisely, let $f : X \to Y$ be a morphism of $\mathcal{C}$ and $I \coloneqq U(f)(U(X)) \subseteq U(Y)$ the (underlying-set) image of f. Then I is a subalgebra of Y, and $f = m \circ e$ is a regular epi-mono factorisation of f, where $e : X \to I$ is the corestriction of f and $m : I \hookrightarrow Y$ is the inclusion.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

***I* is a subalgebra.** For $\omega \in \Omega$ and $y_1,...,y_{n_\omega} \in I$, writing $y_r = U(f)(x_r)$ for $x_r \in U(X)$, since f is a homomorphism $\omega_Y(y_1,...) = \omega_Y(U(f)(x_1),...) = U(f)(\omega_X(x_1,...)) \in I$; so I is a subalgebra (2.6.192). By (2.6.193), I carries a (unique) Ω-algebra structure making the inclusion $m : I \hookrightarrow Y$ a homomorphism, with $I \in \mathcal{C}$ (as $Y \in \mathcal{C}$); as $\mathcal{C}$ is full in Ω-**Alg**, m is a morphism of $\mathcal{C}$, and monic by (2.6.194) (being injective).

Write $e : X \to I$ for the corestriction of f, that is $U(e) \coloneqq U(f)$ regarded as a map into I; since ω_I is just ω_Y restricted to I, e is again a homomorphism, hence a morphism of $\mathcal{C}$, with $m \circ e = f$ and $U(e)$ surjective onto I by construction. By (2.6.195), e is therefore a regular epimorphism, so $f = m \circ e$ is a regular epi-mono factorisation of f (2.6.173). ■

Proposition 2.6.197 (Equalisers in Algebraic Categories)
Consider a commutative diagram in an algebraic category $\mathcal{C}$

$$X \xrightarrow{\ e\ } Y \underset{d_1}{\overset{d_0}{\rightrightarrows}} Z$$

Then e is an equaliser for (d_0,d_1) if and only if $U(e)$ is injective and $\operatorname{Im}(U(e)) = \{y \in U(Y) \mid U(d_0)(y) = U(d_1)(y)\}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $F : \mathbf{2} \to \mathcal{C}$ for the diagram $d_0, d_1 : Y \rightrightarrows Z$, so that e being an equaliser for (d_0,d_1) means precisely that (X,e) is a limit for F.

($\implies$) Suppose (X,e) is a limit for F. Since $\mathcal{C}$ is algebraic, U creates limits (2.6.189), so by (2.6.87), using that **Set** has all limits, U preserves limits. Hence $(U(X),U(e))$ is a limit for $U \circ F$ in **Set**, that is, an equaliser for $(U(d_0),U(d_1))$. By (2.6.139), $U(e)$ is injective and $\operatorname{Im}(U(e)) = \{y \in U(Y) \mid U(d_0)(y) = U(d_1)(y)\}$, as required.

($\impliedby$) Suppose $U(e)$ is injective and $\operatorname{Im}(U(e)) = \{y \in U(Y) \mid U(d_0)(y) = U(d_1)(y)\}$.

By (2.6.139) the diagram $U(Y) \rightrightarrows U(Z)$ has an equaliser in **Set**, that is $\lim(U \circ F)$ exists. Since U creates limits (2.6.189), $\lim F$ exists in $\mathcal{C}$; write (X',e') for this limit, so that e' is an equaliser for (d_0,d_1). By the ($\implies$) direction applied to e', $U(e')$ is injective and

$$\operatorname{Im}(U(e')) = \{y \in U(Y) \mid U(d_0)(y) = U(d_1)(y)\} = \operatorname{Im}(U(e))$$

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using the hypothesis on e for the second equality.

As $d_0 \circ e = d_1 \circ e$, the universal property of the equaliser (X', e') yields a unique morphism $\varphi : X \to X'$ with $e' \circ \varphi = e$.

We claim $U(\varphi)$ is a bijection. If $U(\varphi)(x) = U(\varphi)(y)$ then applying $U(e')$ gives $U(e)(x) = U(e')(U(\varphi)(x)) = U(e')(U(\varphi)(y)) = U(e)(y)$, so $x = y$ as $U(e)$ is injective; hence $U(\varphi)$ is injective. Given $x' \in U(X')$, we have $U(e')(x') \in \mathrm{Im}(U(e')) = \mathrm{Im}(U(e))$, so $U(e')(x') = U(e)(x)$ for some $x \in U(X)$, that is $U(e')(x') = U(e')(U(\varphi)(x))$, whence $x' = U(\varphi)(x)$ as $U(e')$ is injective; hence $U(\varphi)$ is surjective.

So $U(\varphi)$ is a bijection, hence an isomorphism in **Set**. Since $\mathcal{C}$ is algebraic, U reflects isomorphisms (2.6.189), so φ is an isomorphism in $\mathcal{C}$.

Since (X', e') is an equaliser for (d_0, d_1) and $\varphi : X \to X'$ is an isomorphism with $e' \circ \varphi = e$, by (2.6.137) (X, e) is also an equaliser for (d_0, d_1), as required. ∎

Proposition 2.6.198 (Every Object has a Zero Element)
Let $\mathcal{C}$ be an algebraic pointed category. Then a zero object $\mathbf{0}$ consists of precisely one element, denoted by $0_{\mathbf{0}}$.

Every object A has a unique element 0_A such that $U\big(0_{\mathbf{0},A}\big)(0) = 0_A$ for any (resp. all) zero objects $\mathbf{0}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Since $\mathcal{C}$ is an algebraic category, U creates limits (2.6.189), so by (2.6.87) U preserves limits, and this holds unconditionally as the target category is **Set**. In particular U preserves terminal objects, these being the limit over the empty diagram.

As $\mathbf{0}$ is a zero object it is in particular a terminal object of $\mathcal{C}$ (2.6.55), so $U(\mathbf{0})$ is a terminal object of **Set**. A terminal object of **Set** is a singleton, so $U(\mathbf{0})$ consists of precisely one element, which we denote $0_{\mathbf{0}}$.

Now fix an object A and a zero object $\mathbf{0}$, and set

$$0_A \coloneqq U\big(0_{\mathbf{0},A}\big)(0_{\mathbf{0}}) \in U(A)$$

where $0_{\mathbf{0},A} : \mathbf{0} \to A$ is the unique morphism afforded by $\mathbf{0}$ being an initial object (2.6.48), and $0_{\mathbf{0}} \in U(\mathbf{0})$ is the unique element found above.

It remains to check that 0_A does not depend on the choice of zero object. Let $\mathbf{0}'$ be another zero object, with associated element $0'_A \coloneqq U\big(0_{\mathbf{0}',A}\big)(0_{\mathbf{0}'})$. Since $\mathbf{0}, \mathbf{0}'$ are both initial objects (2.6.55), by uniqueness of initial objects (2.6.48) there is a (unique) isomorphism $\varphi : \mathbf{0} \to \mathbf{0}'$. As $\mathrm{Mor}(\mathbf{0}, A)$ is a singleton, being initial, we have

$$0_{\mathbf{0}',A} \circ \varphi = 0_{\mathbf{0},A}$$

so applying U and evaluating at $0_{\mathbf{0}}$

$$U\big(0_{\mathbf{0},A}\big)(0_{\mathbf{0}}) = U\big(0_{\mathbf{0}',A}\big)(U(\varphi)(0_{\mathbf{0}}))$$

Now $U(\varphi) : U(\mathbf{0}) \to U(\mathbf{0}')$ is a map between singletons, so it necessarily sends $0_{\mathbf{0}}$ to the unique element $0_{\mathbf{0}'}$ of $U(\mathbf{0}')$. Hence

$$0_A = U\big(0_{\mathbf{0},A}\big)(0_{\mathbf{0}}) = U\big(0_{\mathbf{0}',A}\big)(0_{\mathbf{0}'}) = 0'_A$$

so 0_A is independent of the choice of zero object, as required. ∎

Proposition 2.6.199 (Kernels in Algebraic Categories)
Let $\mathcal{C}$ be an algebraic pointed category, and consider a diagram

$$\ker(\alpha) \xrightarrow{i} Y \xrightarrow{\alpha} Z$$

Then $(i, \ker(\alpha))$ is a kernel for α if and only if

a) $U(i)$ is injective
b) $\mathrm{Im}(U(i)) = \{y \in U(Y) \mid U(\alpha)(y) = 0_Z\}$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Since $\mathcal{C}$ is pointed, by (2.6.150) $(i, \ker(\alpha))$ is a kernel for α if and only if i is an equaliser for $\big(\alpha, 0_{Y,Z}\big)$, where $0_{Y,Z} : Y \to Z$ denotes the zero morphism (2.6.54).

As $\mathcal{C}$ is algebraic, (2.6.197) (applied with $d_0 = \alpha$, $d_1 = 0_{Y,Z}$) shows this holds if and only if $U(i)$ is injective and

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$$\mathrm{Im}(U(i)) = \{y \in U(Y) \mid U(\alpha)(y) = U(0_{Y,Z})(y)\}$$

It remains to identify $U(0_{Y,Z})(y)$ with 0_Z for every $y \in U(Y)$. Write $0_{Y,Z} = 0_{\mathbf{0},Z} \circ !_Y$, where $!_Y : Y \to \mathbf{0}$ and $0_{\mathbf{0},Z} : \mathbf{0} \to Z$ are the unique morphisms afforded by $\mathbf{0}$ being terminal, resp. initial (2.6.54). By (2.6.198), $U(\mathbf{0})$ is a singleton $\{0_{\mathbf{0}}\}$, so $U(!_Y)(y) = 0_{\mathbf{0}}$ for every $y \in U(Y)$, and therefore

$$U(0_{Y,Z})(y) = U(0_{\mathbf{0},Z})(U(!_Y)(y)) = U(0_{\mathbf{0},Z})(0_{\mathbf{0}}) = 0_Z$$

using the definition of 0_Z from (2.6.198). Hence

$$\{y \in U(Y) \mid U(\alpha)(y) = U(0_{Y,Z})(y)\} = \{y \in U(Y) \mid U(\alpha)(y) = 0_Z\}$$

so combining with the above, $(i, \ker(\alpha))$ is a kernel for α if and only if $U(i)$ is injective and $\mathrm{Im}(U(i)) = \{y \in U(Y) \mid U(\alpha)(y) = 0_Z\}$, as required. ∎

2.6.24 Exact Sequences

Definition 2.6.200 (Exact Sequence)
Let $\mathcal{C}$ be a pointed category with an image of every morphism (2.6.167) – e.g. a pointed algebraic category (2.6.184), which has regular epi-mono factorisations (2.6.196), hence images (2.6.174).

*Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be a pair of composable morphisms and $(\mathrm{Im}(f), e, m)$ an image of f. We say the pair is **exact at Y** if $\mathrm{Im}(f) = \ker(g)$, that is, $(\mathrm{Im}(f), m)$ is (literally) a kernel of g.*

This does not depend on the choice of image of f: images are unique up to (unique) isomorphism (2.6.167), so given any other image $(\mathrm{Im}'(f), e', m')$ of f, there is an isomorphism $\theta : \mathrm{Im}(f) \to \mathrm{Im}'(f)$ with $m' \circ \theta = m$, and by (2.6.137) (applied to θ and to θ^{-1}), $(\mathrm{Im}(f), m)$ is a kernel of g if and only if $(\mathrm{Im}'(f), m')$ is.

Proposition 2.6.201 (Concrete Criterion, Algebraic Case)
Let $\mathcal{C}$ be a pointed algebraic category, and $\mathrm{Im}(f)$ the canonical (literal subalgebra) construction. Then the pair is exact at Y (2.6.200) if and only if $f(X) = g^{-1}(\{0\})$, as subsets of Y.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $I \coloneqq f(X) \subseteq Y$ for the canonical image of f (2.6.196), a subalgebra of Y with the literal inclusion $m : I \hookrightarrow Y$ as structure map, so $f = m \circ e$ (with e the corestriction of f) is a regular epi-mono factorisation of f. By (2.6.200) (independence of representatives), we may use this particular representative: the pair is exact at Y if and only if (I, m) is a kernel of g, that is, by (2.6.150), m is an equaliser of $(g, 0)$. By (2.6.197), since m is (already) injective, this holds if and only if $\mathrm{Im}(m) = I$ equals $\{y \in Y \mid g(y) = 0\} = g^{-1}(\{0\})$. ∎

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2.7 Abelian Categories

Following [Fre64] we define an Abelian category without reference to additivity, however to make the presentation simpler we assume finite limits and colimits exist.

Definition 2.7.1
We say a category $\mathcal{C}$ is Abelian if it

a) is pointed,
b) has finite limits and colimits, and
c) is binormal

Remark 2.7.2 (Abelian is Self-Dual)
If $\mathcal{C}$ is Abelian then so is $\mathcal{C}^{op}$. Indeed $\mathcal{C}^{op}$ is again pointed with the same zero object (2.6.54); finite limits in $\mathcal{C}$ become finite colimits in $\mathcal{C}^{op}$ and vice versa (2.6.92), so $\mathcal{C}^{op}$ again has finite limits and colimits; and $\mathcal{C}^{op}$ is binormal, since monic/epic swap under passing to $\mathcal{C}^{op}$ (2.6.33) while $\ker(f)/\mathrm{coker}(f)$ swap (2.6.154), so a monomorphism being a kernel in $\mathcal{C}$ corresponds to an epimorphism being a cokernel in $\mathcal{C}^{op}$ and vice versa.

Lemma 2.7.3
Let $\mathcal{C}$ be a normal pointed category and $m : I \hookrightarrow Y$ a monomorphism with cokernel $p : Y \to \mathrm{coker}(m)$. Then $f : X \to Y$ factors through m if and only if $p \circ f = 0$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Suppose $f = m \circ g$ for some $g : X \to I$. Then $p \circ f = p \circ m \circ g = 0 \circ g = 0$, since $p \circ m = 0$ by definition of the cokernel (2.6.153).

Conversely, suppose $p \circ f = 0$. As $\mathcal{C}$ is normal and m is monic, m is a normal monomorphism (2.6.163), i.e. (m, I) is the kernel of some morphism (2.6.159). By (2.6.162) it follows that $m = \ker(\mathrm{coker}(m)) = \ker(p)$. So by (2.6.150), $p \circ f = 0$ gives a (unique) morphism $g : X \to I$ such that $m \circ g = f$, that is, f factors through m. ∎

Lemma 2.7.4
Let $\mathcal{C}$ be an Abelian category and $f : X \to Y$ a morphism. Then the following are equivalent

a) f is epic
b) $\mathrm{Im}(f) = Y$ (i.e. $(Y, f, 1_Y)$ is an image for f)
c) $\mathrm{coker}(f) = 0$

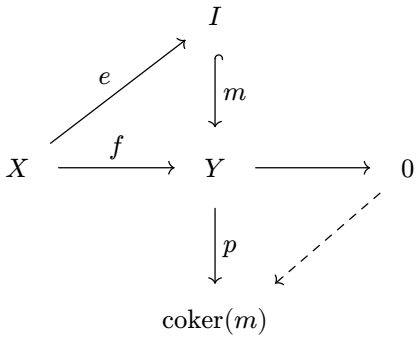
Dually, the following are equivalent

a) f is monic
b) $\mathrm{Coim}(f) = X$ (i.e. $(X, 1_X, f)$ is a coimage for f)
c) $\ker(f) = 0$

Proof. We only prove the first half, with the second following immediately by duality: since $\mathcal{C}$ is Abelian, so is $\mathcal{C}^{op}$ (2.7.2), and by (2.6.169) and (2.6.154) the conditions a)–c) are respectively dual to monic, $\mathrm{Coim}(f) = X$, and $\ker(f) = 0$.

a) $\implies$ c) This is (2.6.151).

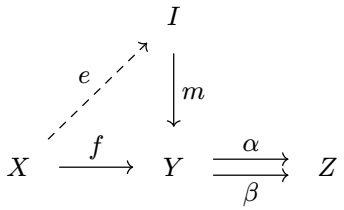
c) $\implies$ b) Suppose $f = m \circ e$ where $m : I \hookrightarrow Y$ is a monomorphism. Consider the cokernel $p : Y \to \mathrm{coker}(m)$



Then $p \circ m = 0 \implies p \circ f = 0$. As $\mathrm{coker}(f) = 0$ there is a morphism $0 \to Z$ such that the bottom right triangle commutes, i.e. $p = 0$. Then $p \circ 1_Y = 0$ and so by (2.7.3) there exists $j : Y \to I$ such that $m \circ j = 1_Y$. As m is monic this factorisation is unique and we conclude that $\mathrm{Im}(f) = Y$.

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b) $\implies$ a) Suppose that $\alpha \circ f = \beta \circ f$ and let $m : \mathrm{eq}(\alpha, \beta) \to Y$ be the equaliser, which is therefore monic (2.6.135).

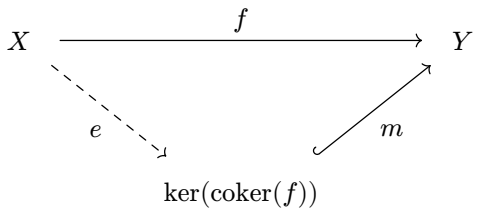


By the universal property of the equaliser there is a (unique) factorisation $f = m \circ e$. Since $(Y, f, 1_Y)$ is the image of f (hypothesis b)) and (I, e, m) is another factorisation of f through a monomorphism, by the universal property of the image (2.6.167) there is a unique morphism $j : Y \to I$ such that $m \circ j = 1_Y$. Therefore m is split-epic so by (2.6.36) it is an isomorphism. Since $(I, m) = \mathrm{eq}(\alpha, \beta)$, by (2.6.136) we conclude $\alpha = \beta$. ∎

2.7.1 Images, Coimages and the First Isomorphism Theorem

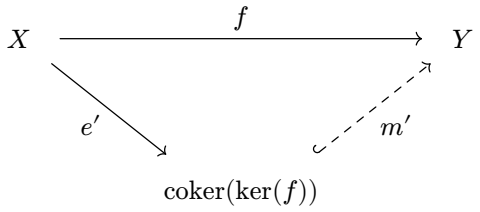
In case of modules we have $\mathrm{coker}(f) = Y/\mathrm{Im}(f)$ and $\ker(\mathrm{coker}(f)) = \mathrm{Im}(f)$ so we might expect this relation to hold in the abstract setting.

Proposition 2.7.5 ($\mathrm{Im}(f) = \ker(\mathrm{coker}(f))$ and $\mathrm{Coim}(f) = \mathrm{coker}(\ker(f))$)
Let $\mathcal{C}$ be a pointed category with kernels and cokernels and $f : X \to Y$ a morphism. Then there is a unique morphism $e : X \to \ker(\mathrm{coker}(f))$ making the following diagram commute



When $\mathcal{C}$ is Abelian e is an epimorphism and $\mathrm{Im}(f) = (\ker(\mathrm{coker}(f)), e, m)$.

Dually, there is a unique morphism $m' : \mathrm{coker}(\ker(f)) \to Y$ such that the following diagram commutes

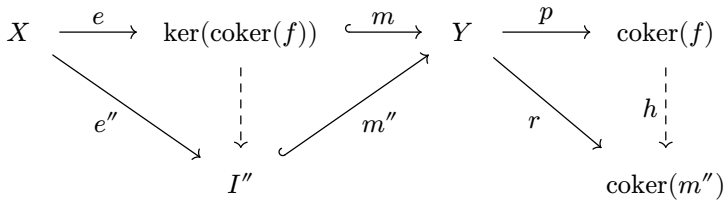


When $\mathcal{C}$ is Abelian m' is a monomorphism and $\mathrm{Coim}(f) = (\mathrm{coker}(\ker(f)), e', m')$.

Proof. Let $p : Y \to \mathrm{coker}(f)$ be the cokernel of f, so by (2.6.153) $p \circ f = 0$. Then by (2.6.150), applied to the kernel $m = \ker(p)$, e exists and is unique.

Dually, let $i : \ker(f) \to X$ be the kernel of f, so by (2.6.150) $f \circ i = 0$. Then by (2.6.153), applied to the cokernel $e' = \mathrm{coker}(i)$, m' exists and is unique.

Now suppose additionally that $\mathcal{C}$ is Abelian. We first show that $(\ker(\mathrm{coker}(f)), e, m) = \mathrm{Im}(f)$. By (2.6.135) m is a monomorphism. Consider another factorisation $f = m'' \circ e''$ with m'' a monomorphism



Let $r := \mathrm{coker}(m'')$. Then $r \circ f = r \circ m'' \circ e'' = 0$. Therefore, by the universal property of $\mathrm{coker}(f)$ (2.6.153), there is a unique morphism $h : \mathrm{coker}(f) \to \mathrm{coker}(m'')$ such that $h \circ p = r$, and we deduce that $r \circ m = h \circ p \circ m = h \circ 0 = 0$. Therefore by (2.7.3) m factors through m'' as required. Uniqueness follows because m'' is monic and therefore $(\ker(\mathrm{coker}(f)), e, m)$ is an image for f

Next we show e is an epimorphism. By (2.7.4) applied to e (with codomain $I := \ker(\mathrm{coker}(f))$), it suffices to show $\mathrm{Im}(e) = I$, i.e. that $(I, e, 1_I)$ is the image of e. Suppose $e = m_1 \circ e_1$ for some monomorphism $m_1 : I_1 \to I$. Then $f =$

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$m \circ e = (m \circ m_1) \circ e_1$ is a factorisation of f through the monomorphism $m \circ m_1$, so by the universal property of the image (I, e, m) of f (2.6.167) (established above) there is a unique morphism $g : I \to I_1$ such that $(m \circ m_1) \circ g = m$; as m is monic this gives $m_1 \circ g = 1_I$. This is precisely the factorisation required for $(I, e, 1_I)$ to be the image of e, so $\mathrm{Im}(e) = I$, and therefore e is an epimorphism.

The remaining claims for the dual statement now follow by duality, applying the above to f^{op} in $\mathcal{C}^{\mathrm{op}}$, which is again Abelian (2.7.2): by (2.6.154), $\ker_{\mathcal{C}^{\mathrm{op}}}(\mathrm{coker}_{\mathcal{C}^{\mathrm{op}}}(f^{\mathrm{op}}))$ is identified with $\mathrm{coker}(\ker(f))$, matching (by uniqueness of the first paragraph, dualized) our m' and e', and by (2.6.33) and (2.6.169) the epic/image conclusions in $\mathcal{C}^{\mathrm{op}}$ translate to m' monic and $(\mathrm{coker}(\ker(f)), e', m') = \mathrm{Coim}(f)$ in $\mathcal{C}$. ∎

In the category of modules we have $\mathrm{coker}(\ker(f)) = X/\ker(f)$ and a canonical isomorphism $X/\ker(f) \xrightarrow{\sim} \mathrm{Im}(f) = \ker(\mathrm{coker}(f))$, so we may expect something similar to hold true in the Abelian setting.

Proposition 2.7.6 (First Isomorphism Theorem)
Let $\mathcal{C}$ be a pointed category with kernels and cokernels and $f : X \to Y$ a morphism. Then there is a unique morphism $\theta : \mathrm{coker}(\ker(f)) \to \ker(\mathrm{coker}(f))$ such that the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ \downarrow{\scriptstyle e'} & & \uparrow{\scriptstyle m} \\ \mathrm{coker}(\ker(f)) & \xrightarrow{\quad \theta \quad} & \ker(\mathrm{coker}(f)) \end{array}$$

When $\mathcal{C}$ is an Abelian category then θ is an isomorphism. Furthermore both $(\mathrm{coker}(\ker(f)), e', m \circ \theta)$ and $(\ker(\mathrm{coker}(f)), \theta \circ e', m)$ are epi-mono factorisations of f and both an image and coimage.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (2.7.5) there is a unique morphism $m' : \mathrm{coker}(\ker(f)) \to Y$ such that $f = m' \circ e'$.

Since $q \circ f = 0$, where $q : Y \to \mathrm{coker}(f)$ is the cokernel of f (2.6.153), we have $q \circ m' \circ e' = q \circ f = 0$; as e' is epic (2.6.135), $q \circ m' = 0$. So by the universal property of $m = \ker(q)$ (2.6.150), there is a unique morphism $\theta : \mathrm{coker}(\ker(f)) \to \ker(\mathrm{coker}(f))$ such that $m \circ \theta = m'$. Combining, $m \circ \theta \circ e' = m' \circ e' = f$, as required.

For uniqueness, suppose θ' also satisfies $m \circ \theta' \circ e' = f$. Then $m \circ (\theta \circ e') = f = m \circ (\theta' \circ e')$, and since m is monic (2.6.135), $\theta \circ e' = \theta' \circ e'$; as e' is epic (2.6.135), $\theta = \theta'$.

Suppose now $\mathcal{C}$ is Abelian, so that $f = m \circ e$ is the image factorisation and $f = m' \circ e'$ is the coimage factorisation (2.7.5). Since $m \circ \theta \circ e' = f = m \circ e$ and m is monic (2.6.135), we may conclude that $\theta \circ e' = e$.

Since $m \circ \theta = m'$ is monic, θ is monic, and since $\theta \circ e' = e$ is epic, θ is epic. As $\mathcal{C}$ is Abelian it is binormal (2.7.1), hence balanced (2.6.164), so θ, being both monic and epic, is an isomorphism.

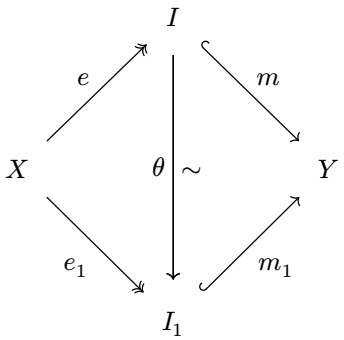
Finally, applying (2.6.172) to the coimage $(\mathrm{coker}(\ker(f)), e', m')$, and dually to the image $(\ker(\mathrm{coker}(f)), e, m)$, via the isomorphism θ shows each factorisation is both an image and a coimage of f. ∎

Proposition 2.7.7 (Epi-Mono Factorisation in an Abelian Category)
Let $\mathcal{C}$ be an Abelian category and $f : X \to Y$ a morphism. Then f factors as $f = m \circ e$ for some epimorphism e and monomorphism m

$$\begin{array}{ccccc} X & & \xrightarrow{\quad f \quad} & & Y \\ & {\scriptstyle e}\searrow & & \nearrow{\scriptstyle m} & \\ & & I & & \end{array}$$

and this factorisation is unique up to isomorphism: if $f = m_1 \circ e_1$ is another such factorisation, with $e_1 : X \to I_1$ epic and m_1 monic, then there is a unique morphism $\theta : I \to I_1$ such that the following diagram commutes, and θ is (moreover) an isomorphism

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Moreover, every epimorphism and every monomorphism of $\mathcal{C}$ is regular, so any such factorisation $f = m_1 \circ e_1$ is simultaneously a regular epi-mono and an epi-regular-mono factorisation of f, and hence (I_1, e_1, m_1) is both an image and a coimage of f (2.6.174).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Regularity. Since $\mathcal{C}$ is Abelian it is binormal (2.7.1), so every monomorphism is normal and every epimorphism is conormal (2.6.163), hence every monomorphism and every epimorphism of $\mathcal{C}$ is regular (2.6.160). In particular, for any epi-mono factorisation $f = m_1 \circ e_1$, e_1 is a regular epimorphism and m_1 is a regular monomorphism, so $f = m_1 \circ e_1$ is simultaneously a regular epi-mono and an epi-regular-mono factorisation of f, and by (2.6.174) (both halves), (I_1, e_1, m_1) is both an image and a coimage of f.

Existence. By (2.7.5), taking $I := \ker(\operatorname{coker}(f))$ with structure maps e, m, we obtain a factorisation $f = m \circ e$ with m monic and, since $\mathcal{C}$ is Abelian, e epic.

Uniqueness up to isomorphism. Let $f = m_1 \circ e_1$ be any other such factorisation, $e_1 : X \to I_1$. By the Regularity paragraph, e and e_1 are regular epimorphisms, so $f = m \circ e$ and $f = m_1 \circ e_1$ are regular epi-mono factorisations of f (2.6.173), and (2.6.178) gives a unique isomorphism $\theta : I \to I_1$ with $\theta \circ e = e_1$ and $m_1 \circ \theta = m$.

Alternatively, uniqueness may be shown directly from the universal property of the image (2.6.167), without invoking regular epi-mono factorisations at all.

Uniqueness up to isomorphism. Let $f = m_1 \circ e_1$ be any other epi-mono factorisation, $e_1 : X \to I_1$. Since $(I, e, m) = \operatorname{Im}(f)$ (2.7.5) and m_1 is monic, the universal property of the image (2.6.167) gives a unique morphism $\theta : I \to I_1$ such that $m_1 \circ \theta = m$, and (as noted there) $\theta \circ e = e_1$.

We show θ is an isomorphism. If $\theta k_1 = \theta k_2$ then $m_1 \theta k_1 = m_1 \theta k_2$, i.e. $m k_1 = m k_2$, and since m is monic, $k_1 = k_2$; so θ is monic. Since $\theta \circ e = e_1$ is epic, θ is epic. As $\mathcal{C}$ is Abelian it is binormal (2.7.1), hence balanced (2.6.164), so θ, being both monic and epic, is an isomorphism. ∎

2.7.2 Exact Sequences

Lemma 2.7.8 (Cancellation for Kernels and Cokernels)
Let $\mathcal{C}$ be a pointed category with kernels and cokernels.

a) If $h = \psi \circ p$ with ψ monic, then (K, i) is a kernel of p if and only if it is a kernel of h.
b) If $f = m \circ e$ with e epic, then (Q, q) is a cokernel of m if and only if it is a cokernel of f.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

a) For any k into the common domain of h, p, $h \circ k = \psi \circ p \circ k$; since ψ is monic, $\psi \circ p \circ k = 0 \iff p \circ k = 0$, so $h \circ k = 0 \iff p \circ k = 0$. In particular $p \circ i = 0 \iff h \circ i = 0$, and for any i', $p \circ i' = 0 \iff h \circ i' = 0$; so the defining conditions of (2.6.150) hold for (K, i) and p if and only if they hold for (K, i) and h, that is, (K, i) is a kernel of p if and only if it is a kernel of h.

b) Dually, for any q' out of the common codomain of f, m, $q' \circ f = q' \circ m \circ e$; since e is epic, $q' \circ m \circ e = 0 \iff q' \circ m = 0$, so $q' \circ f = 0 \iff q' \circ m = 0$; in particular $q \circ m = 0 \iff q \circ f = 0$, and for any q', $q' \circ m = 0 \iff q' \circ f = 0$; so the defining conditions of (2.6.153) hold for (Q, q) and m if and only if they hold for (Q, q) and f, that is, (Q, q) is a cokernel of m if and only if it is a cokernel of f. ∎

Proposition 2.7.9 (Criteria for Exactness)
Let $\mathcal{C}$ be an Abelian category, $X \xrightarrow{f} Y \xrightarrow{g} Z$ a pair of composable morphisms, $f = m \circ e$ an epi-mono factorisation of f and $g = m' \circ e'$ an epi-mono factorisation of g. Then the following are equivalent:

a) $m = \operatorname{Im}(f) = \ker(g)$ (i.e. exactness in the sense of (2.6.200));

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b) $g \circ f = 0$*, and the canonical morphism* $\varphi : \mathrm{Im}(f) \to \ker(g)$ *is epic;*
c) $\mathrm{coker}(f) = \mathrm{Coim}(g) = e'$*.*
d) $g \circ f = 0$*, and the canonical morphism* $\varphi : \mathrm{coker}(f) \to \mathrm{Coim}(g)$ *is monic.*

This does not depend on the choice of factorisation of either f *or* g*.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $k : \ker(g) \to Y$ for a kernel of g.

Independence of factorisation of f. By (2.6.200) (independence of representatives), applied with $\mathcal{C}$ Abelian – so every epi-mono factorisation of f is (in particular) an image of f (2.7.7) – exactness of the pair at Y, in particular the condition $m = \ker(g)$, does not depend on the choice of epi-mono factorisation of f: if $f = m_1 \circ e_1$ is another, then $m = \ker(g)$ if and only if $m_1 = \ker(g)$.

Independence of factorisation of g. If $g = m_1' \circ e_1'$ is another epi-mono factorisation of g, (2.7.7) gives an isomorphism $\phi : I' \to I_1'$ with $\phi \circ e' = e_1'$, so by (2.6.138) (applied to ϕ and to ϕ^{-1}), $\mathrm{coker}(f) = e'$ if and only if $\mathrm{coker}(f) = e_1'$.

Preliminary. By (2.7.7), (I', e', m') is both an image and a coimage of g; in particular $\mathrm{Coim}(g) = e'$.

a) $\Longrightarrow$ b) Taking $\varphi := 1_I$ gives $k \circ \varphi = k = m$; as $g \circ k = 0$ (2.6.150), $g \circ f = g \circ m \circ e = g \circ k \circ e = 0$. So (by uniqueness) $\varphi = 1_I$ is the morphism of b), and is epic, being an isomorphism.

b) $\Longrightarrow$ a) Since $m = k \circ \varphi$ and m is monic, φ is monic: if $\varphi \circ a = \varphi \circ b$ then $m \circ a = k \circ \varphi \circ a = k \circ \varphi \circ b = m \circ b$, so $a = b$. As φ is also epic (b)), and $\mathcal{C}$ is Abelian hence balanced (2.6.164), φ is an isomorphism; so $m = k \circ \varphi$ exhibits $m = \ker(g)$, that is, a).

a) $\Longrightarrow$ c) By (2.7.8) b) (as e is epic), $\mathrm{coker}(f) = \mathrm{coker}(m)$; since $m = \ker(g)$ (a)), $\mathrm{coker}(m) = \mathrm{coker}(\ker(g)) = \mathrm{Coim}(g)$ (2.7.5), which by the Preliminary equals e'. So $\mathrm{coker}(f) = e'$, that is, c).

c) $\Longrightarrow$ a) By hypothesis $\mathrm{coker}(f) = e'$; since $g = m' \circ e'$, (2.7.8) a) (with m' in the role of the monomorphism and $e' = \mathrm{coker}(f)$ in the role of p) gives $\ker(g) = \ker(\mathrm{coker}(f))$. By (2.7.5), $\mathrm{Im}(f) = \ker(\mathrm{coker}(f))$, i.e. (as $\mathrm{Im}(f) = (I, e, m)$) $m = \ker(\mathrm{coker}(f)) = \ker(g)$, that is, a).

c) $\Longleftrightarrow$ d) By duality: apply a) $\Longleftrightarrow$ b), established above for an arbitrary composable pair in an Abelian category, to $g^{\mathrm{op}} : Z \to Y$ and $f^{\mathrm{op}} : Y \to X$ in $\mathcal{C}^{\mathrm{op}}$ (Abelian, by (2.7.2)), using the epi-mono factorisation $g^{\mathrm{op}} = (e')^{\mathrm{op}} \circ (m')^{\mathrm{op}}$ of g^{op} (dual to $g = m' \circ e'$). This gives: $(e')^{\mathrm{op}} = \ker_{\mathcal{C}^{\mathrm{op}}}(f^{\mathrm{op}})$ if and only if $f^{\mathrm{op}} \circ g^{\mathrm{op}} = 0$ and the canonical morphism $\mathrm{Im}_{\mathcal{C}^{\mathrm{op}}}(g^{\mathrm{op}}) \to \ker_{\mathcal{C}^{\mathrm{op}}}(f^{\mathrm{op}})$ is epic in $\mathcal{C}^{\mathrm{op}}$.

Translating via (2.6.154) ($\ker_{\mathcal{C}^{\mathrm{op}}}(f^{\mathrm{op}})$ corresponds to $\mathrm{coker}(f)$), (2.6.169) ($\mathrm{Im}_{\mathcal{C}^{\mathrm{op}}}(g^{\mathrm{op}})$ corresponds to $\mathrm{Coim}(g)$), and (2.6.33) (epic in $\mathcal{C}^{\mathrm{op}}$ corresponds to monic in $\mathcal{C}$), while $f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}}$ vanishes if and only if $g \circ f = 0$: this reads exactly as $e' = \mathrm{coker}(f)$ if and only if $g \circ f = 0$ and the canonical morphism $\mathrm{coker}(f) \to \mathrm{Coim}(g)$ is monic, that is, c) $\Longleftrightarrow$ d).

Independence of choice. a) does not depend on the choice of epi-mono factorisation of f (shown above), nor of g (as a) only references the fixed kernel k, not the chosen factorisation of g); so, by the equivalences just established, neither do b), c), d), each being equivalent to a). ∎

Remark 2.7.10
Conditions b)–d) give further equivalent characterisations of this same notion of exactness at Y*, in particular the self-dual form* $\mathrm{coker}(f) = \mathrm{Coim}(g)$*.*

2.7.3 Biproducts

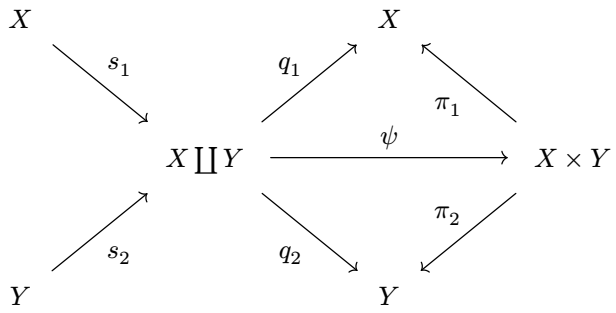
Lemma 2.7.11
Let $\mathcal{C}$ *be an Abelian category,* $X, Y \in \mathcal{C}$*, and* $\psi : X \coprod Y \to X \times Y$ *the canonical comparison morphism (2.6.104), where* π_1, π_2 *are the projections of* $X \times Y$ *and* s_1, s_2 *are the injections of* $X \coprod Y$*. Then* ψ *is monic.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

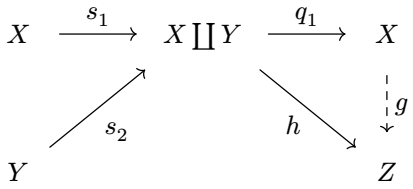
Write $X_1 := X$, $X_2 := Y$.

Maps out of the coproduct. By (2.6.93) let $q_1 := [1, 0] : X \coprod Y \to X$ and $q_2 := [0, 1] : X \coprod Y \to Y$, i.e. the morphisms induced by $q_k \circ s_i = \delta_{ki}$. Since $\pi_i \circ \psi \circ s_j = \delta_{ij} = q_i \circ s_j$ for all j, by the uniqueness clause of (2.6.93), $\pi_i \circ \psi = q_i$, that is, the following diagram commutes (which also displays the canonical injections s_1, s_2)

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$\operatorname{coker}(s_1) = q_2$ and $\operatorname{coker}(s_2) = q_1$. Indeed $q_1 s_2 = \delta_{12} = 0$ by construction. By (2.6.153), given $h : X \coprod Y \to Z$ with $h s_2 = 0$, we must produce a unique $g : X \to Z$ making the following diagram commute

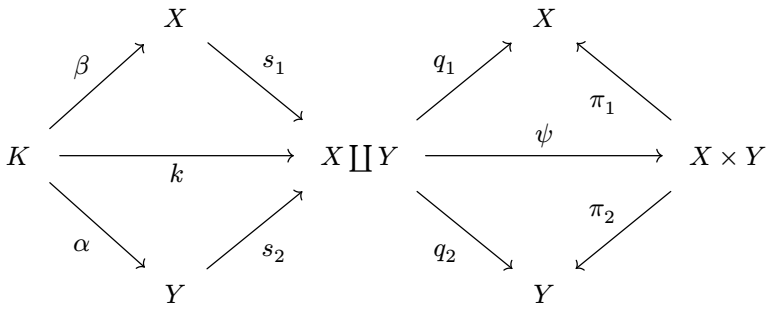


By the uniqueness clause of (2.6.93), $h = [h s_1, h s_2]$ and, for any $g : X \to Z$, $g \circ [1, 0] = [g, 0]$ (both sides restrict to the same maps along s_1, s_2). So since $h s_2 = 0$, $h = [h s_1, 0] = (h s_1) \circ [1, 0] = (h s_1) \circ q_1$. Setting $g := h s_1$ then ensures the diagram commutes. Since $q_1 s_1 = \delta_{11} = 1_X$, q_1 is split-epic, hence epic (2.6.35), so g is the unique such morphism. Therefore $\operatorname{coker}(s_2) = q_1$ as required. By the symmetric argument, $\operatorname{coker}(s_1) = q_2$.

$s_1 = \ker(q_2)$ and $s_2 = \ker(q_1)$. Since $q_1 s_1 = \delta_{11} = 1_X$, s_1 is split-monic, hence monic (2.6.35); similarly s_2 is monic (via $q_2 s_2 = \delta_{22} = 1_Y$). As $\mathcal{C}$ is Abelian it is binormal (2.7.1), so the monomorphism s_2 is normal (2.6.163), i.e. the kernel of some morphism (2.6.159), whence by (2.6.162), $s_2 = \ker(\operatorname{coker}(s_2)) = \ker(q_1)$. Similarly $s_1 = \ker(\operatorname{coker}(s_1)) = \ker(q_2)$.

$\ker(\psi) = 0$. Since π_1, π_2 jointly determine morphisms into $X \times Y$ (2.6.81), $\psi \circ k = 0$ iff $q_1 k = 0$ and $q_2 k = 0$, that is, $\ker(\psi)$ is the intersection, as subobjects of $X \coprod Y$, of $\ker(q_1) = s_2$ and $\ker(q_2) = s_1$. Informally, $\ker(q_1)$ is the "Y-summand" of $X \coprod Y$ (as q_1 is the identity on X and kills Y) and $\ker(q_2)$ is the "X-summand", so this intersection should vanish just as the coordinate subspaces $X \times \{0\}$ and $\{0\} \times Y$ meet only at the origin inside $X \oplus Y$; we verify this directly.

Suppose $k : K \to X \coprod Y$ satisfies $\psi \circ k = 0$. Then $q_1 k = \pi_1 \psi k = 0$ and $q_2 k = \pi_2 \psi k = 0$. Since $s_2 = \ker(q_1)$ there is $\alpha : K \to Y$ with $s_2 \alpha = k$, and since $s_1 = \ker(q_2)$ there is $\beta : K \to X$ with $s_1 \beta = k$, so $s_2 \alpha = s_1 \beta$, that is, the following diagram commutes



Composing with q_1: $q_1 s_2 \alpha = q_1 s_1 \beta$ gives $0 = \beta$. Therefore $k = s_1 \beta = 0$.

By (2.6.166), $\ker(\psi) = 0$, so by (2.7.4), ψ is monic. ∎

Proposition 2.7.12 (Finite Biproducts Exist)
Let $\mathcal{C}$ be an Abelian category. Then $\mathcal{C}$ has all finite biproducts (2.6.103).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (2.6.105) it suffices to show that binary biproducts exist. Fix $X, Y \in \mathcal{C}$; by (2.6.104) there is a unique comparison morphism $\psi : X \coprod Y \to X \times Y$, with π_1, π_2 the projections of $X \times Y$ and s_1, s_2 the injections of $X \coprod Y$, and it suffices to show ψ is an isomorphism.

By (2.7.11), ψ is monic. Since $\mathcal{C}^{\mathrm{op}}$ is again Abelian (2.7.2), in which $X \times Y$ (with π_1, π_2) and $X \coprod Y$ (with s_1, s_2) play the roles of the coproduct and product respectively (2.6.92), the canonical comparison morphism for X, Y in $\mathcal{C}^{\mathrm{op}}$ is

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exactly ψ^{op}, since both are characterised by the same defining equations $\pi_i \circ (-) \circ s_j = \delta_{ij}$. So applying (2.7.11) to $\mathcal{C}^{\mathrm{op}}$, ψ^{op} is monic in $\mathcal{C}^{\mathrm{op}}$, i.e. ψ is epic in $\mathcal{C}$ (2.6.33).

As $\mathcal{C}$ is binormal it is balanced (2.6.164), so ψ, being both monic and epic, is an isomorphism. By (2.6.104), a biproduct $X \oplus Y$ therefore exists. ∎

Proposition 2.7.13 ($\mathrm{Mor}(X,Y)$ is an Abelian Group)
Let $\mathcal{C}$ be an Abelian category and $X, Y \in \mathcal{C}$. Then every $f \in \mathrm{Mor}(X,Y)$ has an additive inverse; combined with (2.6.121) (applicable since $\mathcal{C}$ has all finite biproducts (2.7.12)), $(\mathrm{Mor}(X,Y), +)$ is an abelian group.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino. Following [Fre64] Theorem 2.39.

The underlying idea: in a category of modules, the inverse of the unitriangular matrix $\left(\begin{smallmatrix} 1 & 0 \\ f & 1 \end{smallmatrix}\right)$ is $\left(\begin{smallmatrix} 1 & 0 \\ -f & 1 \end{smallmatrix}\right)$, so the additive inverse of f can be read off from the inverse of this matrix; we therefore try to show that the same matrix M is an isomorphism in any Abelian category.

Fix $f : X \to Y$ and let $M := \left(\begin{smallmatrix} 1_X & 0 \\ f & 1_Y \end{smallmatrix}\right) : X \oplus Y \to X \oplus Y$ (2.6.108). By (2.6.122), matrices now multiply in the usual way, entry-by-entry, dot-product style.

M is monic. For $g = \left(\begin{smallmatrix} g_1 \\ g_2 \end{smallmatrix}\right) : Z \to X \oplus Y$

$$M \circ g = \begin{pmatrix} 1_X & 0 \\ f & 1_Y \end{pmatrix} \circ \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} = \begin{pmatrix} g_1 \\ f \circ g_1 + g_2 \end{pmatrix}$$

so $M \circ g = 0$ gives $g_1 = 0$ and $f \circ g_1 + g_2 = 0$; substituting, $f \circ 0 + g_2 = 0$, i.e. $g_2 = 0$, so $g = 0$. By (2.6.166), $\ker(M) = 0$, so by (2.7.4), M is monic.

M is epic. Dually, for $h = \left(\begin{smallmatrix} h_1 & h_2 \end{smallmatrix}\right) : X \oplus Y \to Z$

$$h \circ M = \begin{pmatrix} h_1 & h_2 \end{pmatrix} \circ \begin{pmatrix} 1_X & 0 \\ f & 1_Y \end{pmatrix} = \begin{pmatrix} h_1 + h_2 \circ f & h_2 \end{pmatrix}$$

so $h \circ M = 0$ gives $h_2 = 0$ and $h_1 + h_2 \circ f = 0$, i.e. $h_1 = 0$, so $h = 0$. Dually, $\mathrm{coker}(M) = 0$, so by (2.7.4), M is epic.

M is invertible. Since $\mathcal{C}$ is Abelian it is binormal (2.7.1), hence balanced (2.6.164), so M, being monic and epic, is an isomorphism.

Extracting the inverse. Write $M^{-1} = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$. Then

$$\begin{pmatrix} 1_X & 0 \\ f & 1_Y \end{pmatrix} \circ \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} p & q \\ f \circ p + r & f \circ q + s \end{pmatrix} = \begin{pmatrix} 1_X & 0 \\ 0 & 1_Y \end{pmatrix}$$

gives $p = 1_X$, $q = 0$, $f \circ p + r = 0$ and $f \circ q + s = 1_Y$. Substituting $p = 1_X$ into the third equation gives $f + r = 0$: so f has additive inverse $-f := r$. ∎

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